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공학박사학위논문

Physics-Informed Machine Learning for
Beam-Type Structures: Multiple Incipient
Crack Assessment in Metal Beams and
Time-Dependent Behavior Prediction of
Prestressed Concrete Beams

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Physics-Informed Machine Learning for Beam-Type
Structures: Multiple Incipient Crack Assessment in Metal
Beams and Time-Dependent Behavior Prediction of
Prestressed Concrete Beams

by
HWANG JIN HO

Submitted to

The Graduate School of Dong-A University in Partial Fulfillment
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in
Engineering

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**Physics-Informed Machine Learning for Beam-Type
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I have examined the final copy of this dissertation for format and content and recommend that it be accepted in partial fulfillment of the requirements for the degree of Doctor of Philosophy in Engineering.

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본 학위 논문은 학위청구자가 주저자 및 공동저자로 참여한 다음의 논문에 기초하여 작성되었으며, 공동저자의 “학술지 게재 논문의 학위논문 작성 동의서”를 득하였음.

This dissertation is written on the basis of the journal published papers in which the current candidate participated as a first author or co-author. The current candidate acquired “Agreement form on the use of journal published papers” from the co-authors of the papers.

- 1) J. H. Hwang and H. W. Park, "Physics-informed machine learning-assisted assessment of multiple incipient cracks using broadband modal frequencies from piezoelectric wafers bonded on a beam," *Structural Control and Health Monitoring* 2026 (2026): 5524571.
- 2) H. W. Park and J. H. Hwang, "Predicting the early-age time-dependent behaviors of a prestressed concrete beam by using physics-informed neural network," *Sensors* 23, (2023): 6649.

Abstract

Physics–Informed Machine Learning for Beam–Type Structures: Multiple Incipient Crack Assessment in Metal Beams and Time–Dependent Behavior Prediction of Prestressed Concrete Beams

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Although machine learning has been increasingly adopted in structural engineering as a flexible alternative to conventional numerical methods, its performance remains highly dependent on the availability of large, high–quality datasets, a condition rarely met in practice. To overcome this limitation, physics–informed machine learning (PIML), which integrates governing physical knowledge into the learning process, has emerged as a promising approach capable of enabling reliable predictions even under data–scarce conditions. This study presents a unified PIML framework for addressing two fundamental machine learning problem categories, classification and regression, through their representative applications in structural

engineering, under conditions where measurement data are scarce and the available physical models are necessarily simplified and cannot fully represent the actual structural behavior, with physical knowledge incorporated into the learning process to compensate for data scarcity in both cases.

In the first application, a PIML-assisted framework is proposed for assessing multiple incipient cracks in beam-type structures as a representative classification problem in structural health monitoring. A closed-form frequency equation derived from the phase-closure principle is employed as a physics-based surrogate model to efficiently generate large-scale training datasets of broadband modal frequencies at minimal computational cost. The augmented data are refined and used to train a classifier capable of identifying the number of cracks. Based on the identified crack count, grid optimization is subsequently applied to assess crack locations and depths. The proposed framework is validated through blind tests using both finite element-based and experimentally measured data, demonstrating reliable crack assessment performance even without prior knowledge of the crack count.

In the second application, a physics-informed neural network (PINN) is introduced for predicting the early-age time-dependent behavior of prestressed concrete girders and reconstructing their camber profiles as a representative regression problem in structural performance prediction. Integro-differential governing equations accounting for concrete creep, shrinkage, tendon relaxation, and the variation of elastic modulus are embedded directly into the neural

network loss function in dimensionless form to predict the time-dependent prestress loss. Based on the predicted prestress force, a second PINN model incorporating the curvature-deflection relationship is subsequently employed to reconstruct the camber profile along the girder span. The prestress loss prediction is validated against reference solutions obtained from the finite difference method, while the camber reconstruction is further validated against experimental measurements from an actual prestressed concrete girder, demonstrating accurate prediction of prestress loss and physically consistent camber reconstruction under limited data conditions.

The two applications collectively establish physics-informed learning as a practical and unified approach for data-scarce structural engineering problems, demonstrating consistent effectiveness across these two distinct problem types.

key word: Physics-informed machine learning (PIML); Physics-informed neural network (PINN); Classification; Regression; Broadband modal frequencies; Multiple incipient cracks; Crack assessment; Time-dependent behavior; Camber reconstruction; Structural health monitoring;

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I. Introduction

1.1 Research Background

Numerical methods for predicting the mechanical behavior of structural members and components have been extensively developed over the past decades, providing fundamental tools for structural analysis and design. Representative methods such as the finite element method (FEM) enable numerical simulation of stress, deformation, and vibration behaviors under various loading conditions. However, their accuracy strongly depends on idealized assumptions regarding material properties, boundary conditions, and external loads, making it difficult to incorporate discrepancies between measured and simulated data into these algorithms.

To overcome these modeling limitations, machine learning (ML) has gained increasing attention as a data-driven alternative. ML techniques identify patterns and correlations within data, enabling the modeling of complex mechanical behaviors without explicitly defining physical models. They have been widely applied to problems such as classifying the presence, location, and severity of structural damage from measured responses, predicting continuous system responses, and estimating material properties based on measured or simulated data. Unlike conventional numerical methods, ML does not require complicated discretization or modeling processes and can efficiently approximate nonlinear relationships once sufficient training data are available. However, the performance of ML strongly

depends on the availability and quality of training data, which limits its applicability under data-scarce conditions. This data dependency becomes particularly evident in structural engineering applications, where the number of measurable specimens is inherently limited and field measurements are often influenced by noise, uncertainty, and accessibility constraints, collectively reducing data availability and consequently restricting the practical applicability of ML.

Such limitations have motivated the development of physics-informed machine learning (PIML), which integrates physical relationships, such as equilibrium, compatibility, and conservation laws, into the learning process, thereby constraining it by both data and physics simultaneously. Unlike purely data-driven approaches, PIML enables reliable and physically consistent predictions even under limited data conditions, as the embedded physical knowledge compensates for the insufficiency of available measurements. Owing to these advantages, PIML has recently gained considerable attention in the engineering community for bridging the gap between conventional numerical simulation and purely data-driven modeling. Figure 1.1 illustrates this evolution from conventional numerical methods to data-driven ML and more recently to PIML frameworks.

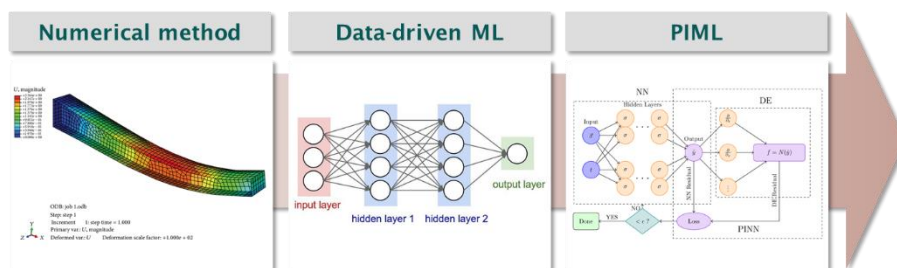


Figure 1.1 Evolution of structural analysis approaches

1.2 Research Motivations

1.2.1 Physics-Informed Machine Learning (PIML)

As mentioned in previous section, PIML has gained attention for its capability to incorporate fundamental physical or mechanical knowledge. Figure 1.2 illustrates the relationship between data availability and the amount of physics knowledge integrated into the learning process [Kar21]. Conventional data-driven ML operates in the rightmost region of Figure 1.2, where learning relies on only data. This dependency makes it difficult to achieve physically consistent predictions, especially when the available data are limited. In practice, obtaining large, high-quality datasets for supervised learning is rarely feasible in structural engineering. Field measurements are often limited and noisy, while numerical simulations required for data generation are computationally expensive and time-consuming. In contrast, PIML extends the learning framework toward the region highlighted in red in Figure 1.2, where the lack of sufficient data is compensated by incorporating physical knowledge into the learning process.

As the relationship between data and physics shifts leftward in Figure 1.2, it becomes increasingly physics-dominant, meaning that physical knowledge plays a greater role in guiding the learning process. When the embedded physics can accurately represent the behavior of the actual structural element, reliable predictions can be achieved even with a minimal amount of data. Conversely, if the physical model is not sufficiently accurate or complete, additional data can be incorporated

to compensate for the missing physical information. Through the interaction between data and physics, PIML alleviates the limitations of purely data-driven learning by incorporating both sources of information within a unified framework.

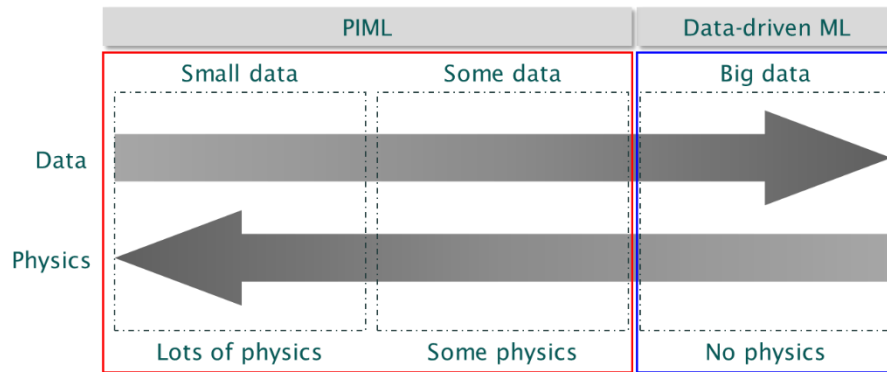


Figure 1.2 Scenarios of data availability and physics knowledge [Kar21]

Owing to these advantages, PIML has been applied across various engineering fields, such as mechanical, electrical, and materials engineering. However, applications in the civil engineering field remain limited, mainly due to the challenges associated with data scarcity and the complexity of accurately modeling structural behavior. Structural members in civil engineering often lie outside the idealized regions illustrated in Figure 1.2, where data availability is low (small data) and the embedded physics cannot fully represent the actual member behavior (some physics). The amount of available data is inherently limited because large-scale structural members are often unique in design and construction, making repeated experiments or standardized datasets difficult to obtain. Field measurements are also restricted to a few sensor locations due to accessibility and maintenance constraints,

which further reduces data availability. In addition, the governing physics of structural members in civil engineering are often incomplete, since simplified analytical or numerical models cannot fully capture nonlinear material behavior or time-dependent effects. Nevertheless, this very limitation makes PIML particularly essential for structural engineering, as it provides a framework that can integrate limited field data with imperfect physical models to assess and predict the mechanical behavior of structural members. The research motivation for applying PIML to civil engineering problems is discussed in the following sections and the problem categories to which PIML can be applied, along with their respective formulations and a unified PIML framework, are discussed in Section 1.3.

1.2.2 Research Trends and Literature Review

Tables 1.1 and 1.2 summarize the publication trends over the past five years (2020–2024), obtained from the Elsevier SciVal database using the Keyword Search function. The searches were performed with the keywords ML, Deep Learning (DL), and PIML in two disciplines. Papers published in the top 10 percent journals (by CiteScore percentile) were taken into account. Mechanical engineering was selected as a reference field because it represents one of the most active domains in applying data-driven ML and PIML.

Table 1.1 presents the publication trends of data-driven ML in the two disciplines, based on the search results using the keywords ML and DL. Mechanical engineering shows a significant increase from 1,449 papers in 2020 to 4,525 in 2024, totaling

14,916 publications over the past five years. In contrast, civil and structural engineering recorded 771 papers in 2020 and 2,066 in 2024, resulting in 6,836 publications in total, which is less than half of mechanical engineering. This result indicates that even data-driven ML, which has been widely adopted in other engineering domains, has been relatively slow to be applied in civil engineering. This delayed adoption is mainly attributed to the nature of civil engineering structures and their constituent members, which are often large-scale and site-specific, making it difficult to secure consistent and high-quality datasets for machine learning

Table 1.1 Number of publications keyword ML+DL

Year	Number of Publications by Discipline	
	Mechanical Engineering	Civil and Structural Engineering
2020	1,449	771
2021	2,159	1,081
2022	3,210	1,282
2023	3,573	1,636
2024	4,525	2,066
Total	14,916	6,836

Table 1.2 presents the publication trends of PIML in the same two disciplines. While the overall number of publications is smaller than that of data-driven ML, the growth rate of PIML has been accelerating in recent years, reflecting the rapid expansion of interest in physics-integrated learning frameworks. Mechanical engineering recorded a total of 641 papers over the past five years, whereas civil and structural engineering produced only 130 during the same period. This large discrepancy confirms that PIML research in the civil-engineering domain is still in

its infancy. Despite the unique characteristics of structural members in civil engineering that make the incorporation of physics knowledge particularly essential, the research activity in this field remains insufficient, clearly revealing the existing gap of knowledge.

Table 1.2 Number of publications keyword PIML

Year	Number of Publications by Discipline	
	Mechanical Engineering	Civil and Structural Engineering
2020	20	9
2021	53	12
2022	100	17
2023	179	36
2024	289	56
Total	641	130

Figure 1.3 illustrates the distribution of recent studies according to the type of problem and learning framework. The horizontal axis distinguishes between data-driven ML and PIML, whereas the vertical axis categorizes studies into classification and regression problems. Most previous studies have been concentrated in the data-driven ML domain, which has been widely applied across various engineering disciplines, including mechanical, electrical, and civil engineering. Representative review papers support this trend. For example, Cen23 reviewed 158 references related to classification problems in mechanical engineering, while Fla21 analyzed 141 studies focusing on classification problems in civil engineering. Similarly, Moe23, Mir24, and Kol25 presented more than one hundred application cases in the context of regression problems, confirming that data-driven ML has already been extensively and maturely applied across diverse engineering disciplines.

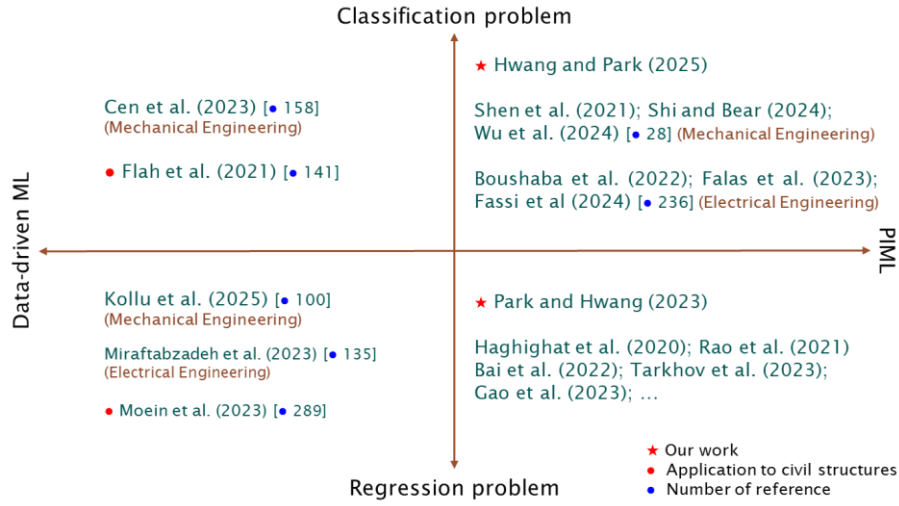


Figure 1.3 Distribution of recent studies

In contrast, studies employing PIML remain relatively scarce. Although the concept of integrating physical laws into learning frameworks has recently gained attention, it has been primarily explored in mechanical and materials engineering, while its application to structural members in civil engineering remains in its early stage. This uneven research distribution clearly shows that, despite the increasing interest in physics-informed approaches, the practical implementation of PIML across engineering disciplines is still far from being established. The representative studies and their application domains are summarized in Table 1.3.

Table 1.3 Representative studies on data-driven and physics-informed machine learning

Framework	Categories	Field	Subfield	References
Data-driven ML	Classification	Mechanical Engineering	Review (*158)	Cen23
		Civil Engineering	Review (*141)	Fla21
	Regression	Electrical Engineering	Review (*135)	Mir24
		Mechanical Engineering	Review (*25)	Kol25
		Civil Engineering	Review (*289)	Moe23
PIML	Classification	Mechanical Engineering	Fault detection	She21
			Fault diagnosis	Shi24
			Review (*28)	Wu24
		Electrical Engineering	Fault diagnosis	Bou22
			Power System Stability	Fal23
			Review (*236)	Fas24
	Classification	Civil Engineering	Crack assessment	Hwa26

Table 1.3 Continued

PIML	Regression	Materials Engineering	Material modeling	Yuc19
				Che20
				Gos20
		Chemical Engineering	Nonlinear diffusivity	Kad20
			Thermophysical systems	Tar23
		Mechanical Engineering	Fluid mechanics	Tar20
				Mao20
				Jin21
			Solid mechanics	Jag22a
				Hag20
				Rao21
			Biological modeling	
				Kis20
		Geoscience	Geophysics	Alk23

Table 1.3 Continued

		Applied Mathematics	Scientific computation	Shu21
PIML	Regression			Jag22b
		Civil Engineering	PSC behavior prediction	Par23

* Number of papers included in each review.

1.3 Problem Categories in PIML

1.3.1 Problem Categorization in ML and Conceptual Reformulation in PIML

In machine learning, problems are broadly categorized into supervised and unsupervised learning based on the availability of labeled data, as illustrated in Figure 1.4.

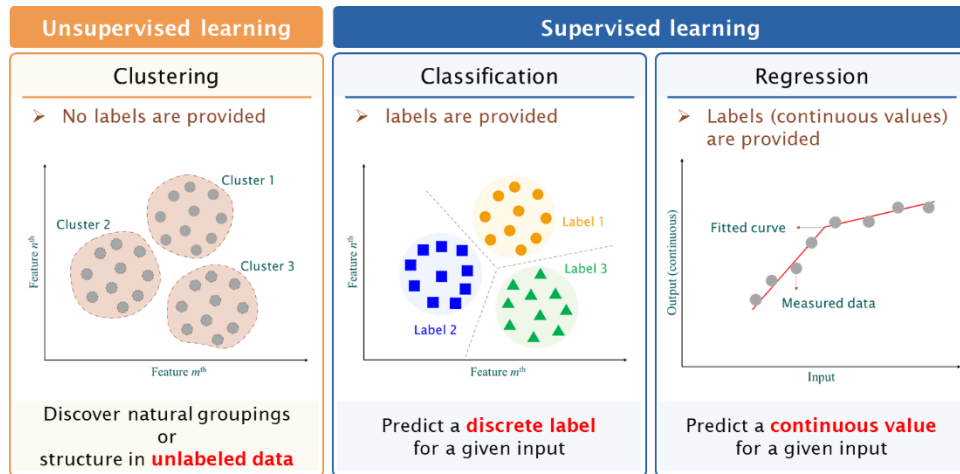


Figure 1.4 Problem categories of traditional data-driven ML

In supervised learning, both input and output data are available, and the learning system aims to approximate the function that maps inputs to outputs. Supervised learning problems are further categorized into classification and regression. Classification assigns each input to a discrete label (e.g., intact or damaged), whereas regression predicts continuous physical quantities, such as displacement or stress.

On the other hand, unsupervised learning is performed using only input data without predefined labels. Typical unsupervised techniques, such as clustering, attempt to group similar data points without labeled information, whereas supervised learning requires predefined labels for training. Representative algorithms commonly used in each category are summarized in Table 1.4.

Table 1.4 Variety of commonly used machine learning algorithms [Moe23]

Learning types	Categories	Algorithm
Unsupervised	Clustering	K-Means
		*Neural networks
		Gaussian mixture
		Hidden markov model
supervised	Classification	Hierarchical
		Support vector machines
		Naïve bayes
		Discriminate analysis
		Nearest neighbor
		*Neural networks
		Decision Trees
		Ensemble method
		*Nonlinear regression
		*Neural networks
	Regression	Decision Trees
		Ensemble method
		*Nonlinear regression

* Duplicate items, applicable to multiple categories

Data-driven ML, illustrated in Figure 1.4, has been applied to various engineering fields, including mechanical engineering [Cen23], electrical engineering [Fla21], and civil engineering [Mir24], but it still has limitations, as discussed in Section 1.1. To address these challenges, PIML has recently emerged as an effective framework. PIML integrates physics-based prior knowledge into the learning process to ensure accurate and physically consistent predictions, even under data-scarce conditions.

When physics-based prior knowledge is integrated into ML, the embedded physics act as implicit labels that transform originally unsupervised learning tasks into supervised ones. As a result, the conventional categorization shown in Figure 1.4 transitions to the structure illustrated in Figure 1.5, where problems traditionally considered unsupervised can effectively be reformulated as supervised ones. While unsupervised algorithms listed in Table 1.4 may still be utilized, the incorporation of physics-based constraints implies that PIML problems can no longer be regarded as purely unsupervised learning. This study demonstrates the application of PIML to the civil engineering field, encompassing two major tasks of classification and regression, which are addressed within a unified PIML workflow.

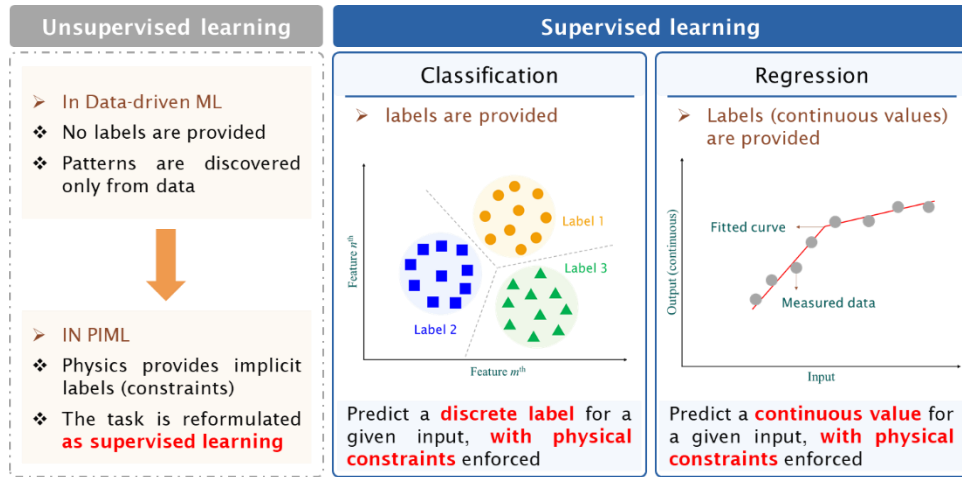


Figure 1.5 Two main problem categories in this study

1.3.2 Classification Problem

Classification problems aim to assign each data sample to one of several discrete labels based on its features. Figure 1.6 schematically illustrates this classification process, where the red circles denote the ground-truth class boundaries and the markers represent samples obtained from the actual structural element. In structural members, these labels represent different states or patterns of behavior (e.g., normal and abnormal conditions), while the data samples correspond to measured or simulated responses of the target elements.

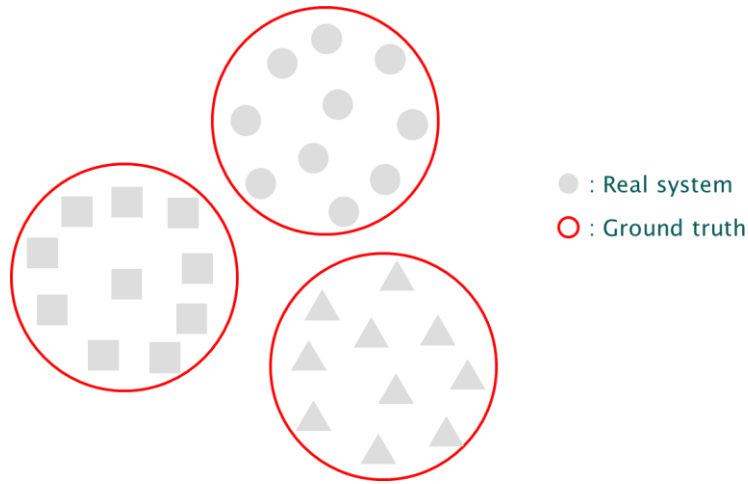


Figure 1.6 Example of classification problem

Classification has been a particularly important topic in structural health monitoring (SHM), as it enables the distinction between different structural conditions based on observed data. However, conventional numerical approaches still have difficulty incorporating the discrepancies with measured data into their analysis procedures. To overcome these limitations, data-driven ML techniques have

been introduced as an alternative for classifying structural states from measured responses. By training on the relationship between features and corresponding labels, ML can learn decision boundaries that accurately separate different structural conditions from the available feature data, as illustrated in Figure 1.6.

Although ML technologies have advanced and various algorithms have been developed to address classification problems, data-driven ML still has a fundamental limitation. Data-driven ML is highly dependent on data quality and quantity. When data-driven ML is applied to classify the state of real structural members, the lack of sufficient and reliable data often causes serious challenges. Reliable data-driven training requires a large amount of high-quality data, but data availability is inherently limited in real engineering applications. Figure 1.7 illustrates the limitations of classification using data-driven ML with limited data.

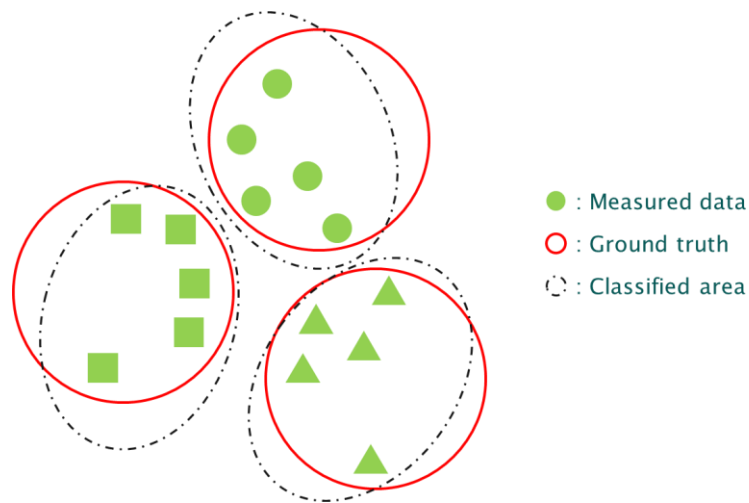


Figure 1.7 Limitations of classification using data-driven ML with limited data

When the data quality is poor or contains measurement noise, the model may learn misleading patterns that do not represent the actual physical behavior, making the classification results unreliable. Moreover, when the available dataset is insufficient, the data distribution tends to become biased, and the labeled samples are often concentrated in a specific region of the feature space. Under such conditions, the representativeness of the training data deteriorates, and the classified regions deviate from the ground truth.

When sufficient data are provided, the classification performance improves significantly, as shown in Figure 1.8. With abundant and diverse training data, the learned decision boundaries become closer to the ideal ones.

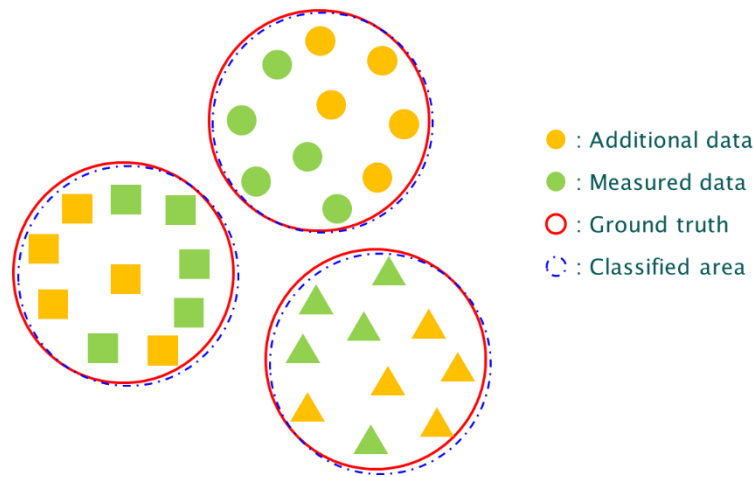


Figure 1.8 Example of classification with data-driven ML under sufficient data availability

Nevertheless, in most real-world field conditions, constructing such sufficient datasets is almost impossible. This difficulty arises because the number of available

specimens is limited, and measurements are often affected by environmental variability, boundary conditions, and inherent experimental errors. Figure 1.9 presents an example of applying PIML to a classification problem. PIML employs data augmentation to compensate for the insufficient amount of measured data by introducing virtual samples generated through a physics-based surrogate model.

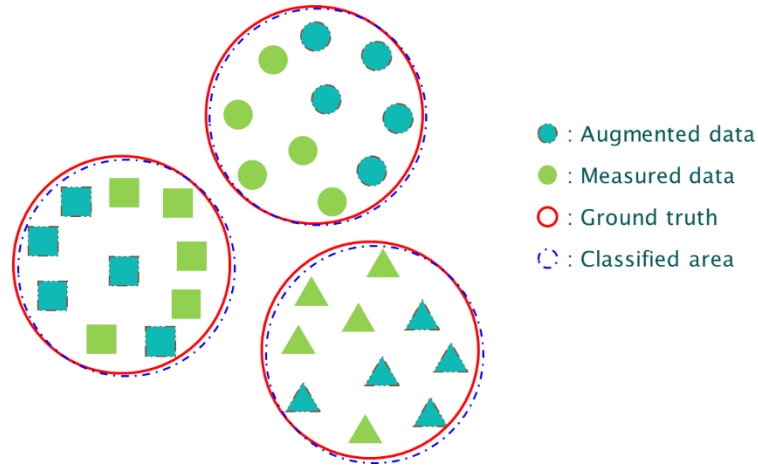


Figure 1.9 Securing sufficient data using physics-informed data augmentation

The surrogate model, which represents the structural behavior, should maintain physical consistency with the real system and possess sufficient fidelity. However, excessively high-fidelity models, such as full finite element analyses, require considerable computational cost and time for data generation, which undermines the purpose of efficient data augmentation. Therefore, the level of model fidelity should be determined by balancing physical accuracy and computational efficiency.

When physics-informed data augmentation is performed using a physics-based

surrogate model, adequate training data can be obtained even without measurements. Since the augmented data already contain physically defined labels, applying unsupervised learning within this framework becomes conceptually inconsistent. In this process, the training becomes comparable to that with sufficient experimental data, as illustrated in Figure 1.8. In other words, the surrogate model alone can provide enough data for the PIML framework to learn decision boundaries close to the ground truth, as shown in Figure 1.9.

However, including measured data in the training process reduces the differences between the surrogate model and the real system, further enhancing the reliability of the classification results. In this study, this PIML-based classification framework is applied to the assessment of multiple incipient cracks in beam-type structures, which is discussed in Chapter 2.

1.3.3 Regression Problem

Regression problems aim to predict continuous responses or physical quantities of a system based on given inputs. Figure 1.10 schematically illustrates this regression process, where the relationship between the input and output is expressed as a continuous mapping function. The figure conceptually represents the problem of identifying the red continuous curve from a few measured green data points that describe the actual physical behavior. In structural engineering, regression analysis is used to construct continuous distributions of quantities such as deflection, stress, and strain based on discrete data obtained from numerical simulations or

experimental measurements. However, conventional numerical approaches rely on extensive parameter calibration and often fail to accurately represent complex nonlinear relationships or interactions among variables observed in real measurements. Consequently, regression methods using ML have been explored as an alternative to learn these complex relationships directly from available data. Representative ML algorithms for regression are listed in Table 1.4.

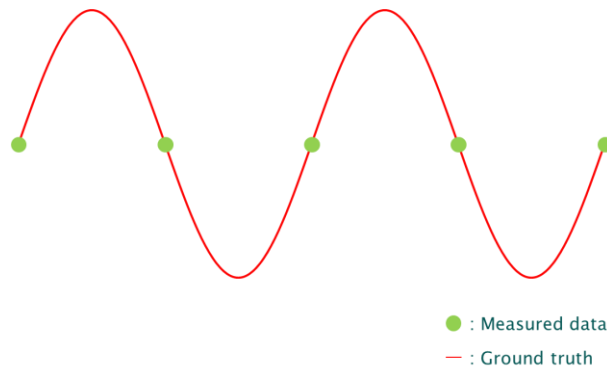


Figure 1.10 Example of regression problem

Despite these advances, ML regression also faces the same limitation. When the number of available data points is small, the model cannot uniquely determine the true functional relationship, leading to multiple plausible but physically inconsistent solutions. Figure 1.11 illustrates this limitation. When only few data points are available in the feature space, numerous possible curves can fit the given data. Among these, one may coincidentally approximate the ground truth, but the probability of obtaining such a curve is extremely low. This example highlights the

inherent non-uniqueness of regression with limited data and the difficulty of identifying the true functional relationship without additional physical constraints.

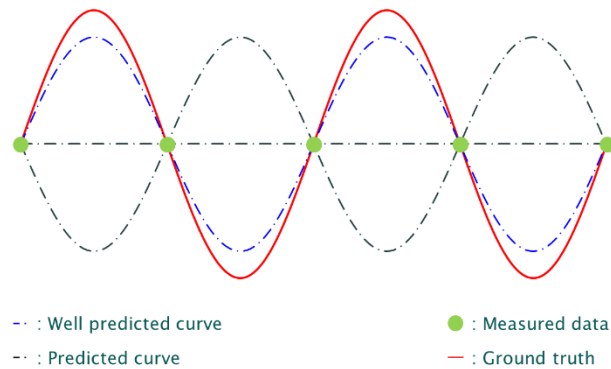


Figure 1.11 Limitations of regression using data-driven ML with limited data

As shown in Figure 1.12, when additional high-quality data are provided and data availability increases, the probability of obtaining a curve close to the ground truth also increases.

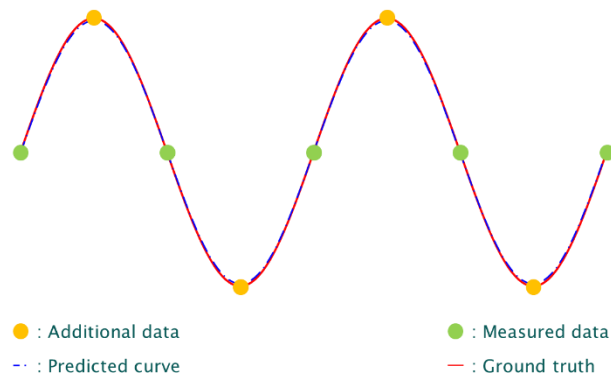


Figure 1.12 Example of regression with data-driven ML under sufficient data availability

With a larger number of data samples, the regression becomes more constrained, and the solution tends to converge toward a unique representation of the actual physical relationship. However, in most practical situations, obtaining sufficient field measurement data is often difficult because of limited measurement periods, sensor capacity, and environmental variability. This limitation can be addressed by introducing PIML, where the insufficient data are augmented with physics-informed samples for training.

Figure 1.13 presents an example of applying PIML to a regression problem. In the PIML framework, the physics-based governing equations are utilized to augment physics-consistent data.

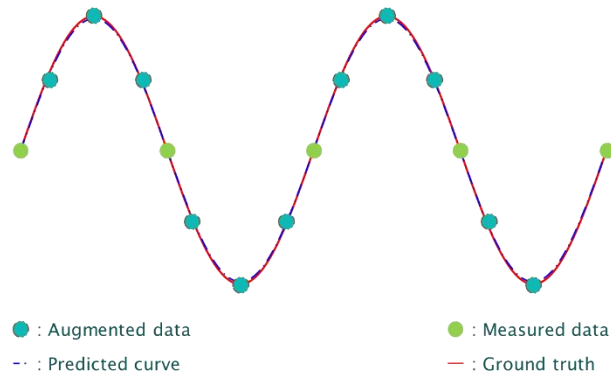


Figure 1.13 Securing sufficient data using physics-informed data augmentation

Through this physics-informed data augmentation, training and regression can be performed under conditions equivalent to having sufficient data, as illustrated in Figure 1.12, even without measured data. Furthermore, by jointly training with both

measured and physics-augmented data, PIML can enhance applicability to field conditions and improve prediction reliability. Therefore, PIML-based regression provides a powerful framework for predicting continuous or time-dependent responses of structural elements while maintaining consistency with the underlying physical behavior. In particular, when the governing equations are expressed in the form of differential or integral equations, a physics-informed neural network (PINN) can be employed, in which the governing equations are directly embedded into the loss function of the neural network and enforced at collocation points. In this study, PINN is applied as an example of PIML-based regression in the field of civil engineering to predict the time-dependent behavior and camber evolution of a prestressed concrete (PSC) girder, which is discussed in Chapter 3.

As discussed in Sections 1.3.2 and 1.3.3, both classification and regression problems in structural engineering share a common challenge in that reliable measurement data are inherently scarce, and PIML provides an effective framework for addressing this limitation in both problem categories. The current state of PIML research across engineering disciplines is examined in the following section.

1.4 Objectives and Scope

As discussed in the preceding sections, the limitations of purely data-driven ML under data-scarce conditions have motivated the emergence of PIML as an effective approach. Figure 1.14 illustrates a unified PIML framework applicable to both problem categories addressed in this study

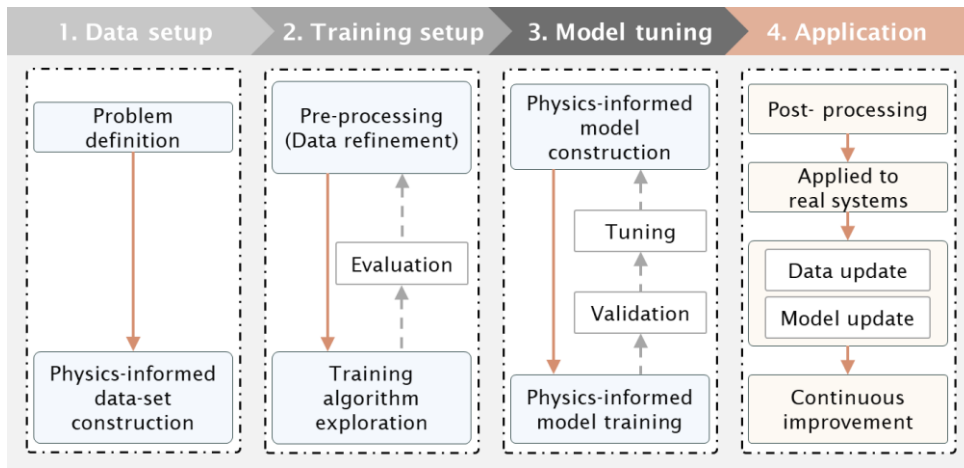


Figure 1.14 General framework of PIML

The overall scope of this dissertation is divided into two parts according to the problem type, both of which are addressed by applying the unified PIML framework proposed in Figure 1.14. In Chapter 2, crack assessment in beam-type structures is considered as a representative classification problem in structural engineering. This problem is inherently ill-posed, as multiple unknown parameters, such as crack locations and depths, must be identified from limited and often noisy measurement data. The difficulty becomes more pronounced for incipient cracks, whose influence on structural response is subtle and difficult to distinguish from measurement

uncertainty. Under such conditions, conventional data-driven approaches often struggle to achieve reliable identification. Physics-informed machine learning (PIML), which integrates governing physical principles into the learning process, provides an effective approach to address these challenges. For this reason, the assessment of multiple incipient cracks in beam-type structures is selected as a representative example of classification problems in this study.

A closed-form frequency equation derived from the phase-closure principle is employed as a physics-based surrogate model to augment the dataset for the assessment of multiple incipient cracks in beam-type structures. The augmented data are then refined and used to train a PIML model capable of identifying the number of cracks. The proposed approach introduces an observational bias into the learning process, as defined in the PIML taxonomy of Kar21. Observational bias is defined as the introduction of physical knowledge through carefully crafted data augmentation procedures, and the proposed approach is therefore classified as a form of physics-informed machine learning within this framework.

Based on the crack-count information identified by the PIML, a grid optimization technique is applied to assess the crack locations and depths. The proposed PIML-assisted method is validated using two datasets: broadband modal frequencies measured using surface-bonded piezoelectric wafers on actual cracked beams, and those computed from finite element simulations. In this chapter, a representative application of PIML for crack assessment in beam structures, one of the most common components in civil infrastructure, is presented. This study demonstrates

the applicability of PIML to classification tasks in beam-type structural elements.

In Chapter 3, predicting the early-age, time-dependent behavior of prestressed concrete (PSC) girders is considered as a representative example of regression problems in structural engineering, where strongly coupled physical phenomena must be modeled from limited measurement data over time. This problem involves complex interactions among concrete creep, shrinkage, tendon relaxation, and the variation of elastic modulus, making accurate prediction challenging, particularly in early-age stages where rapid changes occur. In addition, these time-dependent behaviors are difficult to measure directly in practice, despite their critical importance in quality control and serviceability of PSC structures. Based on the predicted time-dependent behavior, a subsequent model is introduced to reconstruct the camber response of PSC girders, which is a key indicator of structural performance. These characteristics make the problem well suited for physics-informed machine learning, which enables reliable prediction by incorporating governing physical laws into the learning process under limited data conditions.

Among various PIML approaches, a physics-informed neural network (PINN) is introduced as a representative method that introduces a learning bias by directly embedding governing differential equations into the loss function of the neural network [Kar21]. Two sets of governing equations are considered in the proposed framework. The first describes prestress loss through integro-differential equations that account for concrete creep, shrinkage, tendon relaxation, and the variation of elastic modulus. These equations are formulated in a dimensionless form and

embedded into the loss function of the network. The second represents the geometric relationship between deflection and curvature, expressed as a differential relation and incorporated to reconstruct the camber response of PSC girders over time. By embedding these two physical relationships, the proposed PINN captures both the time-dependent stress variation and the camber evolution of PSC girders without numerical discretization. The model is validated using numerical analyses and experimental data obtained from actual PSC girders. This chapter demonstrates that PINN can serve as an effective framework for modeling complex, time-dependent behaviors of PSC girders, demonstrating its potential for broader regression problems involving complex, time-dependent behaviors.

Through these two representative problem types, this dissertation demonstrates that physics-informed learning can be formulated as a unified framework for structural engineering applications. While different model architectures are adopted for classification and regression tasks, the overall workflow remains consistent across both applications. This unified formulation, in which Chapter 2 employs an observational bias and Chapter 3 employs a learning bias within the PIML framework, enables physically consistent and interpretable solutions while reducing dependency on large datasets, highlighting the potential of physics-informed learning for reliable assessment and prediction under practical data limitations.

1.5 Notations

1.5.1 Nomenclature Used in Chapter II

E	Elastic modulus of beam
L	Length of the beam
c	An arbitrary integer
h	Height of beam
k	Wave number
m	Crack-count
n_{cr}	The total number of cracks
n_{mode}	The total number of bending modes
δ	Crack depth ratio vector
η	Analytically computed NMFS vector
$\bar{\eta}$	Measured NMFS vector
ν	Poisson's ratio
ξ	Crack location vector
ξ_L	Lower bound of crack location vector
ξ_U	Upper bound of crack location vector
ρ	Mass density
ω_n	The modal frequencies of cracked beam
$\bar{\omega}_n$	The modal frequencies of intact beam
ϕ_A	The phase shifts at the left ends of the beam

ϕ_B	The phase shifts at the right ends of the beam
ϕ_m^{cb}	Phase shifts caused by the m^{th} crack in the backward directions
ϕ_m^{cf}	Phase shifts caused by the m^{th} crack in the forward directions
$\bar{\psi}$	The NMFS factors
$\ \cdot\ _2$	L_2 -norm of a functions or a vector

1.5.2 Nomenclature Used in Chapter III

A_c	Net section area of concrete of PSC beam
A_p	Gross section area of tendons
$E_c(t)$	Elastic modulus of concrete at time t .
E_{c0}	Elastic modulus of concrete at $t = 0$ (i.e. $E_c(0)$)
E_{ci}	Elastic modulus of concrete at the age of 28 days. (i.e. $E_c(28 - d_0)$)
E_p	Elastic modulus of the tendon
M_d	Bending moment induced by the self-weight of the beam
P_i	Initial effective prestress force of tendon after elastic shortening of concrete at $t = 0$
$P(t)$	Effective prestress force of tendon at time t
d_0	The age of concrete at $t = 0$
e	Vertical distance from the centroid of the net concrete section of the beam
f_{ck}	Characteristic compressive strength of concrete

f_{cu}	Characteristic ultimate strength of concrete
f_{pi}	Initial stress of the tendon
f_{py}	Yield stress of the tendon
h	Height of the PSC beam
$n(t)$	Modular ratio at time t (i.e. $E_p/E_c(t)$)
t	Time (day) beginning from the initial prestress transfer
w_c	Specific weight of concrete per unit volume
y	Vertical coordinate with the origin at the centroid of the concrete section of the beam
Γ	$\gamma(-e) = n(0)A_p \left(\frac{1}{A_c} + \frac{e^2}{I_c} \right)$
$\Phi_{cr}(t, \tau)$	Creep coefficient at time t for unit strain of which the corresponding stress is E_{c0} applied at time τ (i.e. $\frac{E_{c0}}{E_{ci}} \tilde{\Phi}_{cr}(t, \tau)$)
$\Phi_{el}(t)$	The ratio of E_{c0} to $E_c(t)$ at time t (i.e. $\frac{E_{c0}}{E_c(t)}$)
$\Phi_{re}(t, \tau)$	Relaxation function of the tendon at time t .
$\tilde{\Phi}_{cr}(t, \tau)$	Creep coefficient at time t for the unit strain of which the corresponding stress is E_{ci} applied at time τ .
$\gamma(y)$	$n(0)A_p \left(\frac{1}{A_c} - \frac{e}{I_c} y \right)$
$\sigma(y, t)$	Axial-bending stress at location y and time t
$\varepsilon_{cr}(y, t)$	Creep strain of concrete at location y and time t
$\varepsilon_{el}(y, t)$	Elastic strain of concrete at location y and time t

$\varepsilon_{\text{net}}(y, t)$	Net strain of concrete at location y and time t
$\varepsilon_{\text{sh}}(t)$	Shrinkage strain of concrete at time t .
$\varepsilon_{\text{p}}(t)$	Net strain of tendon at time t
$\varepsilon_{\text{p-d}}(t)$	Elastic strain of concrete at the location of the tendon due to self-weight of the beam at time t (i.e. $\frac{\Phi_{\text{el}}(t)}{E_{\text{co}}} \sigma_{\text{p-d}}(t) = \Phi_{\text{el}}(t) \frac{M_{\text{d}}}{E_{\text{co}} I_{\text{c}}} e$)
$\varepsilon_{\text{p-el}}(t)$	Elastic strain of tendon at time t
$\varepsilon_{\text{p-re}}(t)$	Relaxed strain of tendon at time t
$\varphi(t)$	Prestress transfer coefficient at time t i.e. $\frac{1}{1+n(t)A_{\text{p}}\left(\frac{1}{A_{\text{c}}}+\frac{e^2}{I_{\text{c}}}\right)}$ or $\frac{1}{1+\Gamma\Phi_{\text{el}}(t)}$
$\hat{}$	Symbol for defining the associated dimensionless variables normalized by P_{i} , $\frac{P_{\text{i}}}{E_{\text{p}}A_{\text{p}}}$ and $E_{\text{co}}\left(\frac{P_{\text{i}}}{E_{\text{p}}A_{\text{p}}}\right)$ for forces, strains and stresses, respectively.
E	Elastic modulus *
I	Second moment of inertia *
L	Length of PSC girder *
P_0	Prestress force*
w	Deflection of PSC girder*
ξ	Normalized length coordinate*
ϕ_0	Curvature of PSC girder *

* Variables used only in camber reconstruction

II. PIML-Based Classification for Multiple Incipient Crack Assessment in Beams

2.1 Assessment of Multiple Incipient Cracks in Beams

2.1.1 Limitations of Previous Studies

Crack evaluation is an important research area in structural health monitoring. Cracks reduce the rigidity of a structure without affecting its mass, leading to a reduction in the natural frequencies of the affected structure. Numerous studies have used this phenomenon to detect crack locations and sizes through fractional reduction of modal frequencies [Dim96; Fan11; Sal97]. Over the past few decades, continuous research has been conducted on crack assessment using modal characteristics, such as modal frequencies and mode shapes. While methods for identifying a single crack based on changes in modal characteristics have been successfully studied [Khi14a], studies on the assessment of multiple cracks are relatively limited [Gil21]. To provide a comprehensive overview, previous studies on crack assessment ranging from single to multiple cracks are summarized in Table 2.1 and categorized based on the type of input data and methodological approach.

Table 2.1 Summary of previous studies on crack assessment in beams

Type	Input data	Method / Model	References
Single crack	Narrowband modal frequency	E-B beam based Analytical model	Kam92
		E-B beam based Analytical model	Dim96
		E-B beam based Analytical model	Fan11
		Analytical model	Lee00
		Analytical model	Kim03a
		T-beam based Analytical model	Zhe01
		T-beam based Analytical model	Lel02
		T-beam based Analytical model	Car02
		T-beam based Analytical model	Loy06
		Experimental modal testing	Sal97
		Closed-form equation	Gud83
		Closed-form equation	Lin04
Multiple Cracks	Narrowband modal frequency	Closed-form equation	Kha09
		Spring / continuum model	Hu93
		Scenario-based identification	Ves00
		Nonlinear frequency equation	Khi14b
		Modal strain energy	Zha17
		Frequency shift method	Gil21
		Analytical	Shi20
		T-beam based Analytical model	Tor14
		FE updating with experimental validation	Sin02
		Closed-form equation with experimental validation	Owo03
		Dynamic stiffness matrix with experimental validation	Khi04
		Rotational spring model with experimental validation	Pat05
		Finite Element and Newton–Raphson with experimental validation	Lee09

Table 2.1 Continued

Multiple Cracks	Narrowband modal frequency	Bayesian fusion with experimental validation	Sha19
		Experimental modal testing	Riz90
		Experimental modal testing	Ruo97
		Experimental modal testing	Kim03b
		Experimental modal testing	Che05
	Mode shape	Experimental modal testing	Dil02
		Experimental modal testing	Qua16
		Experimental modal testing	Dah19
		Hybrid model	He19
		Hybrid model	Chi21
		Analytical model	Dil10
		Analytical model	Cao14
		Analytical model	Ngu14
		Signal processing	Wan21a
	Broadband modal frequency	T-beam based analytical model	Par18
		T-beam based analytical model	Par20
		T-beam based closed-form equation with experimental validation	Par22
		T-beam based closed-form equation with experimental validation	Lim24

Although several studies have been conducted, a major limitation in assessing multiple cracks through modal-frequency reduction is the insufficient number of measurable modal frequencies relative to the number of cracks to be identified. Consequently, increasing the number of cracks can create a non-uniqueness problem, where multiple crack scenarios yield almost identical fractional modal frequency reductions [Khi14a]. Mitigating this issue often involves supplementing modal frequency data with additional information, particularly mode shapes [Khi04; Dil02; Dil10; He19; Wan21a; Ngu14; Qua16; Dah19; Chi21; Cao14]. However, employing

mode shapes poses challenges as the number of sensors, such as accelerometers, required for dynamic response measurement increases with the mode number [Qua16; Dah19]. Additionally, mode shape changes are often localized near the incipient cracks and remain insignificant elsewhere, limiting the information available for crack assessment [Ngu14].

Regarding experimental validation methods, most experimental studies on crack assessment [Ruo97; Sin02; Khi04; Pat05; Lee09; Sha19; Owo03; Kim03b; Riz90; Che05] have utilized modal tests with impact hammers or shakers. However, these methods are limited to frequencies below 20 kHz. Consequently, previous studies have used narrowband modal frequencies within the sixth bending mode for crack assessment. The limited number of available modal frequencies limits the information available for effective crack assessment, especially when multiple cracks are involved, as identifying n cracks necessitates $2n$ modal frequencies [Pat05]. The use of surface-bonded piezoelectric wafers made of lead zirconate titanate (PZT) has enhanced the acquisition of modal frequencies from cracked beams in a higher frequency range compared to conventional methods using impact hammers and shakers [Par18]. These wafers can measure all modal frequencies below the cut-off frequency, significantly increasing the number of available modal frequencies for assessing multiple cracks in a beam [Par18; Par20; Par22; Lim24]. This study harnesses the advantages of surface-bonded piezoelectric wafers, particularly their ability to capture broadband modal frequencies below the cut-off frequency, to address the challenge of limited information when evaluating an

increasing number of cracks. The Euler–Bernoulli beam (E-B beam) model, commonly employed in conventional methods using the first few narrowband modal frequencies [Dim96; Fan11; Sal97; Ves00; Pat05; Sha19; Kam92; Kim03a; Lel02; Lee00], becomes less effective as modal frequencies approach the cut-off due to the effects of rotational inertia and shear deformation, which are overlooked in the E-B beam model, become more pronounced when the modal frequencies approach the cut-off frequency. Although FE analysis (FEA) using solid elements can accurately predict all modal frequencies below the cut-off, it requires considerable computational resources for fine mesh refinement. Therefore, adopting the Timoshenko beam (T-beam) model is crucial for utilizing all modal frequencies below the cut-off frequency. Several studies have investigated the dynamic behavior of cracked T-beams [Lel02; Zhe01; Car02; Loy06]. Nevertheless, these approaches require iterative procedures for solving the nonlinear frequency equation. Thus, a closed-form equation based on T-beam theory is required for predicting modal frequencies of cracked T-beams. [Gud83; Lin04; Kha09]. However, these studies have largely focused on single cracks, with limited research on predicting the dynamic behavior of multiple cracks using T-beam theory [Tor14].

The limitations identified above collectively reveal that the conventional approach to multiple-crack assessment is constrained by three fundamental gaps. First, narrowband testing conditions restrict the number of measurable modal frequencies, providing insufficient information to uniquely resolve the inverse problem as the number of cracks increases. Second, the E-B beam model becomes increasingly

inaccurate at higher-order bending modes, where the effects of rotational inertia and shear deformation can no longer be neglected, necessitating the adoption of Timoshenko beam theory for reliable frequency prediction across the broadband range. Third, even when Timoshenko beam-based models are employed, generating the large number of forward solutions required for inverse crack assessment remains computationally prohibitive, rendering brute-force inversion impractical for multiple-crack scenarios. Notably, previous studies have largely addressed this inverse problem under the assumption that the number of cracks is known a priori, leaving the more practical and challenging case in which the crack count must itself be identified from measured data largely unresolved. Overcoming these gaps therefore requires a framework that simultaneously expands the measurable frequency range through broadband sensing, replaces computationally expensive simulations with an efficient closed-form surrogate, and leverages physics-informed learning to determine the number of cracks as well as their locations and depths from measured data alone. The specific challenges associated with applying such a physics-informed machine learning framework to this problem are discussed in the following subsection.

2.1.2 Challenges in Applying PIML

While the general concept of PIML-based classification was introduced in Section 1.4.2, this section addresses the specific challenges that arise when this framework is applied to the assessment of multiple incipient cracks. A representative challenge

is the severe imbalance between normal and abnormal data, as illustrated in Figure 2.1. Normal data can be easily collected during regular operation, whereas abnormal data representing damage or defects are extremely limited because such conditions rarely occur and are difficult to reproduce experimentally. This imbalance causes the learning process to be dominated by normal data, resulting in a biased model that tends to misclassify or overlook abnormal states. A representative example is the crack assessment problem in structural health monitoring.

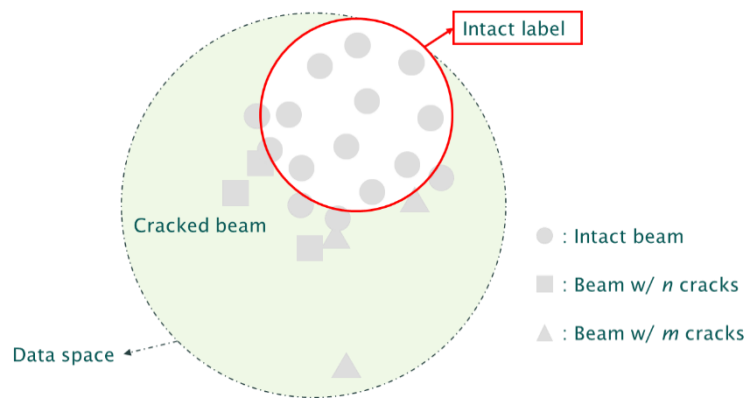


Figure 2.1 Application limits of data-driven machine learning to multiple incipient crack assessment

In this case, measurements from intact beams are abundant, while data from cracked beams are scarce. As shown in Figure 2.1, intact-beam data form a dense cluster in the data space, whereas cracked-beam data are sparsely distributed. Because labeled data are unavailable in most structural monitoring cases, applying supervised classification methods is difficult. To address this limitation, unsupervised clustering has been adopted to separate anomalous data from the intact

domain. However, this approach is typically limited to a binary decision of whether the structure is cracked or not, and it cannot be extended to multiple-crack cases with different numbers of cracks. To overcome the imbalance between intact and cracked data, a physics-informed data augmentation strategy can be introduced. As illustrated in Figure 2.2, a physics-based surrogate model generates additional labeled data corresponding to various crack scenarios.

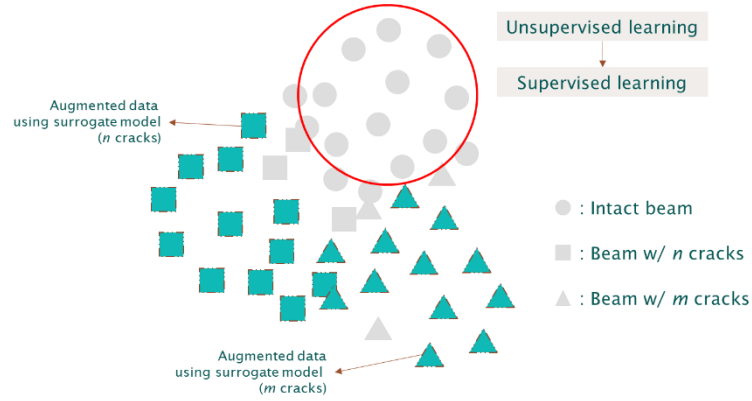


Figure 2.2 Labeled data augmentation using surrogate models

By augmenting the limited cracked-beam data with physics-consistent samples, the learning process transitions from unsupervised clustering to supervised classification. As shown in Figure 2.3, groups corresponding to each crack -count appear emerge in the data space, indicating that the model can now be trained to identify the number of cracks rather than merely detecting the presence of damage. This transformation establishes the foundation for the subsequent physics-informed machine learning framework that classifies multiple incipient cracks.

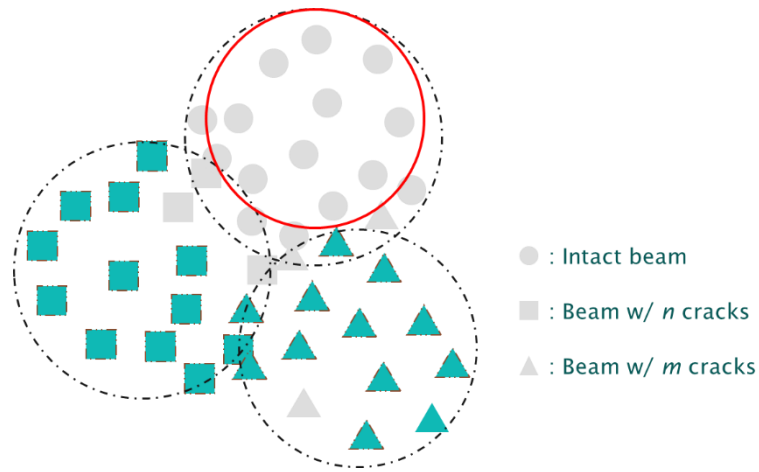


Figure 2.3 Crack-count classification using physics-informed machine learning

While PIML provides a powerful framework for incorporating physical constraints into learning, its application to classification problems is not without challenges. As illustrated in Figure 2.4, label ambiguity often arises even when physical knowledge is embedded in the model.

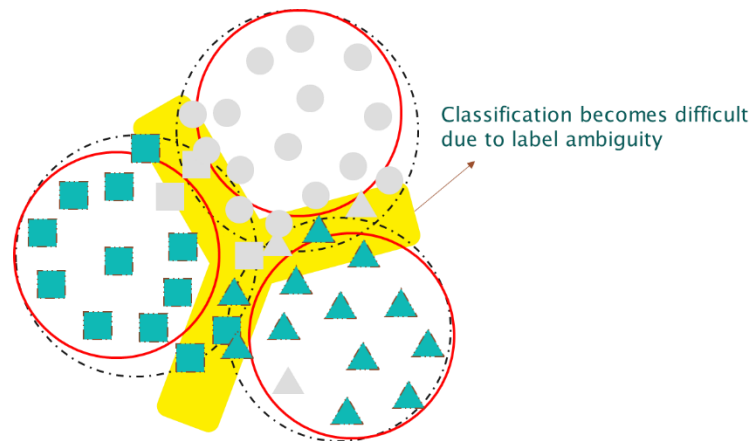


Figure 2.4. Limitations of crack-count classification using physics-informed machine learning

Although surrogate models can generate large amounts of augmented data, the overlap of input features across different labels causes label conflicts during supervised learning. In crack assessment, this challenge becomes more critical because the modal frequency changes induced by cracks are relatively small compared with the overall dynamic response. As the number of cracks increases, different crack configurations can generate almost indistinguishable frequency-shift patterns. In particular, for incipient cracks, the frequency changes are extremely subtle, resulting in greater label ambiguity among similar crack scenarios. Consequently, the data distributions of multiple crack scenarios tend to overlap, causing label conflicts in the augmented dataset and reducing the classification accuracy of the PIML model.

Even when PIML is applied, data refinement remains essential because label ambiguity can still occur in augmented datasets. In this process, the data are reorganized according to specific criteria, and ambiguous samples are removed or reclassified. These criteria are problem-dependent, and establishing appropriate ones tailored to each application is crucial for reliable data refinement. As shown in Figure 2.5, applying data refinement mitigates label ambiguity and improves the reliability of the data distribution used for learning. The refined dataset allows the PIML model to more accurately distinguish between different labels.

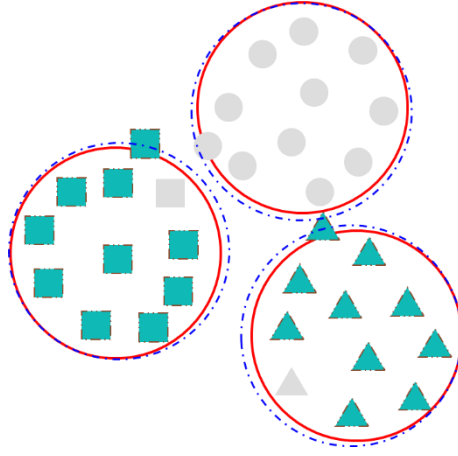


Figure 2.5 Crack-count classification with physics-informed machine learning after data refinement

The three challenges in applying PIML identified above collectively define the design requirements that any PIML-based crack assessment framework must satisfy. These challenges are not merely numerical inconveniences; they reflect the fundamental ill-posedness of the inverse problem, in which distinct physical configurations can produce nearly indistinguishable measurement signatures. To satisfy these requirements, this chapter presents a novel PIML-based framework for assessing multiple incipient cracks as a representative example of a classification problem in structural engineering. In this framework, the closed-form frequency surrogate based on the phase-closure principle [Par22], which is capable of accurately predicting fractional reductions in modal frequencies across the broadband range including higher-order modes, directly addresses the three fundamental gaps identified in Section 2.1.1. This surrogate is employed for physics-informed data augmentation, simultaneously alleviating the data imbalance between

intact and cracked specimens by supplying abundant labeled samples across all crack-count categories at minimal computational cost. Problem-specific refinement criteria are subsequently applied to resolve label conflicts arising from overlapping NMFS patterns across different crack-count categories. A PIML model trained on this refined dataset identifies the number of incipient cracks from the measured frequency data, even without prior knowledge of the crack count. Based on the identified crack count, grid optimization is then applied to assess crack locations and depths by sequentially evaluating predefined combinations and selecting the optimal one that minimizes the least-squares error between the measured and analytically computed NMFS. The optimal resolution for grid optimization is determined using a bilinear fitting method, reducing computational time and avoiding convergence to local minima. The validity of the proposed framework is statistically verified through blind tests incorporating both finite element-based and experimentally measured NMFS data, as detailed in Sections 2.3 through 2.6.

2.2 Broadband Modal Frequency Measurement

Figure 2.6 shows a free-free aluminum beam with multiple incipient cracks. The beam specimen measures 198 mm in length, 10.20 mm in height, and 10 mm in width. Specimens with one to four cracks were tested, with corresponding crack scenarios detailed in Appendix (A1).

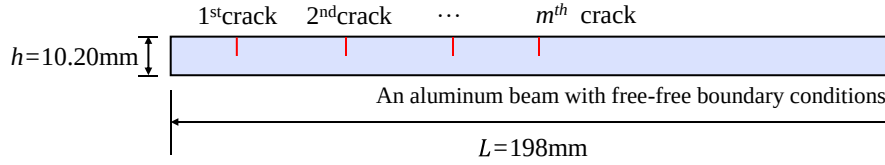


Figure 2.6 Free-free aluminum beam with multiple cracks used in the experiment ($E = 69 \text{ GPa}$, $\nu = 0.34$, $\rho = 2750 \text{ kg/m}^3$).

Modal testing employed surface-bonded piezoelectric wafers to gather experimental data on the fractional reduction of modal frequencies (hereinafter referred to as normalized modal frequency shift (NMFS)). Aluminum beams with one to four cracks (Figure 2.7(a)) were used in the experiment. Each crack has a width of 0.3 mm, designed to simulate the mechanical behavior of an open crack while minimizing mass loss and avoiding contact between crack surfaces during vibration [Lim24, Mon90]. As shown in Figure 2.7(b), piezoelectric wafers from Piezo Systems Inc (United States) are placed on the top and bottom surfaces of each beam to measure dynamic responses using the pitch-catch method [Pie25]. Each wafer, cut to size for proper attachment, measures $10 \text{ mm} \times 10 \text{ mm} \times 0.19 \text{ mm}$ (length $\times$ width $\times$ thickness). The sensing wafer is positioned at the top of the beam,

while the actuating wafer is positioned at the bottom.

Because PZT wafers positioned near nodal points of a bending mode exhibit significantly reduced sensitivity for that mode, the mode shapes of the intact beam over the entire frequency range of interest were first computed through FE analysis. The actuating and sensing PZT wafers were then placed at selected locations to avoid nodal regions of all bending modes below the cut-off frequency. This placement strategy ensures sufficient modal strain at each sensor location and improves the reliability of broadband frequency excitation. Consequently, the piezoelectric wafers are positioned 48 mm from each end of the beam, as shown in Figure 2.7(c), to capture the maximum number of bending modes within the beam's cut-off frequency [Lim24]. The modal frequency measurement environment, presented in Figure 2.7(d), is controlled in a temperature chamber set to 25 °C to minimize the influence of temperature variations on frequency shifts. Two strings are suspended 10 mm from each end of the beam to simulate free-free boundary conditions.

To assess potential sensor–structure interaction and mass loading effects, a FE eigenvalue analysis was conducted by comparing the bare beam with the beam containing surface-bonded PZT wafers. The relative differences in bending natural frequencies for all modes up to the 19th mode, including those near the cut-off frequency, were found to be less than 0.22%. Furthermore, because the PZT wafers are present in both the intact and cracked states, their influence largely cancels out in the definition of NMFS. These results demonstrate that PZT-induced mass loading and local stiffening effects do not significantly perturb the inherent dynamics of the

specimen and do not affect the NMFS-based crack assessment.

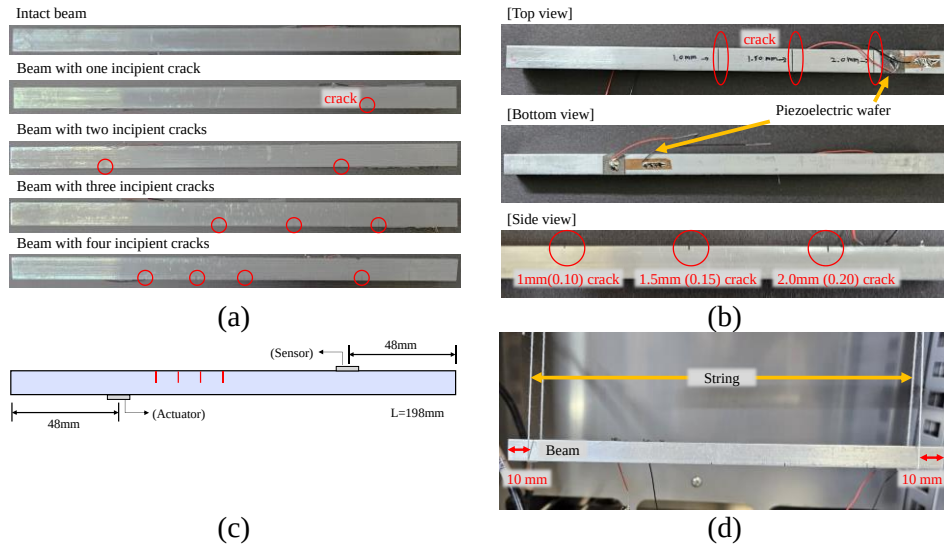
A chirp-voltage input was applied to the actuating piezoelectric wafer using a 16-bit arbitrary waveform generator (NI PXI-5422), and the resulting output voltage was recorded from the sensing piezoelectric wafer using a 14-bit high-speed digitizer (NI PXIe-5122). Ten measurements of the output voltage signal were recorded with LabVIEW™, with the average of the results taken as the representative value. The output was converted into a frequency-response spectrum using a fast Fourier transform.

Peak frequencies were determined by comparing them with FEA results. Specifically, the bending-mode frequencies and associated mode shapes were first extracted from the FE model of the intact beam and used as references for mode identification. For the intact beam, each experimental resonance peak corresponding to a bending-type response was matched to the closest bending-mode frequency obtained from the FEA, thereby establishing a one-to-one correspondence between numerical and experimental bending modes.

Figure 2.7(e) presents the averaged output voltage signal from ten repeated measurements on an intact beam specimen. The dotted vertical lines mark the time instances at which the driving frequency of the chirp input aligns with the natural frequencies of the bending modes. Label 'B' indicates a bending mode; for example, B5 represents the fifth bending mode. The fast Fourier transform applied to the time-domain signal shown in Figure 2.7(e) yielded the corresponding frequency response

spectrum shown in Figure 2.7(f).

For cracked beams, the bending-mode frequencies were identified by selecting resonance peaks closest to the corresponding intact-beam bending-mode frequencies. This procedure is justified by the observation that the maximum observed crack-induced fractional reduction in bending-mode frequencies remains below 0.046 across all crack scenarios considered, which is significantly smaller than the separation between adjacent bending-mode frequencies. The validity of this correspondence was further confirmed by computing the mean absolute percentage error between the FEA-predicted and experimentally measured bending-mode frequencies across all crack configurations ($1.16 \pm 0.37\%$). Each resonance peak in the spectrum is matched to a mode based on comparisons with the FEA results.



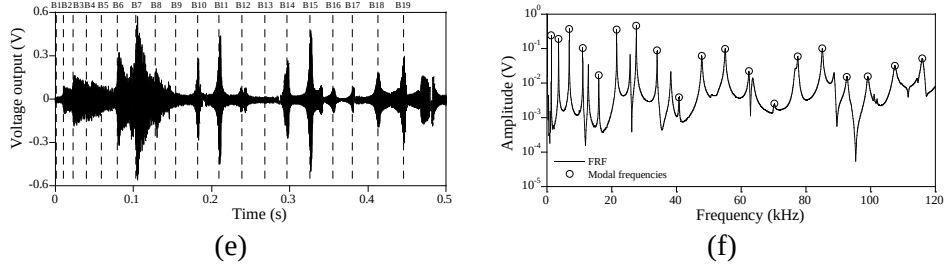


Figure 2.7 Experimental validation setup: (a) aluminum beams with incipient cracks; (b) sensing and actuating piezoelectric wafers mounted on the top and bottom of an aluminum beam; (c) wafer attachment positions; (d) beam suspended using two strings to simulate free-free conditions; (e) average output voltage signal of an intact beam; (f) frequency response function (FRF) obtained via fast Fourier transformation of the measured signal in (e), with modal frequencies matched via finite element analysis (FEA) results.

Cracks can reduce the stiffness of a structure without changing its mass, resulting in decreased natural frequencies of the cracked beam compared to the intact beam. Figure 2.8 shows the frequency response spectra from representative cracked beam cases with one to four cracks. We selected cases where the decrease in modal frequencies due to cracking is easily observable across multiple modes. The modal frequencies extracted from the intact beam, shown in Figure 2.7(f), serve as the baseline for quantifying the frequency reduction caused by cracks. The baseline modal frequencies are averaged over five intact beam specimens to improve accuracy. Although this averaged baseline does not exactly match the pre-cracked condition of each beam, prior studies [Par20, Lim24] indicate that specimen-to-specimen thickness tolerances lead to bounded variations in intact modal frequencies, which in turn introduce bounded baseline-induced variability in NMFS. However, baseline-induced variability alone does not fully account for the remaining diagnostic errors, which primarily arise from limited separability of NMFS patterns

associated with multiple incipient cracks under experimental uncertainty. Accordingly, while the use of an averaged baseline may increase NMFS scatter when systematic differences exist between baseline and test specimens, it provides a practical reference that preserves the overall validity of the proposed crack assessment framework.

The vertical dashed lines in Figure 2.8 represent the averaged modal frequencies, while the red '+' markers indicate modal frequencies measured from each cracked beam. The vertical axis scales in Figure 2.8(a)–(d) maintain a consistent logarithmic scale, mirroring that of Figure 2.7(f). As the crack count increases (from Figure 2.8(a) to (d)), the shift in the resonant peaks becomes increasingly evident, reflecting the reduction in modal frequencies due to cracks. This impact is particularly pronounced in higher modes where the sensitivity to local stiffness changes becomes more evident. NMFS is derived by normalizing the modal frequency of a cracked beam against the corresponding baseline frequency from an intact beam [Par22]. The reductions in modal frequencies depicted in Figure 2.8 are used to calculate the normalized NMFS values shown in Figure 2.9. As the number of cracks increases, the magnitude of the NMFS tends to increase due to the cumulative effect of frequency reductions from each crack. Each subplot in Figure 2.9 includes a small inset graph showing the true crack configuration of the corresponding beam. In each inset, the horizontal and vertical axes represent the crack location (normalized by the beam length up to $0.5L$) and depth (normalized by the beam height up to $0.2h$), respectively. The PIML-assisted assessment of multiple incipient cracks was

conducted using the NMFS values from all bending modes below the cut-off frequency.

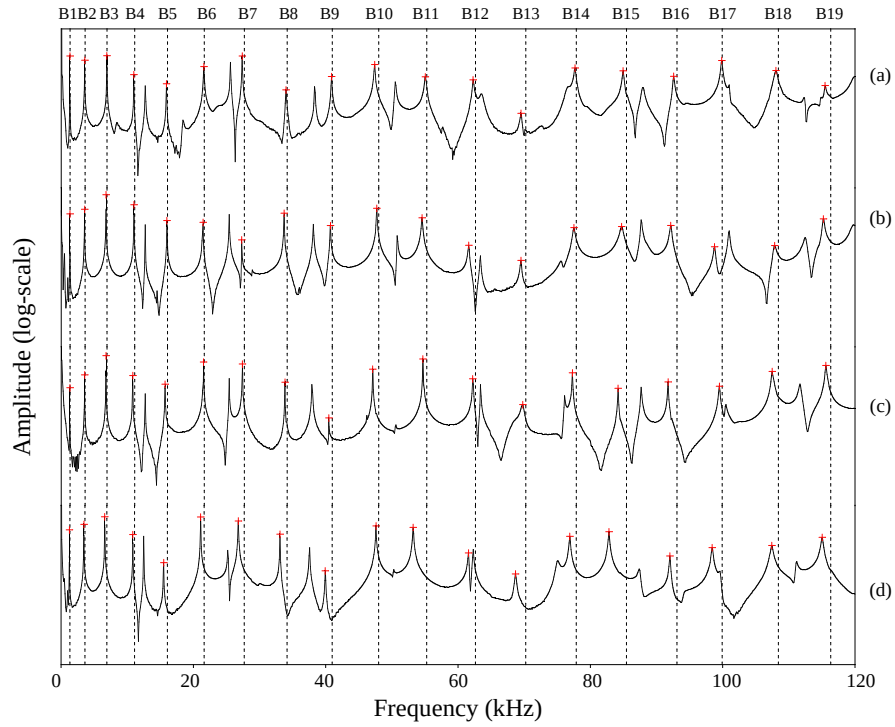


Figure 2.8 Representative frequency response spectra from experiments: (a) one-crack scenario (Case 7); (b) two-crack scenario (Case 1); (c) three-crack scenario (Case 5); (d) four-crack scenario (Case 6) (+: modal frequencies; —: frequency response spectrum; - - : average modal frequencies of the five intact beams). Note: The "cases" discussed here refer to the crack scenarios outlined in Appendix (A1). Each case number uniquely identifies a specific configuration of crack number, location, and depth used for blind-test validation.

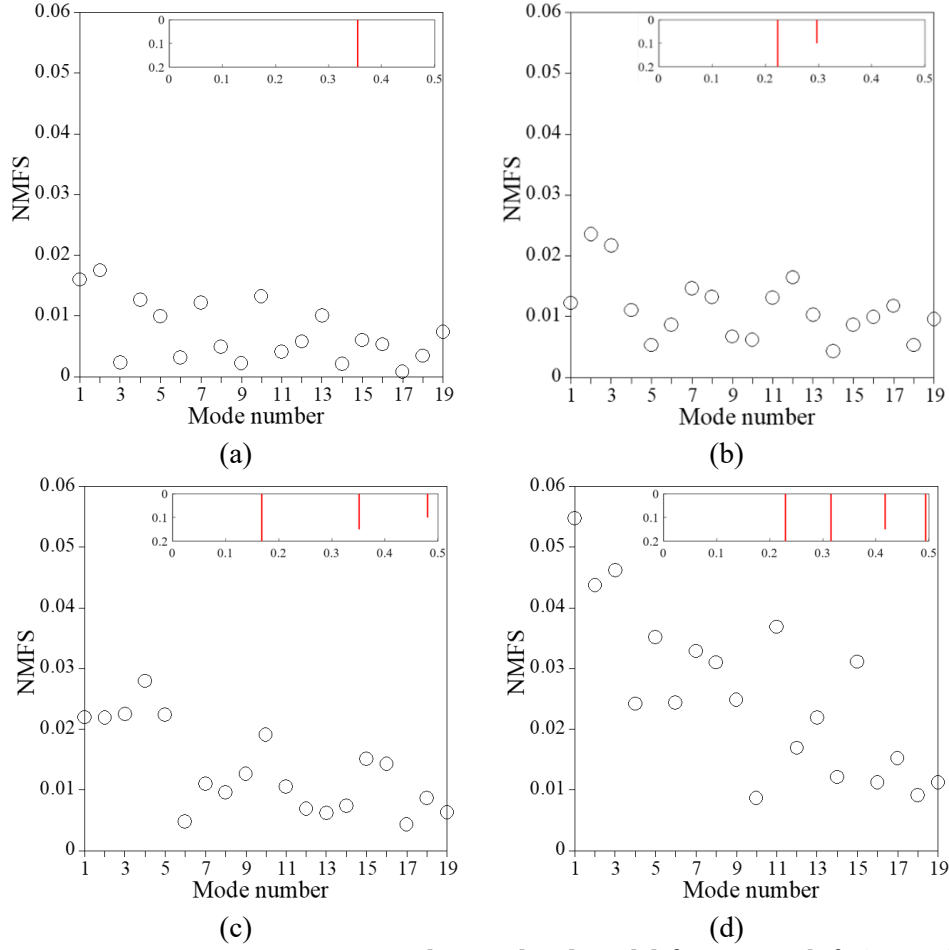


Figure 2.9 Representative measured normalized modal frequency shift (NMFS) data obtained from Figure 2.8: (a) one-crack scenario (Case 7); (b) two-crack scenario (Case 1); (c) three-crack scenario (Case 5); (d) four-crack scenario (Case 6).

2.3 Surrogate Model for PIML

2.3.1 Closed-Form Solution Based on the Phase-Closure Principle

The free vibration mode of a 1-D waveguide arises from the superposition of standing wave motions [Fre71]. According to the phase-closure principle, a natural mode occurs when the total phase change equals an integer multiple of 2π , with the corresponding frequency defined as the natural frequency [Mea94].

In [Par22], we extended the phase-closure principle to beams with multiple cracks, as shown in Figure 2.10.

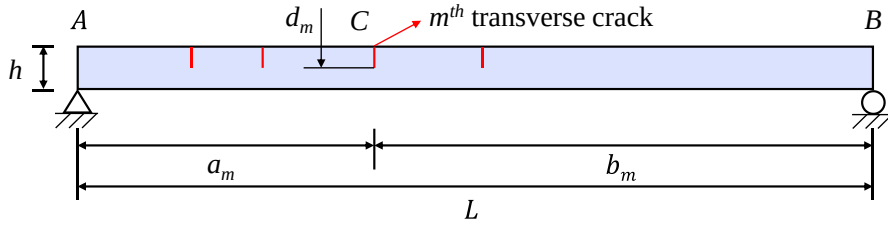


Figure 2.10 Simply-supported beam structure with multiple incipient cracks [Par22].

The extended phase-closure condition is given as follows:

$$-2kL + \phi_A + \phi_B + \sum_{m=1}^{n_{cr}} (\phi_m^{cf} + \phi_m^{cb}) = 2c\pi, \quad (2.1)$$

where n_{cr} denotes the total number of cracks in the waveguide. k and L represent the wave number and length of the beam, respectively; c is an arbitrary integer. The phase shifts at the left and right ends of the beam are denoted by ϕ_A and ϕ_B , respectively. ϕ_m^{cf} and ϕ_m^{cb} represent phase shifts caused by the m^{th} crack in the

forward and backward directions, respectively.

From Equation (2.1), we derive a closed-form equation predicting the fractional reduction between the modal frequencies of an intact beam ($\bar{\omega}_n$) and a cracked beam (ω_n) as [Par22].

$$\eta_n = \frac{\bar{\omega}_n - \omega_n}{\bar{\omega}_n} = \bar{\psi} \sum_{m=1}^{n_{cr}} \cos^2 \left(\bar{k}_n a_m - \frac{\bar{\phi}_A}{2} \right) \frac{F_m}{L}, \quad (2.2)$$

where η_n represents NMFS, which quantifies the decrease in natural frequency caused by cracks. The NMFS factor $\bar{\psi}$ is expressed as $\bar{\psi} = \frac{(1-\bar{\alpha}_p)}{(1+\bar{\gamma}^2)(1+\bar{\zeta})} \frac{\bar{c}_g}{\bar{c}_p}$ [Par22].

The frequency equation employs the T-Beam theory for precise prediction of high-order modal frequencies, which are particularly sensitive to incipient cracks. This allows the use of broadband modal frequencies below the cut-off frequency for accurate and reliable assessments of multiple incipient cracks.

In this context, the closed-form solution used in this study is applied strictly within its asymptotic validity range. The formulation intentionally omits the evanescent wave components that are dominant in the 1st and 2nd bending modes. Although a full Timoshenko beam model could capture these effects, low-order modes are excluded from the inverse problem domain because they are highly sensitive to boundary condition uncertainties in practical applications. Consequently, the closed-form equation serves as a computationally efficient surrogate that accurately describes the propagating-wave behavior of the higher-order modes employed for crack detection. Accordingly, a large dataset of NMFS values for various damage

scenarios with an arbitrary number of cracks can be constructed, enabling a PIML model with minimal computing costs.

2.3.2 Finite Element Analysis for Validation

The FEA methodology adopted in this study follows the modeling approach established in [Par22]. The FE model was implemented in ABAQUS 2022/Foundation, in which the beam was discretized into 807,840 eight-node solid elements ($L50\ \mu\text{m} \times T50\ \mu\text{m}$), where L and T denote the longitudinal and through-thickness dimensions, respectively. This mesh density was previously verified in [Par22] to provide sufficient spatial resolution for accurate prediction of NMFS up to the cut-off frequency. Free-free boundary conditions were imposed by leaving all external surfaces unconstrained, consistent with the experimental setup described in Section 2.2. Transverse open cracks were modeled by introducing double nodes along the crack interface, which permits free separation of the crack faces while precluding interpenetration, thereby replicating the mechanical behavior of an open crack without contact nonlinearity [Lim24]. This open-crack assumption is physically consistent with the crack geometry employed in the experiments, where each crack has a finite width of 0.3 mm designed to prevent crack-face contact during vibration. Natural frequencies and mode shapes were extracted via eigenvalue analysis and subsequently used to compute the FEA-based NMFS for each crack scenario.

Figure 2.11 presents a comparative analysis of NMFS data generated using the

surrogate model alongside experimental data measured using surface-bonded piezoelectric wafers, described in Section 2.2, and the numerical NMFS data calculated using FEA.

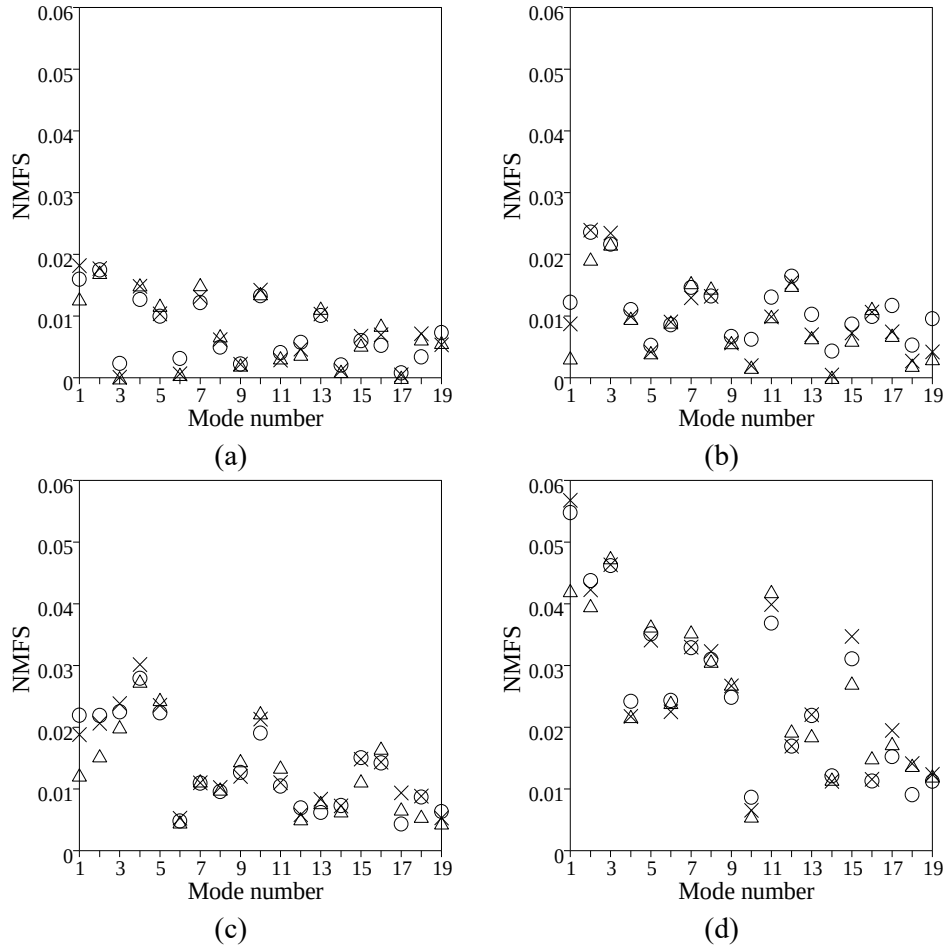


Figure 2.11 Representative examples of NMFS comparison among the surrogate model, FEA, and experiments: (a) one-crack scenario (Case 7); (b) two-crack scenario (Case 1); (c) three-crack scenario (Case 5); (d) four-crack scenario (Case 6) (Δ : surrogate model using Eq. (2.2); $\times$: FEA; $\circ$: experiment). All crack scenarios are identical to those presented in Figure 2.10

This comparison verifies the validity of the NMFS dataset generated using the surrogate model and its suitability for training the PIML model efficiently and accurately. The results demonstrate that the surrogate model can serve as a reliable alternative to computationally expensive simulations for training data generation. Furthermore, the modeling error between the closed-form NMFS given by Eq. (2.2) and the FE-based NMFS was quantified across all blind-test crack scenarios to assess the impact of simplified assumptions, including the omission of axial-bending modal coupling (ABMC) effects. The quantitative comparison yielded a mean absolute error (MAE) of 0.093 ± 0.049 percentage points (pp). The maximum observed error across all cases was 0.20 pp, which represents an empirical upper bound on the combined effects of the modeling assumptions. Because this bounded error is substantially smaller than the crack-induced NMFS variations, Eq. (2.2) serves as an accurate and robust physics-informed surrogate for the inverse problem.

2.4 PIML for Crack-count Identification

2.4.1 Data Augmentation

In this section, we describe the implementation of the PIML model, followed by numerical and experimental validation results. For the randomly generated crack scenarios, we augmented 5,000 NMFS data points per crack-count label, as shown in Figure 2.12 ranging from an intact beam (zero cracks) to four cracks, using Eq. (2.2).

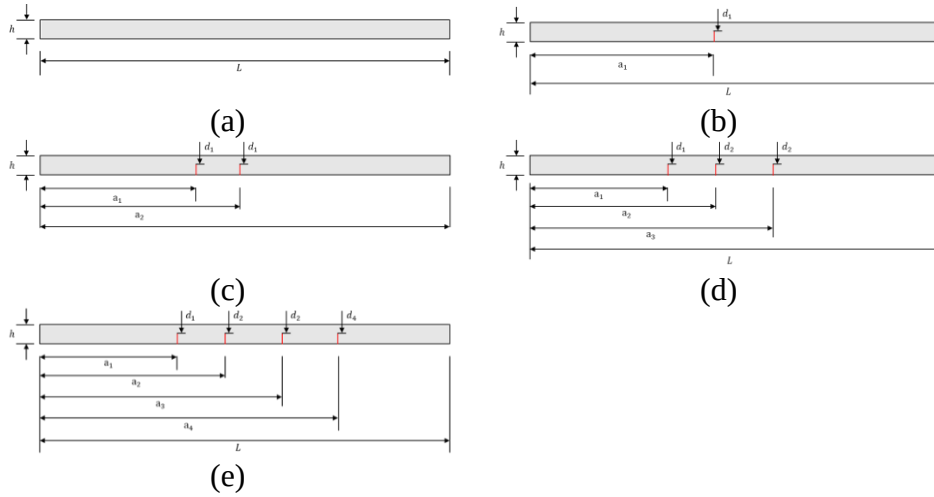


Figure 2.12 Schematic representation of crack-count labels: (a) intact (b) one-crack; (c) two-crack; (d) three-crack (e) four-crack; randomly generated ($E = 69$ GPa, $\nu = 0.34$, $\rho = 2750$ kg/m³). Note: All crack scenarios were randomly generated.

Crack locations were randomly generated between 5% and 50% of the beam's length, considering the symmetric modal behavior of a free-free beam. This range captures all unique crack scenarios because the modal behavior of the beam is

symmetric with respect to its center. Scenarios with cracks near the free ends of the beam were excluded due to their negligible effect on the NMFS. In each scenario, the spacing between cracks was at least 5% of the total beam length, avoiding adjacency. The crack-depth ratio was randomly assigned from three predefined values (0.1, 0.15, and 0.2), corresponding to the three assumed progression stages of incipient cracking.

The modal frequencies for generating blind-test data ranged from the 3rd to the highest bending mode, immediately below the cut-off frequency. The exclusion of the 1st and 2nd modes is motivated by two complementary considerations. First, from a theoretical standpoint, Eq. (2.2) neglects evanescent wave effects on the modal behavior of the cracked beam. While evanescent waves significantly affect lower modes, their influence is negligible for higher modes (3rd and above), rendering Eq. (2.2) inaccurate for predicting the 1st and 2nd mode frequencies [Par22]. Second, and more critically for practical implementation, the strategic exclusion of these low-order modes mitigates experimental uncertainties. In experimental and operational environments, the 1st and 2nd bending modes are significantly influenced by support stiffness variability. Seminal studies [Doe98, Far98] have shown that frequency shifts induced by changes in boundary conditions often mask the subtle deviations caused by incipient cracks. Sohn [Soh07] also highlighted that filtering out such environmental variability is essential for reliable damage identification. By restricting the inversion to higher-order modes, for which crack-induced propagating-wave effects dominate and boundary-related noise is minimized, the

framework achieves a higher Signal-to-Noise Ratio (SNR) and ensures that the diagnosis is governed by structural damage rather than boundary artifacts.

2.4.2 Data Refinement

Label ambiguity often occurs in physics-informed approaches, particularly in complex problems or when input similarity is high. A prevalent challenge is conflicting labeling, where similar data points are assigned different labels, leading to reduced model accuracy during training. Therefore, refining the data through tailored criteria specific to the problem is essential for enhancing PIML model performance. Figure 2.13 presents examples of label ambiguity in the PIML model used to identify the number of cracks. In Figure 2.13(a), cracks with insufficient depth may not be recognized, causing adjacent data points to have different labels, increasing the classification difficulty. In Figure 2.13(b), closely spaced cracks could be misinterpreted as a single crack, introducing label ambiguity. Two refinement criteria were introduced to address these issues: identifiable crack depth (ICD) and identifiable crack separation (ICS). The ICD criterion defines the minimum crack depth detectable by the model, while ICS specifies the minimum spacing required to distinguish adjacent cracks. Based on engineering judgment, ICD was set to 10% of the beam height and ICS to 5% of the total beam length.

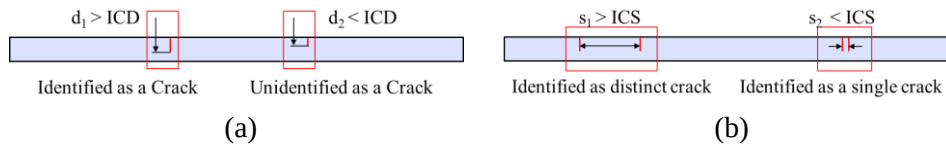


Figure 2.13 Examples of label ambiguity and data refinement criteria: (a) Identifiable crack depth (ICD); (b) Identifiable crack separation (ICS).

2.4.3 Implementation of PIML

The entire 25,000-point training dataset augmentation was completed in significantly less time using a closed-form analytical solution than with FEA [Hwa23]. The augmented data were provided to MATLAB's Classification Learner [Mat25a], facilitating the testing and comparison of various machine learning models on the same dataset. All classification models available in MATLAB were tested on the same 25,000-point training dataset using a five-fold cross-validation [Mat25b]. Figure 2.14 presents the model performance comparison results obtained.

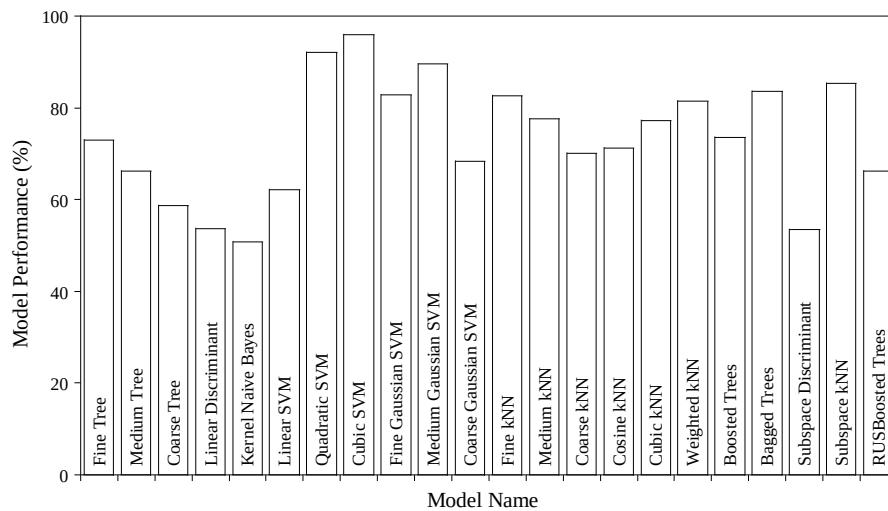


Figure 2.14 Model performance comparison using MATLAB Classification Learner.

. The quadratic discriminant and Naive Bayes models, which could not be directly trained using the initial settings, were excluded. Additionally, because the study aims to provide a straightforward approach, models requiring additional configurations

were not considered. Comparison values for the tested models are presented in Appendix (A2). Accordingly, the cubic support vector machine (SVM), which achieved the highest accuracy was selected for the PIML model implementation. Although this model demands substantial training time, it can be repeatedly used to identify the number of cracks in real time once trained.

Subsequently, to rigorously validate the engineering judgment regarding ICD and ICS introduced in Section 2.4.2, an additional sensitivity analysis was performed using the trained cubic SVM. As detailed in Appendix (A3), Monte Carlo simulations confirmed that the classification performance of the model degrades sharply below $ICD=0.1h$ and saturates beyond $ICS=0.05L$. This quantitative validation demonstrates that the selected thresholds represent the physical observability limits of the proposed framework rather than arbitrary assumptions.

2.4.4 Blind-Test Data Generation for Numerical and Experimental Validation

To validate the PIML model implemented in Section 2.4.3, blind-test data were generated via FEA and modal testing using surface-bonded piezoelectric wafers, as illustrated in Figure 2.10. Table 2.2 lists the number of crack scenarios per crack-count label used to generate the blind-test data. Specifics about crack locations and depths for each scenario can be found in Appendix (A2). Crack locations for blind-test data generation were randomly assigned between 5% and 95% of the beam length and rounded to the nearest 0.1 mm due to practical constraints. A minimum

spacing of 5% of the beam length was enforced for multiple crack scenarios to prevent clustering. The crack depths followed three predefined progression stages, consistent with the setup used in the PIML training process. The resulting blind-test data consisted of two types: (1) NMFS data obtained from FEA and (2) experimental NMFS data acquired through modal testing using surface-bonded piezoelectric wafers.

Table 2.2 Number of crack scenarios per crack-count label for blind-test data generation.

Crack-count label	FEA	Experimental
One-crack	12	12
Two-crack	8	8
Three-crack	10	10
Four-crack	7	7
Overall	37	37

Figure 2.15 presents the validation results of the proposed PIML model, categorized into correct predictions, false negatives (FN), and false positives (FP), and visualized through a stacked bar plot. The misidentified cases were further subdivided based on the deviation in the predicted number of cracks. In Figure 2.15(a), the predicted crack-count labels from the trained PIML model, using the NMFS data measured from the cracked beams in Table 2.2, are displayed. For each of the 37 cracked beam cases, the plot shows the proportions of correct and incorrect identifications by comparing the predicted labels with the ground truth. FN(2+) and FP(2+) indicate cases where the PIML model underestimates and overestimates crack counts by two or more, respectively, while FN(1) and FP(1) represent one-

crack discrepancies. The prediction performance was compared based on the type of blind-test data used. The training accuracy of the PIML model serves as a reference in the first bar, with seventeen bending modes from the 3rd to the 19th evaluated against blind-test data.

The validation results using the FE-based blind-test data exhibited a trend similar to that of the PIML model, but reflected an approximately 8% increase in FP cases. The PIML model achieved 81.1% accuracy in identifying the number of cracks. In contrast, the experimental blind-test data included uncertainties from the manufacturing and measurement processes, compromising the reliability of the NMFS values. Consequently, the FP and FN rates increased relative to the FE-based results. Under these conditions, the PIML model achieved a crack count identification accuracy of 54.1%.

The observed discrepancy between the high training accuracy (96.0%) and the experimental blind-test accuracy (54.1%) is attributed to the Sim-to-Real gap. While the classifier effectively learns the theoretical topology of crack-induced NMFS, experimental measurements are inevitably affected by boundary condition variability and manufacturing tolerances that are not represented in the physics-informed training data. This mismatch between the idealized numerical domain and the physical experimental domain poses a challenge for classification. However, it is important to note that the PIML classifier is not intended as a standalone diagnostic tool but rather as an auxiliary component used to constrain the inversion search space. The framework is designed with an inherent safety mechanism, whereby the final

crack locations and depths are determined through subsequent physics-based optimization, which exhibits strong robustness to imperfect crack-number estimates. Consequently, the reduction in performance at the intermediate classification step does not translate into a failure of the overall damage assessment.

With the FE-based blind-test data, misidentifications occurred only when the number of cracks was overestimated (or underestimated) by one. However, the experimental blind-test data showed deviations of up to two cracks. Most errors were attributed to incipient cracks at shallow depths (i.e., 0.1). Although the PIML model did not always predict the crack count accurately, it consistently provides reliable estimates of the crack locations. This observation is discussed in detail in Section 2.6, including the fundamental reasons behind the PIML model's limited accuracy in predicting the number of cracks and its broader implications.

In addition, Figure 2.15 compares the impact of the number of bending modes used in the training on the PIML model's predictive performance. The model was trained as described in Section 2.4.3 using three sets of modal information: three modes (3rd–5th) and six modes (3rd–8th), corresponding to narrowband modal frequencies; twelve modes (3rd–14th), representing broadband modal frequencies; and seventeen modes (3rd–19th) for the full set within the cut-off frequency in this study. As demonstrated in Figures 2.15(a) and (b), the use of broadband modal frequencies significantly improved the prediction accuracy and reduced large misidentifications. In contrast, the use of narrowband frequencies, as shown in Figures 2.15(c) and (d), resulted in poor performance and included cases where the

predicted crack count deviated by more than two. Thus, increasing modal information is shown to improve prediction accuracy and generalization to unseen data. These findings underscore the importance of incorporating broadband modal frequencies into the crack assessment process, highlighting their role in supporting accurate and robust predictions using the PIML model.

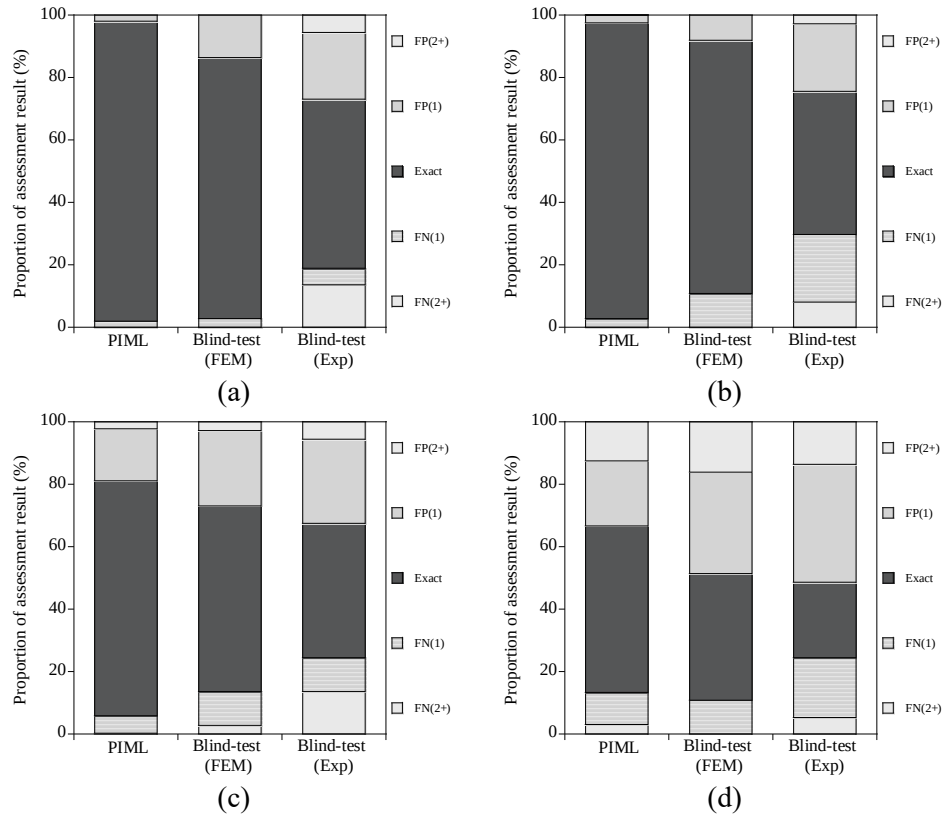


Figure 2.15 Comparison of validation results with respect to the amount of modal information used in the physics-informed machine learning (PIML) model: (a) seventeen modes (3rd–19th); (b) twelve modes (3rd–14th); (c) six modes (3rd–8th); (d) three modes (3rd–5th) (Exact: predicted crack-count label matches the true label; FN(1) and FP(1): one-crack underestimation and overestimation, respectively; FN(2+) and FP(2+): underestimation and overestimation by two or more crack, respectively).

2.5 Crack Location and Depth Assessment

2.5.1 Grid Optimization

After identifying the number of cracks (n_{cr}) using the PIML model described in Section 2.4, the crack location vector $\boldsymbol{\xi} \in \mathbb{R}^{n_{cr} \times 1}$ and crack depth ratio vector $\boldsymbol{\delta} \in \mathbb{R}^{n_{cr} \times 1}$ were estimated by minimizing the 2-norm of the difference between the measured and analytically computed NMFS vectors as follows:

$$\min_{\mathbf{x}, \mathbf{F}} \Pi = \frac{1}{2} \|\boldsymbol{\eta}(\boldsymbol{\xi}, \boldsymbol{\delta}) - \bar{\boldsymbol{\eta}}\|_2^2 \quad \text{subject to } \boldsymbol{\xi}_L \leq \boldsymbol{\xi} \leq \boldsymbol{\xi}_U, \boldsymbol{\delta} > \mathbf{0}, \quad (2.3)$$

where $\boldsymbol{\eta} \in \mathbb{R}^{n_{mode} \times 1}$ and $\bar{\boldsymbol{\eta}} \in \mathbb{R}^{n_{mode} \times 1}$ denote the NMFS vectors calculated from Eq. (2.3), and is derived from the measured modal frequencies. n_{mode} is the total number of bending modes used in the optimization. $\boldsymbol{\xi}_L \in \mathbb{R}^{n_{cr} \times 1}$ and $\boldsymbol{\xi}_U \in \mathbb{R}^{n_{cr} \times 1}$ denote the component-wise lower and upper bounds of $\mathbf{x}$, respectively.

Eq. (2.3) formulates a nonlinear optimization problem subject to linear inequality constraints on $\mathbf{x}$ and $\mathbf{F}$. The objective function in Eq. (2.3) has multiple local minima with respect to the crack location and local flexibility vectors. Given that gradient-based optimization methods are prone to converging to local minima depending on the initial guess, they are not well-suited for solving Eq. (2.3). Therefore, a grid-based optimization scheme is adopted to seek the optimal solution while avoiding suboptimal outcomes.

In grid optimization, predefined combinations of crack locations and local flexibilities are sequentially evaluated, and the optimal combination that minimizes the objective function is selected. Although computing NMFS values for all grid

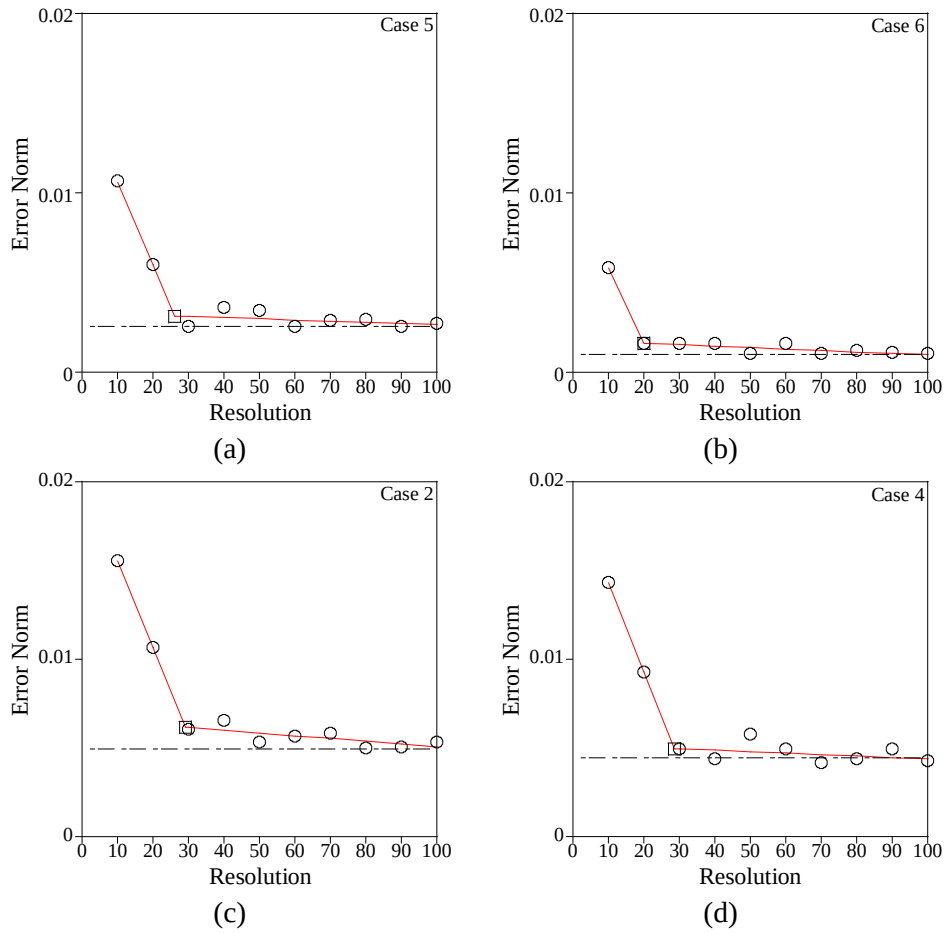
points using FEA is computationally intensive, it is made manageable by utilizing the closed-form solutions from Eq. (2.2) to instantaneously calculate $\boldsymbol{\eta}(\boldsymbol{\xi}, \boldsymbol{\delta})$ in Eq. (2.3). The upper and lower bounds for crack locations were set to 5% and 50% of the beam length, respectively, considering the symmetry of the free-free boundary conditions.

2.5.2 Determining Optimal Resolution

The number of intervals used to discretize the search space for each parameter is termed the resolution. Identifying the crack location is more critical than estimating the crack-depth ratio for incipient cracks. Therefore, the resolution of the crack-depth ratio was fixed at three discrete levels: 10%, 15%, and 20% of the beam height. Determining the appropriate resolution of the crack location is critical in grid optimization because it directly affects computational efficiency and prediction robustness. A finer resolution exponentially increases the number of grid points as the number of cracks grows, thereby raising computational costs. Conversely, an overly fine resolution may cause overfitting, where the optimization process begins to capture the physical response and modeling errors inherent in the approximate solution of Eq. (2.2). In such cases, the estimated crack configuration may significantly deviate from the actual configuration.

To balance computational cost and accuracy while preventing overfitting, the optimal resolution was empirically determined using synthetic NMFS vectors generated from FEA for the crack scenarios described in Appendix (A1). The

resolution was gradually increased in steps of 10, with grid optimization performed at each step. For each cracked beam case, the error norm at each resolution level was fitted via bilinear regression, as shown in Figure 2.16. The error norm was computed using the NMFS from Eq. (2.2), based on the exact crack configuration, which served as a reference for comparison with the NMFS obtained from FEA.



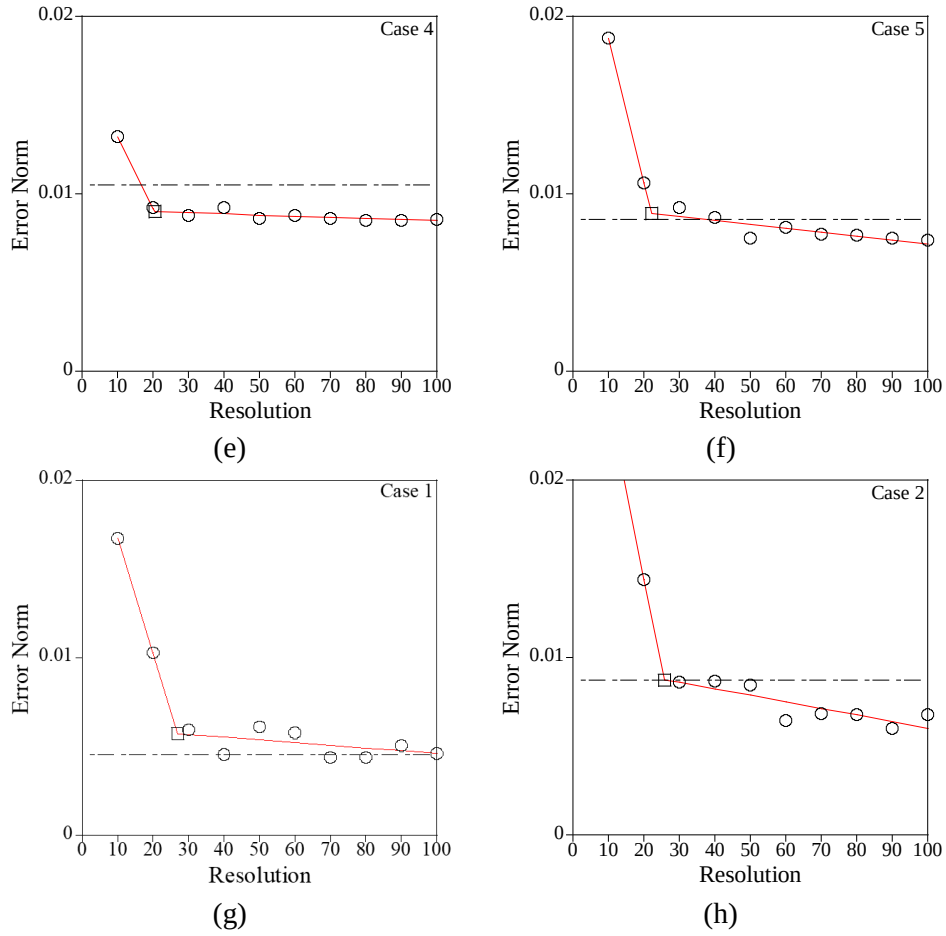
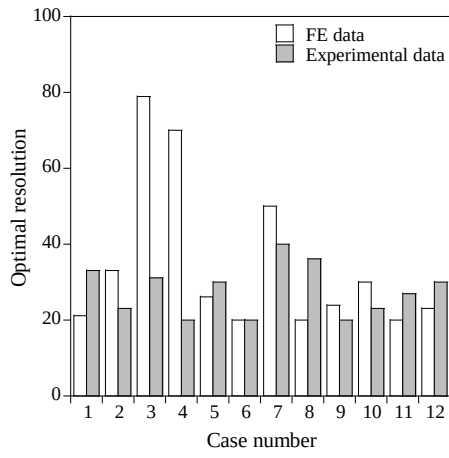
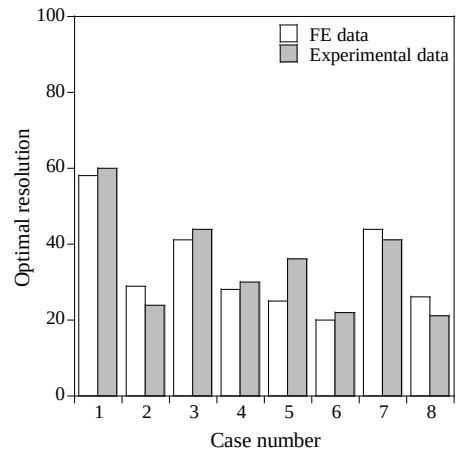


Figure 2.16 Representative examples of determining the optimal resolution via bilinear fitting: (a), (b) one-crack scenario; (c), (d) two-crack scenario; (e), (f) three-crack scenario; (g), (h) four-crack scenario ($\circ$: error norm for each resolution step; $\square$: determined optimal resolution; —: fitted curve from bilinear regression; - -: reference error norm).

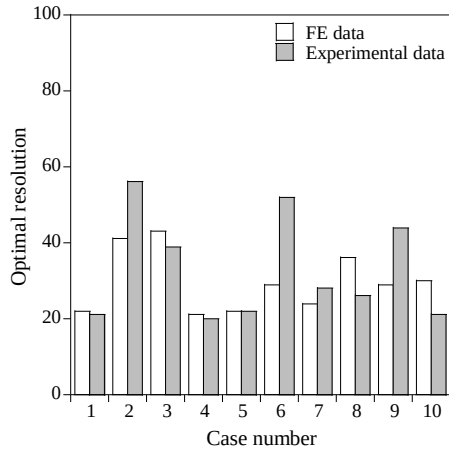
The inflection points of the fitted curve, determined without knowledge of the exact crack configuration, were used to select the optimal resolution, aligning closely with the reference error level from the exact crack configuration, as shown in Figure 2.16. The optimal resolution for each case, categorized by the crack-count label, is shown in Figure 2.17.



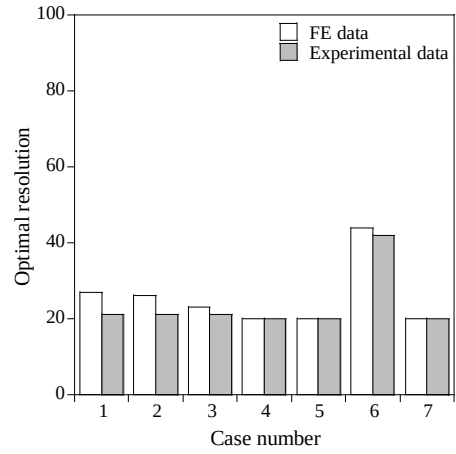
(a)



(b)



(c)



(d)

Figure 2.17 Determined optimal resolution for beams with cracks: (a) one-crack scenario; (b) two-crack scenario; (c) three-crack scenario; (d) four-crack scenario.

2.6 Validation Results

Using the optimal resolution determined in Section 2.5.2, grid optimization was applied to the measured NMFS corresponding to the crack scenarios in Appendix (A1) for crack location identification. In this study, the criteria for assessing crack locations relative to the true crack locations are defined as follows: a prediction is considered (1) exact if the identified location falls within a 5% margin of error ($\pm 2.5\%$ from the true location), (2) an error if it falls outside this margin, and (3) FN if no crack is identified.

Figure 2.18(a) shows the assessment results assuming prior knowledge of the number of cracks. In the numerical validation using the optimal resolution described in Section 2.5.2, approximately 94.2% of the 86 cracks were accurately identified. The lower graph depicts the crack assessment results based on the NMFS data measured from vibration experiments using surface-bonded piezoelectric wafers (as shown in Figure 2.7) for the cracked beams listed in Appendix (A1). The crack location identification accuracy reached 82.6% across 86 cracks. Detailed results for each crack scenario are presented in Appendix (A4 and A5). Additional analysis focused on FN and error cases is presented in Figure 2.18(a). In the numerical validation, all cracks in these categories had a depth ratio of 0.1. For experimental validation, shallow cracks with the same depth ratio accounted for 73.3% of the cases, while 26.7% had a depth ratio of 0.15. These results indicate that crack identification becomes increasingly difficult as crack depth decreases, particularly for crack-depth

ratios below 0.1 [Khi14a]. The minor influence of shallow cracks on the modal frequencies leads to a low signal-to-noise ratio in the NMFS, resulting in different crack scenarios producing nearly indistinguishable NMFS patterns. The indistinguishable patterns complicate the unique identification of true crack configurations, leading to multiple local optima in the grid optimization space, where even the global minimum may correspond to a configuration that overfits the noise or modeling error of the surrogate model. Thus, while the NMFS can be accurately predicted using a surrogate model for small crack-depth ratios, identifying crack locations from experimental NMFS becomes challenging, especially at crack-depth ratios below 0.1.

Figure 2.18(b) presents the crack assessment results when the number of cracks is unknown and estimated using the PIML approach described in Section 2.4. Based on the predicted crack count, grid optimization was applied to the crack scenarios presented in Appendix (A1). In the numerical validation, 94.2% of the 86 cracks are accurately located, while only 81.4% are correctly identified during the experimental validation. These results are comparable to those obtained when true crack counts are known a priori, suggesting that the proposed method maintains reliable assessment performance even without prior knowledge of the crack count. The detailed case-by-case results are presented in Appendices (A4) and (A5). Additionally, the depth distribution for cracks corresponding to FN and error cases in Figure 2.18(b) was examined. In the numerical validation, all cracks in these categories had a depth ratio of 0.1. In the experimental validation, shallow cracks

with this depth ratio comprised 75% of cases, while 25% had a depth ratio of 0.15. This trend is consistent with the findings reported for the known crack-count condition in Figure 2.18(a), indicating that small incipient cracks with low depth ratios predominantly contribute to assessment errors. This confirms that identifying these incipient cracks remains challenging, regardless of the availability of crack count information. Although the accuracy in estimating crack counts by PIML decreases with increasing crack count, primarily due to the masking effect of the shallow cracks, this did not significantly degrade the overall crack assessment performance. After estimating the crack count, grid optimization effectively identified plausible crack locations based on NMFS matching, achieving results similar to those where the true crack count is known.

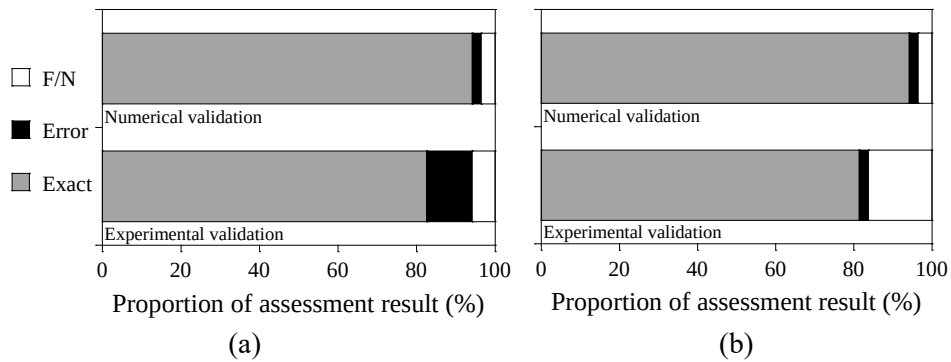


Figure 2.18 Comparison of crack assessment results: cumulative crack assessment ratio when the number of cracks is (a) assumed to be known a priori and (b) identified using PIML (Exact: assessed crack location is within a $\pm 2.5\%$ range of the true crack location; Error: assessed location is outside this range; FN: failure to identify the crack).

To further investigate the robustness of the framework against classification errors, misclassified cases were analyzed in detail. The results indicate that a low modeling

error does not necessarily guarantee perfect crack-count classification, suggesting that misclassifications arise from the intrinsic ill-posedness of the inverse problem rather than inaccuracies in the surrogate model. For instance, Figure 2.19 illustrates a 'false-positive' scenario where the crack count is overestimated. However, the PIML classifier predicted an excessive number of cracks, the subsequent grid optimization clustered the predicted cracks within a narrow region surrounding the true damage location. This result indicates that the framework is robust, as the final assessed configuration converges to the single effective crack, thereby mitigating the initial classification error. Conversely, Figure 2.20 shows a 'false-negative' scenario, typically associated with shallow cracks (e.g., depth ratio of 0.1). While the surrogate model accurately predicts the modal shifts for such cracks, their physical influence on the NMFS is minimal. This observation indicates that the limitation stems from the physical observability limit of the sensing features rather than from deficiencies in the theoretical approximation of the surrogate model.

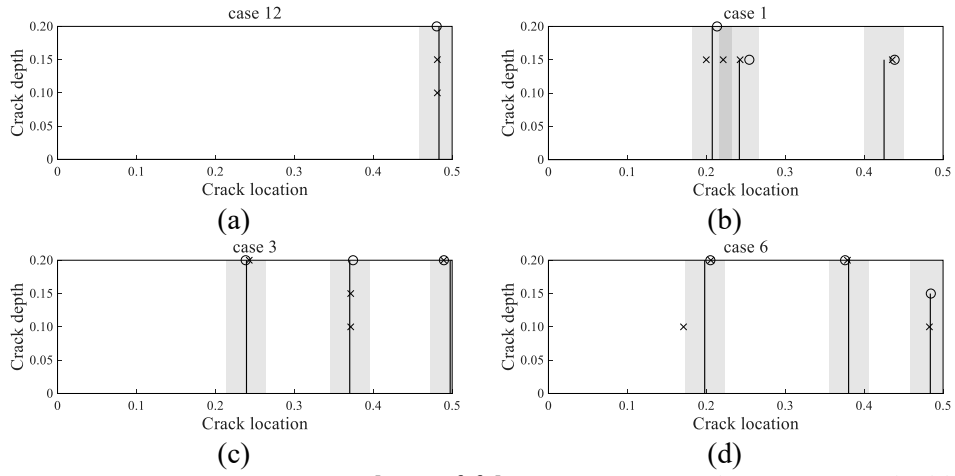


Figure 2.19 Representative analysis of false-positive cases (overestimation): (a) one-crack (case 12); (b), (c), (d) three-crack (case 1,3,6). The markers 'O' and 'X' denote the inversion results obtained using the true number of cracks and the PIML-predicted number of cracks, respectively.

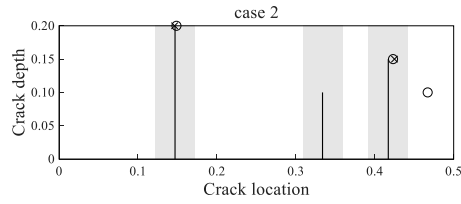


Figure 2.20 Representative analysis of false-negative cases (underestimation) – three-crack (case 2): The markers 'O' and 'X' denote the inversion results obtained using the true number of cracks and the PIML-predicted number of cracks, respectively.

2.7 Additional Validation: Fixed-Free Beam

In this section, the crack assessment method described in Section 2.5 was applied to cracked beams with a fixed-free support condition to validate its effectiveness. For comparison, we adopted the same beam specifications and crack location scenarios from Pat05 as illustrated in Figure 2.21 and listed in Tables 2.3 and 2.4. Although the beam specifications were identical, their lengths varied based on the number of cracks. Their modal testing employed an accelerometer and impact hammer to measure the modal frequencies of the cracked beams up to the 6th mode. Moreover, as mentioned in Section 2.1.1, frequencies above 20 kHz outside the measurable range were excluded, limiting their crack assessment to narrowband modal frequencies. This limitation hindered the assessment of more than three cracks because identifying n cracks requires $2n$ mode frequencies. Furthermore, they focused on assessing deep cracks with crack-depth ratios of approximately 0.15–0.65, primarily above 0.3. As discussed in Section 2.6, assessing incipient cracks is challenging. Consequently, while we retained the crack locations, all crack depths were fixed at 0.2, as listed in Tables 2.3 and 2.4. This crack-depth ratio is smaller than that used in the original study, making identifying multiple crack locations even more challenging than that in Pat05.

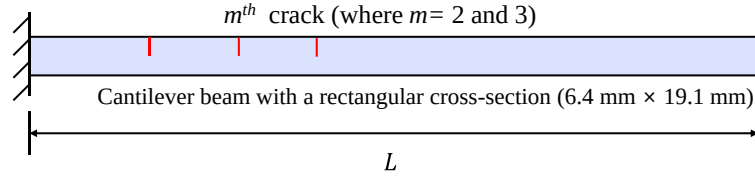


Figure 2.21 Validation example ($E = 70.06$ GPa, $\nu = 0.33$, $\rho = 2645.19$ kg/m³).

Table 2.3 Two-crack scenarios.

Case No.	L (mm)	a_1/L	d_1/h	a_2/L	d_2/h
Case 1	240	0.20	0.20	0.50	0.20
Case 2		0.10	0.20	0.50	0.20
Case 3		0.15	0.20	0.50	0.20
Case 4		0.25	0.20	0.50	0.20

Note: a_m/L and d_m/h are the normalized crack location and depth of the m^{th} crack, relative to the total beam length (L) and height (h), respectively.

Table 2.4 Three-crack scenarios.

Case No.	L (mm)	a_1/L	d_1/h	a_2/L	d_2/h	a_3/L	d_3/h
Case 1	340	0.35	0.20	0.55	0.20	0.75	0.20
Case 2		0.10	0.20	0.40	0.20	0.62	0.20
Case 3		0.28	0.20	0.50	0.20	0.75	0.20
Case 4		0.10	0.20	0.45	0.20	0.73	0.20

Note: a_m/L and d_m/h are the normalized crack location and depth of the m^{th} crack, relative to the total beam length (L) and height (h), respectively.

Figures 2.22 and 2.23 compare the NMFS results from Eq. (2.2) and FEA for the crack scenarios represented in Tables 2.3 and 2.4, respectively. Similar to the free-free beam case, bending modes above the 3rd were used for crack assessment in the cantilever beam. In this analysis, all cracks have a crack-depth ratio of 0.2, where the influence of ABMC begins to appear, which is unaccounted for in Eq. (2.2) [Par18]. Thus, discrepancies between the FEA and NMFS results from Eq. (2.2) tend to become greater than those observed in Figure 2.11, where multiple cracks with depth ratios of 0.1 and 0.2 are present simultaneously, reducing the ABMC dominance. Despite these discrepancies, employing broadband modal frequencies enables robust crack assessment by compensating for modeling errors, improving accuracy even under ABMC effects compared to previous studies using the first few modal frequencies [Pat05].

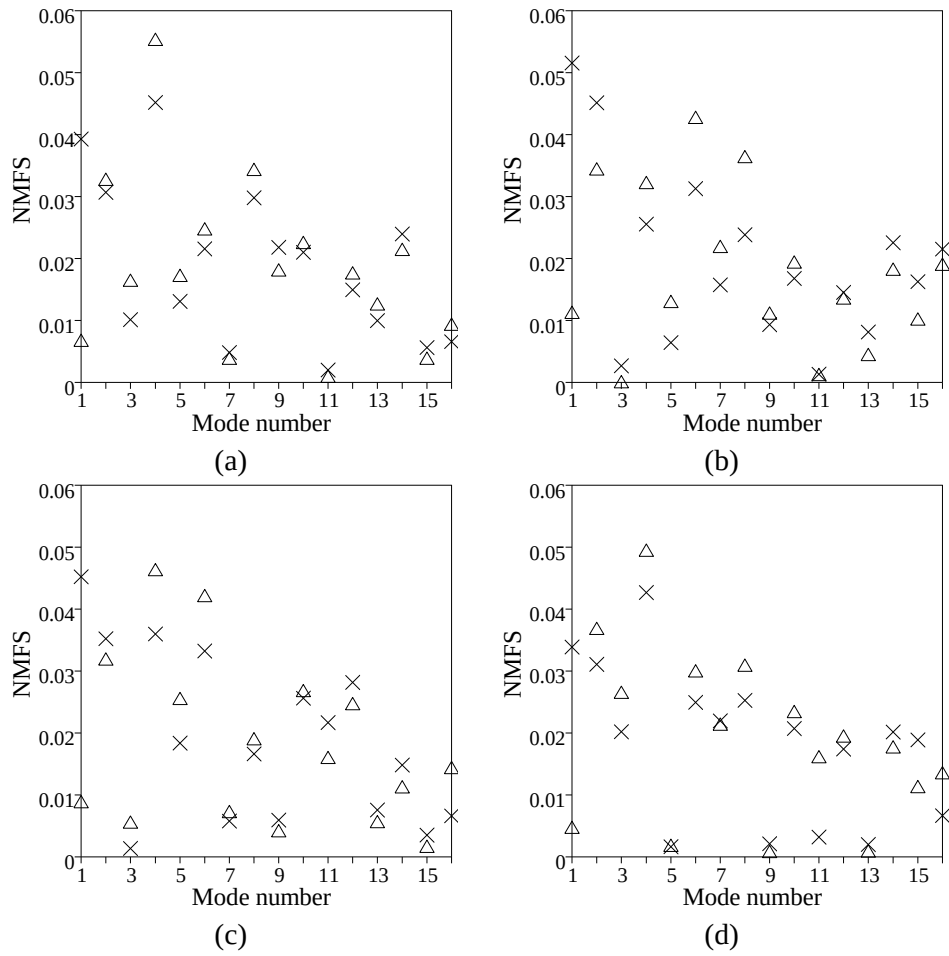


Figure 2.22 NMFS comparison results for a fixed-free beam with two incipient cracks between the surrogate model and FEA: (a) Case 1; (b) Case 2; (c) Case 3; (d) Case 4 (Δ : surrogate model using Eq. (2.2); $\times$: FEA).

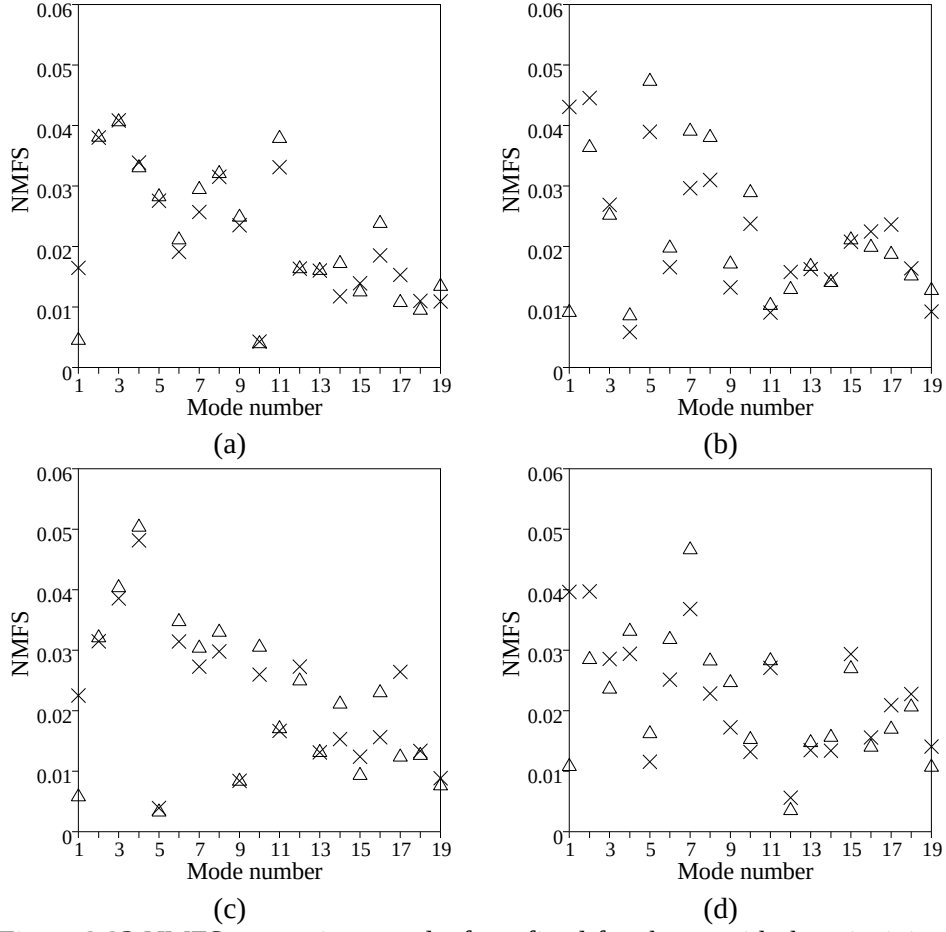


Figure 2.23 NMFS comparison results for a fixed-free beam with three incipient cracks between the surrogate model and FEA: (a) Case 1; (b) Case 2; (c) Case 3; (d) Case 4 (Δ : surrogate model using Eq. (2.2); $\times$: FEA).

In contrast to previous studies that focused on a few lower-order modes, this study incorporated all bending modes within the cut-off frequency. The 1st and 2nd modes were excluded due to inaccuracies in the surrogate model's prediction. This approach aimed to assess the incipient cracks in the beam, as shown in Figure 2.21. For the beam with two incipient cracks, the defined specifications allowed bending modes up to the 16th order within the cut-off frequency range. Conversely, for the beam

with three incipient cracks, where the beam was longer, bending modes up to the 23rd order were theoretically available. However, similar to the free-free condition, only modes up to the 19th order were considered for crack assessment. Figure 2.24 presents the raw data for the crack assessment results of the fixed-free beam with two cracks, as listed in Table 2.3. The crack locations are accurately predicted in all the cases, except for that in Figure 2.24(a). In Figures 2.24 (a)–(d), the central crack is consistently identified with high accuracy in terms of location and depth. However, prediction errors are observed near the fixed end. For example, in Figure 2.24(a), the crack near the fixed end is incorrectly identified as a smaller one located symmetric to the true crack with respect to the center of the beam. In other cases, although the crack location is correctly identified, the depth is underestimated.

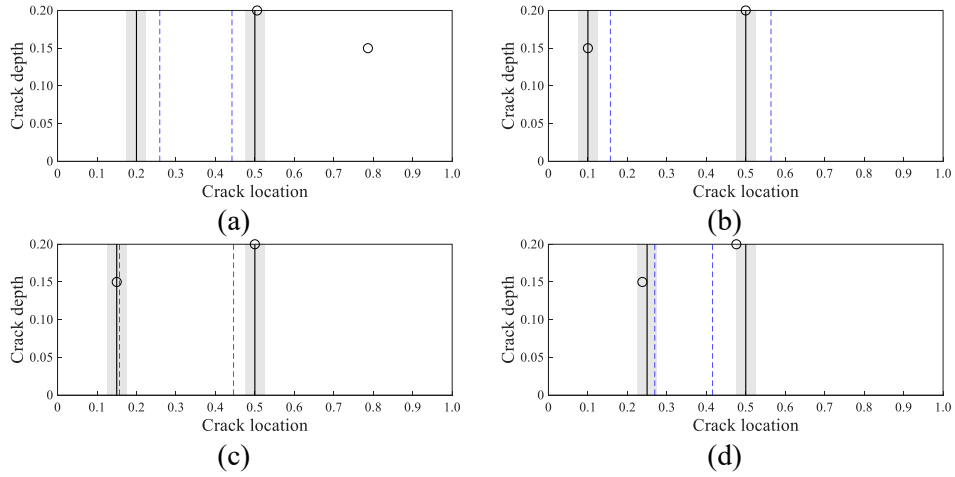


Figure 2.24 Crack assessment results for fixed-free beam with two incipient cracks: (a) Case 1; (b) Case 2; (c) Case 3; (d) Case 4 (○: Predicted cracks under known crack-count condition; —: true crack; - -: Pat05; shaded area: tolerance range for correct assessment). Note: Crack location and depth are normalized with respect to the total beam length and height.

Figure 2.25 presents the raw data for the three-crack scenario in Table 2.4. In Figures 2.25(a) and (c), the location and depth of all three cracks are accurately predicted. By contrast, in Figures 2.25(b) and (d), although two cracks are accurately predicted, the remaining crack is identified as having a smaller crack-depth ratio and at a location outside the tolerance range. This inconsistency originated from the grid optimization process, where the NMFS generated from the predicted crack scenario more closely matches the measured NMFS than that from the true crack scenario.

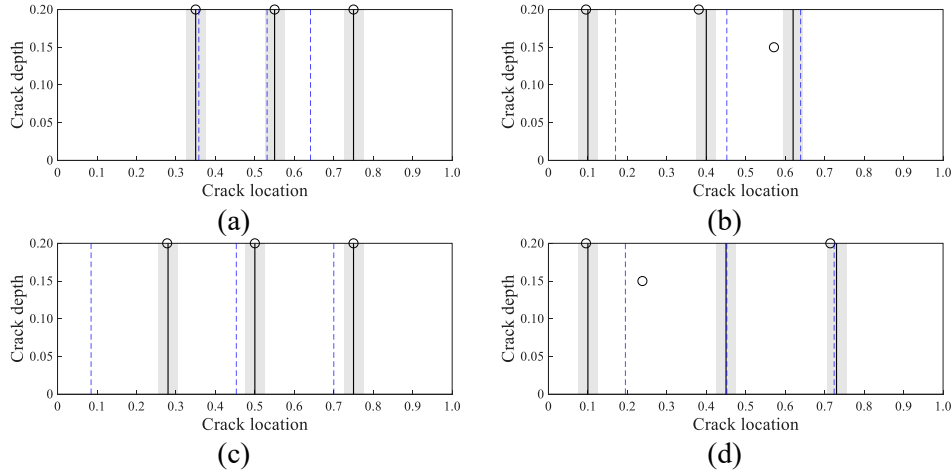


Figure 2.25 Crack assessment results for fixed-free beam with three incipient cracks: (a) Case 1; (b) Case 2; (c) Case 3; (d) Case 4 (○: Predicted cracks under known crack-count condition; —: true crack; - -: Pat05; shaded area: tolerance range for correct assessment). Note: Crack location and depth are normalized with respect to the total beam length and height.

Figure 2.26 presents the results of crack location assessment using grid optimization. The upper graph compares the findings from Pat05 with the outcomes obtained through the proposed grid optimization method in the lower graph. In Figure 2.26 (a), stacked bar plots for the two-crack fixed-free beam scenarios, shown in Figure 2.24, demonstrate an accuracy of 87.5% with optimal resolutions. Similarly,

Figure 2.26(b) presents results for the three-crack scenario depicted in Figure 2.25, also as a stacked bar plot, achieving an accuracy of 83.33%.

These findings reveal that employing broadband modal frequencies significantly improves crack assessment accuracy compared to methods relying solely on a few low-order modal frequencies, as reported in Pat05. As discussed in Section 2.6, assessing smaller cracks is more challenging from an inverse analysis perspective. Although Eq. (2.2) does not account for modeling errors due to the ABMC effect, the application of broadband modal frequencies still allows for an accurate assessment of crack locations, offering a clear advantage over conventional approaches limited to low-order modes. Overall, the results in Figure 2.26 highlight enhanced assessments of smaller cracks at comparable locations to those reported by Pat05, validating the effectiveness of the broadband mode frequency-based assessment method.

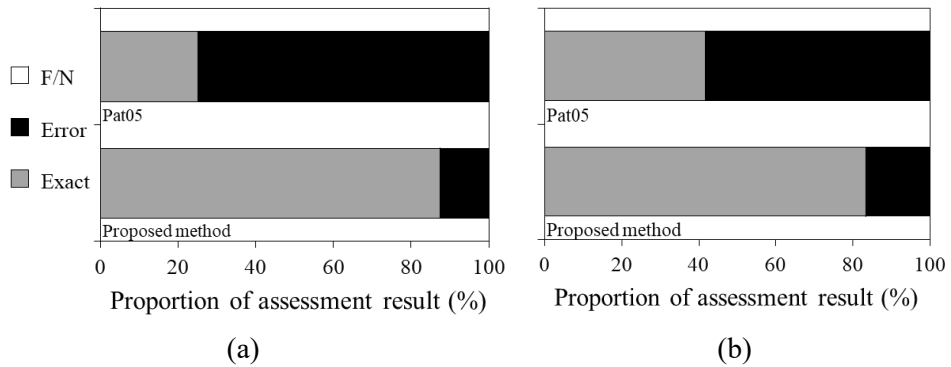


Figure 2.26 Comparison of crack assessment results for fixed-free beam with (a) two incipient cracks and (b) three incipient cracks (exact: assessed crack location is within a $\pm 2.5\%$ range of the true crack location; error: assessed location is outside this range; FN: failure to identify the crack).

2.8 Chapter Summary

In this chapter, PIML-assisted framework was presented for assessing multiple incipient cracks using broadband modal frequencies obtained from surface-bonded piezoelectric wafers. The framework identifies the number of cracks and assesses their locations and depths through grid optimization, and its effectiveness was validated using blind-test data from both finite element analysis and experimental measurements. The main findings are summarized as follows.

- Broadband modal frequencies up to the cut-off frequency were measured using surface-bonded piezoelectric wafers. The NMFS extracted from this broadband range showed enhanced sensitivity to multiple incipient cracks compared to studies using only the first three to five modal frequencies within the narrowband range.
- To address the difficulty in constructing large training datasets from measured or FE-simulated data, we introduced a surrogate model to efficiently augment the NMFS data. A PIML model trained on 25,000 augmented NMFS samples achieved an accuracy of 81.1% on the blind-test data from FEA and 54.1% on the experimental data in identifying the number of cracks. The performance improved significantly when broadband modal frequencies were used instead of the first few modal frequencies.
- After identifying the number of cracks using the PIML model, grid optimization was applied to assess the crack locations and depths. In grid

optimization, considering that the crack location is critical than the depth for incipient cracks, the optimal resolution for location assessment was derived using bilinear fitting. This approach mitigated overfitting and balanced accuracy with computational cost. Grid optimization with optimal resolution accurately identified crack locations, achieving 94.2% and 82.6% accuracy for the FE-based and experimental blind tests, respectively, assuming the crack count was known. When the crack count was estimated via the PIML model, the identification accuracies were 94.2% and 81.4% for the FE-based and experimental blind tests, respectively, with most errors occurring in cases involving shallow cracks with a depth ratio of 0.1.

- Cracks with a depth ratio of 0.1 accounted for all misidentified cases in the FE-based blind test and approximately 73% of errors in experimental tests when the crack count was known, rising to 75% when identified using the PIML model. Although small incipient cracks can be accurately predicted in forward analysis, they remain challenging to assess in inverse analysis. Moreover, the accuracy of crack location assessment remained consistent, regardless of whether the crack-count label was known or estimated by the PIML model. This consistency underscores the difficulty in assessing cracks with a depth ratio of 0.1, even when the true crack-count label is provided. Furthermore, these shallow cracks impact the accuracy of crack count identification by the PIML model.
- The proposed crack assessment through grid optimization was tested using

a fixed-free beam example from a prior study to evaluate its applicability and reliability. The beam specifications and crack locations were taken from a previous crack assessment study, with the crack-depth ratio uniformly set at 0.2. The method achieved accuracies of 87.5% and 83.3% for the two- and three-crack scenarios, respectively. Moreover, the proposed method outperformed the reference study in location accuracy, despite assessing more challenging, smaller depth ratios. This highlights the advantages of using broadband modal frequencies and demonstrates the effectiveness of the proposed method in assessing multiple incipient cracks.

- While the closed-form surrogate neglects evanescent wave effects in the low-frequency regime, this limitation is effectively mitigated by restricting the inversion to higher-order modes. This strategy enhances robustness to boundary-related noise while maintaining computational feasibility for large-scale training.

The results of this study highlight the potential of broadband modal frequencies combined with PIML-assisted analysis to significantly advance crack detection and assessment in beam-type structures. Unlike traditional approaches requiring prior knowledge of damage conditions, the proposed method enables accurate crack assessment even when the number of cracks is unknown. This capability supports early intervention, especially in safety-critical beam-type structures where incipient cracks may evolve into major failures if left undetected. The validation through both numerical and experimental blind tests underscores the method's robustness and

practical applicability. In the long term, the approach offers a pathway toward automated, cost-effective inspection systems that reduce maintenance efforts and enhance safety in real-world engineering practice.

For future studies, we aim to develop an advanced PIML model that simultaneously identifies the number of cracks and assesses their locations and depths, utilizing shifts in broadband modal frequencies measured from surface-bonded piezoelectric wafers. In parallel, the robustness of the proposed method under environmental variations, particularly temperature changes, will be investigated through controlled heat-chamber experiments. This effort aims to evaluate and enhance the applicability of the method to real-world SHM scenarios, in which temperature effects often dominate measurement uncertainty.

In addition to environmental factors, addressing practical challenges associated with sensor deployment is essential for field implementation. Although sensor placement in this study was carefully designed based on prior modal analysis, the requirement for precise PZT positioning poses a significant logistical challenge for field-scale SHM. Unlike laboratory conditions, real structures may exhibit variability in boundary conditions and modal characteristics, making millimeter-level placement accuracy difficult to achieve. Moreover, the sensitivity of broadband frequency excitation to placement errors introduces an additional source of uncertainty. Therefore, future research will include quantitative tolerance analysis of PZT placement errors, along with the development of strategies to enable robust sensor deployment in operational environments.

Finally, it should be noted that the present study was conducted on small-scale aluminum beams considering only open cracks, serving as a preliminary experimental study toward the eventual application to actual civil engineering structural members. Future work should therefore address the extension to concrete materials, diverse cross-sectional geometries such as I-shaped sections with web cracks, full-scale structural members, and various crack types including fatigue and closed cracks.

III. PINN-Based Regression for Time-Dependent Behaviors Prediction and Camber Reconstruction of PSC Girders

3.1 Time-Dependent Behaviors of PSC Girder

3.1.1 Early-Age Behavior and Limitations of Previous Approaches

Prestressed concrete (PSC) beams are becoming more and more common in construction owing to their advantages over reinforced concrete (RC) beams, particularly in terms of crack control and the reduction of long-term deflections [Mar12; Kum14]. Figure 3.1 shows an example of such precast PSC girders, illustrating the typical fabrication process and tendon arrangement used in practice.



Figure 3.1 Fabrication of the 50 m-long PSC I-girder before prestressing: (a) fabrication process before concrete casting; (b) anchorage zone for prestressing

The use of high-strength concrete in PSC girders provides additional benefits such as reduced deflection and increased span capacity, enabling longer spans and more

slender cross-sections in bridge design. As construction technology advances, precast PSC girders with span lengths ranging from 40 m to 60 m have become standard elements for highway bridges and segmental structures.

Accordingly, the demand for prefabricated mass production has increased in recent years.

Despite these advancements, accurate prediction of early-age prestress loss remains a key challenge for ensuring the safety and serviceability of PSC girders. In particular, reliable estimation of prestress loss and camber before the secondary prestressing stage is critical for maintaining constructability and geometric compatibility during the deck–slab composite process. An underestimated prestress loss may cause cracking or excessive deflection under service loads, whereas an overestimated value can lead to unnecessary additional prestressing and excessive camber [Par24]. Both cases may compromise long-term performance or increase construction cost. The early-age time-dependent behavior of PSC girders is primarily governed by the combined effects of concrete creep and shrinkage, tendon relaxation, and the age-dependent evolution of the concrete elastic modulus. These coupled mechanisms cause a gradual reduction in the effective prestress and lead to geometric changes such as camber growth and long-term deflection. Extensive studies have been conducted to model these phenomena through experimental, numerical, and analytical approaches, as summarized in Table 3.1. Further details and comprehensive reviews of these methods can be found in Par24.

Table 3.1 Previous studies on time-dependent behavior of PSC structures

Approach	Method	References
Experimental	Experimental observation and monitoring	Ye20
		Alm21
		Bal22
		Che23
Numerical	Finite Element Method (FEM),	Rob05
		Li19a
		Bir 20
		Kra21
		DE21
		Kra22
		Joy22
		Luo22
		Gal 23
Analytical	Effective modulus method	Gil88
		Li19b
		Kim20
	Age-adjusted effective modulus method	Alm22
		Baz72
		P��e18
		Baz84
	Integro-differential equation method	Au11
		Ran13
		Par24

While the approaches summarized in Table 3.1 have contributed significantly to understanding the time-dependent behavior of PSC structures, each has inherent limitations. Experimental investigations provide reliable physical data but are costly and time-consuming, and the results are often affected by environmental conditions. Numerical models based on the finite element (FE) or finite difference method (FDM) can capture detailed nonlinear coupling among creep, shrinkage, and relaxation;

however, they are computationally expensive and require iterative coupling procedures at each time step. Analytical formulations, such as the effective modulus method and the integro-differential equation approach, offer clear physical interpretation and computational efficiency, but their applicability is limited under complex boundary conditions or nonlinear interactions due to the simplifying assumptions required for closed-form solutions. Consequently, a unified framework that rigorously captures these coupled effects with reasonable computational efficiency is required to evaluate the early-age time-dependent behavior of PSC girders. PIML is suitable for this purpose. Among PIML techniques, PINN directly embeds governing equations into the training process and is particularly suitable for regression-type problems, such as predicting the time-dependent behavior of PSC girders.

3.1.2 Challenges in Applying PINN to PSC Girders

PINN have recently gained attention for solving complex physical problems by embedding governing equations directly into neural-network architectures. Unlike purely data-driven ML, PINN combine measured data with physical constraints, enabling accurate predictions even when experimental data are limited. This feature makes them particularly attractive for civil engineering applications, where data acquisition is often costly or infeasible. Civil structures are typically one-off systems, making data acquisition and model transferability difficult. In contrast, precast PSC girders offer a practical platform for implementing and validating PINN frameworks.

Their standardized fabrication allows consistent boundary and material conditions, enabling repeated measurements of prestress loss and camber. This repeatability provides a suitable foundation for applying this modeling approach and allows the developed models to be extended to future bridge projects.

However, despite these advantages, several technical challenges remain in applying PINN to PSC girders. Figure 3.2 illustrates two representative challenges in applying PINN to PSC girders. As shown in Figure 3.2(a), a stiff problem arises when approximating the time-dependent behavior of PSC girders, such as creep and relaxation. The regression function representing this behavior exhibits a steep gradient during the early-age stage, which causes instability and convergence difficulty in the training of PINN. [Wan21b]. The regression function representing this behavior exhibits a steep gradient during the early-age stage, leading to instability in the training process of PINN. This stiffness-dominated behavior makes the convergence highly sensitive to hyperparameters (e.g., number of domains, activation function), and proper tuning can improve training stability, as schematically illustrated in Figure 3.2(a). Figure 3.2(b) illustrates a representative example of the multi-scale disparity, exemplified by the typical camber profile of a PSC girder during the storage stage before secondary prestressing. This challenge arises because the governing equations to be learned by the PINN contain dimensional quantities, which make network convergence difficult [Moe23]. To address this issue, a data refinement process is required in which the governing equations are reformulated into a dimensionless form. As shown in Figure 3.2(b),

incorporating the dimensionless governing equations into the PINN improves convergence stability and accelerates learning. In this chapter, a PINN-based regression framework is proposed to predict two representative early-age responses of PSC girders: (1) the time-dependent loss of effective prestress and (2) the reconstruction of camber.

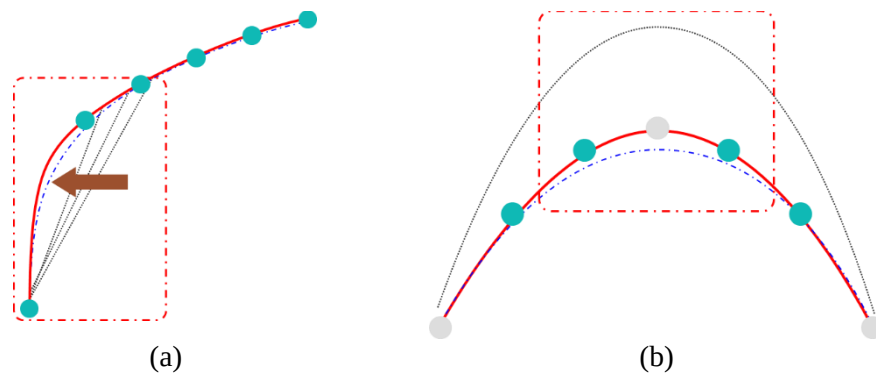


Figure 3.2 Representative challenges in applying PINN to PSC girders: (a) stiffness-induced training difficulty; (b) scale imbalance due to non-dimensional inconsistency.

3.2 Governing Equations of PSC Girder

3.2.1 Prestress Loss

Figure 3.3 presents a simplified PSC beam model to analyze the early-age time-dependent behaviors of a precast prestressed concrete beam. The period of interest for the analysis spans approximately one year beginning from the initial prestress transfer to the installation of the beam at the intended a target construction site. The assumptions for the time-dependent analysis of this study can be described as follows:

- (i) Multiple tendons of the PSC beam can be substituted with a single equivalent tendon as shown in Figure. 3.3(a).
- (ii) The tendon is perfectly bonded with concrete; thus, section analysis for equilibrium between the tendon and the concrete is possible.
- (iii) The beam model follows the Euler–Bernoulli beam theory.
- (iv) The time-dependent behaviors of concrete include the change of elastic modulus, creep and shrinkage.
- (v) The creep of concrete follows the standard linear model of viscoelasticity.
- (vi) The relaxation of the tendon is assumed to be intrinsic.
- (vii) The shrinkage of concrete occurs uniformly throughout the gross section of concrete.
- (viii) The net strain of concrete consists of elastic strain, creep strain and shrinkage strain.
- (ix) The self-weight of the PSC beam is introduced at the moment of initial

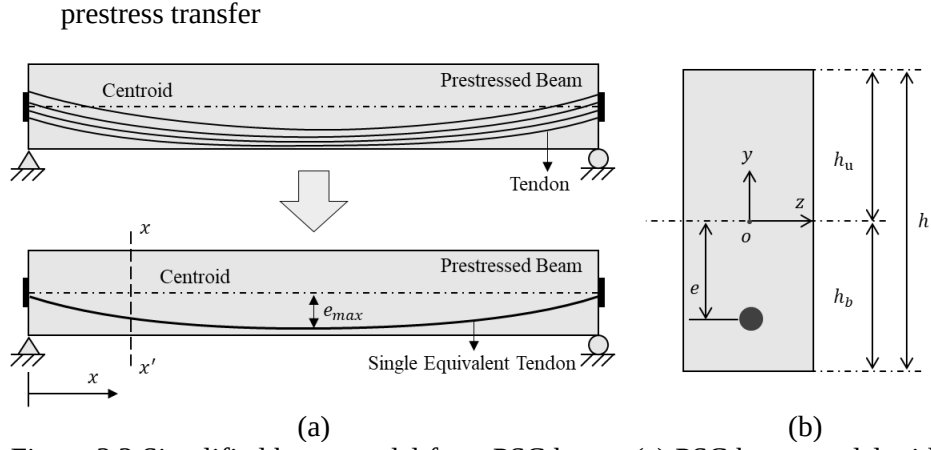


Figure 3.3 Simplified beam model for a PSC beam: (a) PSC beam model with a single equivalent tendon; (b) Cross section at AA'

In the simplified model for a PSC beam in Figure 3.3, time-dependent analysis should be conducted at each section of interest to satisfy the equilibrium between the concrete and the tendon. For the simplicity of description, the spatial variable x corresponding to section AA' in Figure 3.3 is omitted in all field variables hereafter.

The governing equations for time-dependent behaviors can be expressed as follows:

(i) Axial-bending stress equation of concrete

$$\sigma(y, t) = -P(t) \left(\frac{1}{A_c} - \frac{e}{I_c} y \right) - \frac{M_d}{I_c} y \quad (3.1)$$

where t and M_d denote the time beginning from the initial prestress transfer and the bending moment induced by the self-weight of the beam, respectively.

(ii) Constitutive equation of elastic strain of concrete [Gil90]

$$\begin{aligned}\varepsilon_{el}(y, t) &= \frac{\sigma(y, t)}{E_c(t)} = \Phi_{el}(t) \frac{\sigma(y, t)}{E_{c0}} \\ &= -\Phi_{el}(t) P(t) \left(\frac{1}{E_{c0} A_c} - \frac{e}{E_{c0} I_c} y \right) - \Phi_{el}(t) \frac{M_d}{E_{c0} I_c} y\end{aligned}\quad (3.2)$$

where E_{c0} is the elastic modulus of concrete at $t = 0$ and $\Phi_{el}(t) = \frac{E_{c0}}{E_c(t)}$.

(iii) Constitutive equation of creep strain of concrete [Liu21]

$$\varepsilon_{cr}(y, t) = \int_0^t \frac{\tilde{\Phi}_{cr}(t, \tau)}{E_{ci}} \frac{\partial \sigma(y, \tau)}{\partial \tau} d\tau + \frac{\tilde{\Phi}_{cr}(t, 0)}{E_{ci}} \sigma(y, 0) \quad (3.3)$$

where E_{ci} is the elastic modulus of concrete at the age of 28 days while $\tilde{\Phi}_{cr}(t, \tau)$ is the creep coefficient at time t for the unit strain of which the corresponding stress is E_{ci} applied at time τ .

(iv) Net strain of concrete

$$\varepsilon_{net}(y, t) = \varepsilon_{el}(y, t) + \varepsilon_{cr}(y, t) + \varepsilon_{sh}(t) \quad (3.4)$$

(v) Constitutive equation of elastic strain of the tendon

$$\varepsilon_{p_el}(t) = \frac{P(t)}{E_p A_p} + \varepsilon_{p_d}(t) \quad (3.5)$$

where $\varepsilon_{p_d}(t) = \Phi_{el}(t) \frac{M_d}{E_{c0} I_c} e$ representing the elastic strain of concrete at the location of the tendon due to the self-weight of beam at time t .

(vi) Constitutive equation of relaxed strain of the tendon

$$\varepsilon_{p_re}(t) = \Phi_{re}(t, 0) \varepsilon_{p_el}(0) \quad (3.6)$$

(vii) Net strain of the tendon

$$\varepsilon_p(t) = \varepsilon_{p_el}(t) + \varepsilon_{p_re}(t) \quad (3.7)$$

(viii) Compatibility between the concrete and tendon at $y = -e$

$$\frac{\partial \varepsilon_p(t)}{\partial t} = \frac{\partial \varepsilon_{net}(-e, t)}{\partial t} \quad (3.8)$$

For the simplicity of formulation governing equations from Eq. (3.1) to Eq. (3.8) are rewritten by introducing dimensionless variables normalized by respective mechanical quantities. Force, strain and stress variables are normalized by P_i , $\frac{P_i}{E_p A_p}$ and $E_{c0}(\frac{P_i}{E_p A_p})$, respectively. The normalized variables will be denoted by using $\hat{}$ otherwise mentioned hereafter. For example, substituting $\sigma(y, t) = E_{c0}(\frac{P_i}{E_p A_p})\hat{\sigma}(y, t)$ and $P(t) = P_i\hat{P}(t)$ into Eq. (3.1) results in

(i) Dimensionless axial-bending stress equation of concrete

$$\hat{\sigma}(y, t) = -\gamma(y)\hat{P}(t) - \hat{\sigma}_{p_d}\frac{y}{e} \quad (3.9)$$

where $\gamma(y) = n(0)A_p(\frac{1}{A_c} - \frac{e}{I_c}y)$ and $\hat{\sigma}_{p_d} = (\frac{M_d}{I_c}e)/(\frac{E_{c0}P_i}{E_p A_p})$ implying the normalized stress at the location of the tendon due to the self-weight of the beam.

In a similar fashion, Eq. (3.2) to Eq. (3.8) can be rewritten as follows

(ii) Dimensionless constitutive equation of elastic strain of concrete

$$\hat{\varepsilon}_{el}(y, t) = -\Phi_{el}(t)\gamma(y)\hat{P}(t) - \hat{\varepsilon}_{p_d}(t)\frac{y}{e} \quad (3.10)$$

where $\hat{\varepsilon}_{p_d}(t) = \varepsilon_{p_d}(t)/(\frac{P_i}{E_p A_p})$.

(iii) Dimensionless constitutive equation of creep strain of concrete

$$\hat{\varepsilon}_{cr}(y, t) = \int_0^t \Phi_{cr}(t, \tau) \frac{\partial \hat{\sigma}(y, \tau)}{\partial \tau} d\tau + \Phi_{cr}(t, 0) \hat{\sigma}(y, 0) \quad (3.11)$$

where $\Phi_{cr}(t, \tau) = \frac{E_{c0}}{E_{ci}} \tilde{\Phi}_{cr}(t, \tau)$ implying the creep coefficient at time t for unit strain of which the corresponding stress is E_{c0} applied at time τ

(iv) Dimensionless net strain of concrete

$$\hat{\varepsilon}_{net}(y, t) = \hat{\varepsilon}_{el}(y, t) + \hat{\varepsilon}_{cr}(y, t) + \hat{\varepsilon}_{sh}(t) \quad (3.12)$$

(v) Dimensionless constitutive equation of elastic strain of the tendon

$$\hat{\varepsilon}_{p_el}(t) = \hat{P}(t) + \hat{\varepsilon}_{p_d}(t) \quad (3.13)$$

(vi) Dimensionless constitutive equation of relaxed strain of the tendon

$$\hat{\varepsilon}_{p_re}(t) = \Phi_{re}(t, 0) \hat{\varepsilon}_{p_el}(0) \quad (3.14)$$

(vii) Dimensionless net strain of the tendon

$$\hat{\varepsilon}_p(t) = \hat{\varepsilon}_{p_el}(t) + \hat{\varepsilon}_{p_re}(t) \quad (3.15)$$

(viii) Dimensionless compatibility between the concrete and tendon at $y = -e$

$$\frac{\partial \hat{\varepsilon}_p(t)}{\partial t} = \frac{\partial \hat{\varepsilon}_{net}(-e, t)}{\partial t} \quad (3.16)$$

Because there are eight unknown dimensionless variables $\hat{\sigma}(y, t)$, $\hat{P}(t)$, $\hat{\varepsilon}_{el}(y, t)$, $\hat{\varepsilon}_{cr}(y, t)$, $\hat{\varepsilon}_{net}(y, t)$, $\hat{\varepsilon}_{p_el}(t)$, $\hat{\varepsilon}_{p_re}(t)$ and $\hat{\varepsilon}_p(t)$ in eight governing equations from Eq. (3.9) to Eq. (3.16), a single-closed form governing equation with respect to $\hat{P}(t)$

Integrating Eq. (3.16) with respect to t results in

$$\hat{\varepsilon}_p(t) = \hat{\varepsilon}_{\text{net}}(-e, t) + C_1 \quad (3.17)$$

where C_1 can be determined by using two initial conditions $\hat{\varepsilon}_p(0) = \hat{\varepsilon}_{p_el}(0)$ and $\hat{\varepsilon}_{\text{net}}(-e, 0) = \hat{\varepsilon}_{el}(-e, 0)$ as follows

$$C_1 = \hat{\varepsilon}_p(0) - \hat{\varepsilon}_{\text{net}}(-e, 0) = \hat{\varepsilon}_{p_el}(0) - \hat{\varepsilon}_{el}(-e, 0) \quad (3.18)$$

Substituting Eqs. (3.10) and (3.13) at $y = -e$ and $t = 0$ into Eq. (3.18) yields

$$C_1 = \hat{P}(0) + \hat{\varepsilon}_{p_d}(0) - \left\{ -\Phi_{el}(0)\gamma(-e)\hat{P}(0) - \hat{\varepsilon}_{p_d}(0)\frac{-e}{e} \right\} = 1 + \Gamma \quad (3.19)$$

where $\hat{P}(0) = 1$, $\Phi_{el}(0) = 1$ and $\Gamma = \gamma(-e)$.

Substituting Eq. (3.13) into Eq. (3.15) produces

$$\hat{P}(t) = \hat{\varepsilon}_p(t) - \hat{\varepsilon}_{p_re}(t) - \hat{\varepsilon}_{p_d}(t) \quad (3.20)$$

Substituting Eq. (3.17) with Eq. (3.18) into Eq. (3.20) results in

$$\hat{P}(t) = \hat{\varepsilon}_{\text{net}}(-e, t) + 1 + \Gamma - \hat{\varepsilon}_{p_re}(t) - \hat{\varepsilon}_{p_d}(t) \quad (3.21)$$

Substituting Eq. (3.12) into Eq. (3.21) produces

$$\hat{P}(t) = \hat{\varepsilon}_{el}(-e, t) + \hat{\varepsilon}_{cr}(-e, t) + \hat{\varepsilon}_{sh}(t) + 1 + \Gamma - \hat{\varepsilon}_{p_re}(t) - \hat{\varepsilon}_{p_d}(t) \quad (3.22)$$

$\hat{\varepsilon}_{el}(-e, t)$ in Eq. (3.22) can be expressed as follows by using Eq. (3.10) at $y = -e$:

$$\hat{\varepsilon}_{el}(-e, t) = -\Phi_{el}(t)\gamma(-e)\hat{P}(t) + \hat{\varepsilon}_{p_d}(t) = -\Gamma\Phi_{el}(t)\hat{P}(t) + \hat{\varepsilon}_{p_d}(t) \quad (3.23)$$

Introducing a prestress transfer coefficient $\varphi(t) = \frac{1}{1+\Gamma\Phi_{el}(t)}$ at time t , Eq. (3.23)

is rewritten as

$$\hat{\varepsilon}_{el}(-e, t) = -\left\{ \frac{1}{\varphi(t)} - 1 \right\} \hat{P}(t) + \hat{\varepsilon}_{p_d}(t) \quad (3.24)$$

Substituting Eq. (3.24) into Eq. (3.22) results in the following equation:

$$\hat{P}(t) = \frac{\varphi(t)}{\varphi(0)} + \varphi(t)\{\hat{\varepsilon}_{\text{cr}}(-e, t) - \hat{\varepsilon}_{\text{p_re}}(t) + \hat{\varepsilon}_{\text{sh}}(t)\} \quad (3.25)$$

where $\varphi(0) = \frac{1}{1+\Gamma}$.

$\hat{\varepsilon}_{\text{cr}}(-e, t)$ in Eq. (3.25) can be expressed as follows by substituting Eq. (3.9) into Eq. (3.11) at $y = -e$

$$\hat{\varepsilon}_{\text{cr}}(-e, t) = -\Gamma \left\{ \int_0^t \Phi_{\text{cr}}(t, \tau) \frac{\partial \hat{P}(\tau)}{\partial \tau} d\tau + \Phi_{\text{cr}}(t, 0) \right\} + \Phi_{\text{cr}}(t, 0) \hat{\sigma}_{\text{p_d}} \quad (3.26)$$

Substituting Eq. (3.13) into Eq. (3.14), $\hat{\varepsilon}_{\text{p_re}}(t)$ in Eq. (3.25) can be expressed as follows:

$$\hat{\varepsilon}_{\text{p_re}}(t) = \Phi_{\text{re}}(t, 0) + \Phi_{\text{re}}(t, 0) \hat{\varepsilon}_{\text{p_d}}(0) \quad (3.27)$$

Substituting Eqs. (3.26) and (3.27) into Eq. (3.25) produces an integro-differential equation for $\hat{P}(t)$ as follows:

$$\begin{aligned} & \hat{P}(t) \\ &= \frac{\varphi(t)}{\varphi(0)} \\ &+ \varphi(t) \left[-\Gamma \left\{ \int_0^t \Phi_{\text{cr}}(t, \tau) \frac{\partial \hat{P}(\tau)}{\partial \tau} d\tau + \Phi_{\text{cr}}(t, 0) \right\} + \Phi_{\text{cr}}(t, 0) \hat{\sigma}_{\text{p_d}} \right. \\ &\quad \left. - \Phi_{\text{re}}(t, 0) - \Phi_{\text{re}}(t, 0) \hat{\varepsilon}_{\text{p_d}}(0) + \hat{\varepsilon}_{\text{sh}}(t) \right] \end{aligned} \quad (3.28)$$

The numerical solution of Eq. (3.28) for the dimensionless effective prestress force $\hat{P}(t)$ can be obtained using the finite difference method [Par24]. The detailed computational procedure is presented in the Appendix (A6).

The advantage of the PINN over conventional numerical methods lies in solving

partial differential equations in strong forms without discretization through the automatic differentiation. When the governing equation is expressed as an integro-differential equation like Eq. (3.28), the PINN can be employed to solve it through automatic differentiation for integer-order derivatives and approximate integral operators based on Gaussian quadrature [Lu21a].

Substituting $\hat{Z}(t) = \frac{\hat{P}(t)}{\varphi(t)}$ into Eq. (3.28) results in the integro-differential equation in terms of $\hat{Z}(t)$ as follows:

$$\begin{aligned} & \hat{Z}(t) \\ &= \frac{1}{\varphi(0)} \\ &+ \left[-\Gamma \left\{ \int_0^t \Phi_{\text{cr}}(t, \tau) \left\{ \varphi(\tau) \frac{\partial \hat{Z}(\tau)}{\partial \tau} + \hat{Z}(\tau) \frac{\partial \varphi(\tau)}{\partial \tau} \right\} d\tau + \Phi_{\text{cr}}(t, 0) \right\} \right. \\ &\quad \left. + \Phi_{\text{cr}}(t, 0) \hat{\sigma}_{\text{p.d}} - \Phi_{\text{re}}(t, 0) - \Phi_{\text{re}}(t, 0) \hat{\varepsilon}_{\text{p.d}}(0) + \hat{\varepsilon}_{\text{sh}}(t) \right] \end{aligned} \quad (3.29)$$

The right-hand side of Eq. (3.29) can be rearranged by separating the integral term from the non-integral terms as follows:

$$\begin{aligned} & \hat{Z}(t) \\ &= \underbrace{-\Gamma \int_0^t \Phi_{\text{cr}}(t, \tau) \left\{ \varphi(\tau) \frac{\partial \hat{Z}(\tau)}{\partial \tau} + \hat{Z}(\tau) \frac{\partial \varphi(\tau)}{\partial \tau} \right\} d\tau}_{\text{RHS1}} \\ &\quad + \underbrace{\left[\frac{1}{\varphi(0)} - \Gamma \Phi_{\text{cr}}(t, 0) + \Phi_{\text{cr}}(t, 0) \hat{\sigma}_{\text{p.d}} - \Phi_{\text{re}}(t, 0) - \Phi_{\text{re}}(t, 0) \hat{\varepsilon}_{\text{p.d}}(0) + \hat{\varepsilon}_{\text{sh}}(t) \right]}_{\text{RHS2}} \end{aligned} \quad (3.30)$$

where the integral term and non-integral terms are denoted as RHS1 and RHS2, respectively. Then, RHS1 is approximated through Gauss quadrature:

$$\begin{aligned}
\text{RHS1} &= -\Gamma \int_0^t \Phi_{\text{cr}}(t, \tau) \left\{ \varphi(\tau) \frac{\partial \hat{Z}(\tau)}{\partial \tau} + \hat{Z}(\tau) \frac{\partial \varphi(\tau)}{\partial \tau} \right\} d\tau \\
&\approx -\Gamma \sum_{i=1}^{n_{\text{gq}}} w_i \Phi_{\text{cr}}(t, \tau_i(t)) \left\{ \varphi(\tau) \frac{\partial \hat{Z}(\tau)}{\partial \tau} + \hat{Z}(\tau) \frac{\partial \varphi(\tau)}{\partial \tau} \right\} \Big|_{\tau=\tau_i(t)}
\end{aligned} \tag{3.31}$$

where n_{gq} , w_i and $\tau_i(t)$ represent the number of Gauss quadrature points, weight factors, and Gauss points associated with the integral interval from 0 to t , respectively. Note that automatic differentiation is used to analytically derive $\frac{\partial \hat{P}(\tau)}{\partial \tau} \Big|_{\tau=\tau_i(t)}$ at each Gauss point. Substituting Eq. (3.31) into Eq. (3.30) yields the approximate integro-differential equation that is solved through the PINN. Once $\hat{Z}(t)$ is obtained by the PINN, $\hat{P}(t)$ can be calculated by multiplying $\varphi(t)$ to $\hat{Z}(t)$.

3.2.2 Camber Reconstruction

The displacement that occurs after the introduction of tension in a pre-tensioned PSC beam was shown in Figure 3.4. By superimposing the curvature caused by eccentricity and the curvature due to the beam's own weight, the overall curvature of the PSC beam can be determined. Eq. (3.32) shows the displacement-curvature relationship of a Euler–Bernoulli beam.

$$\frac{d^2 w}{dx^2} = \phi_0 \quad x \in [0, L] \tag{3.32}$$

Where $\phi_0 = -\frac{1}{EI}(P_0 e - M^d)$

In this study, to ensure the convergence of the PINN, Eq. (3.32) was normalized to have values between -1 and 1, as shown in Eq. (3.33). By combining the

normalized governing equation with neural network, A PINN model was implemented to predict the camber shape of a PSC beam.

$$\frac{d^2w}{dx^2} = \frac{d^2w}{d\xi^2} \left(\frac{d\xi}{dx}\right)^2 = \phi_0 \quad \xi \in [-1,1] \quad (3.33)$$

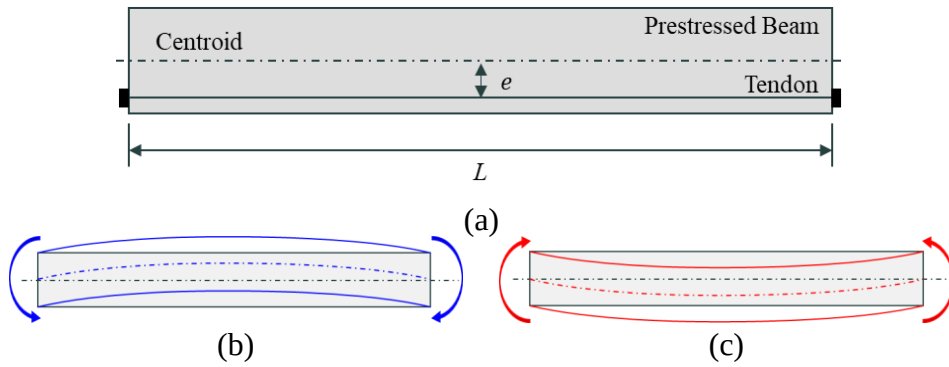


Figure 3.4 Curvature after the introduction of tension in the PSC beam (a) Configuration of a single-tendon post-tensioned PSC girder; (b) Curvatures from prestress force; (c) Curvatures from self-weight of the beam

3.3 PINN Configuration

3.3.1 Prestress Loss

The PINN can directly solve the integro-differential governing equation without numerical discretization by employing automatic differentiation and Gaussian quadrature [Lu21a] for efficient implementation, Eq. (3.28) was reformulated into its approximate form, Eq. (3.29), which was then embedded into the PINN framework as the governing loss function. Table 3.2 presents the software used for designing the PINN model. In this study, DeepXDE was adopted because it can

handle not only ordinary differential equations but also partial differential equations. DeepXDE is optimized for NVIDIA GPUs, making it suitable for solving engineering problems with complex domains. Furthermore, its user code closely resembles mathematical formulations, allowing researchers to implement physics-based models with ease.

Table 3.2 Representative software for implementing PINN [Kar21]

Software	Type	Language	Environment
DeepXDE	Solver	Python	TensorFlow
SimNet	Solver	Python	TensorFlow
PyDEns	Solver	Python	TensorFlow
NeuroDiffEq	Solver	Python	PyTorch
NeuralPDE	Solver	Julia	Julia
SciANN	Wrapper	Python	TensorFlow
ADCME	Wrapper	Julia	TensorFlow
GPyTorch	Wrapper	Python	PyTorch
Neural Tangents	Wrapper	Python	JAX

Figure 3.5 illustrates the architecture of the proposed PINN for predicting early-age prestress loss. The neural network takes the dimensionless time t as input and predicts the time-dependent effective prestress loss. The predicted output is constrained by the integro-differential equation and the initial condition, as shown in Eq. (3.29). The loss function is designed to minimize the residuals of these physical constraints. The implemented PINN can predict the time-dependent variable $\hat{Z}(t)$, from which the effective prestress force and the corresponding concrete sectional stresses can be reasonably estimated.

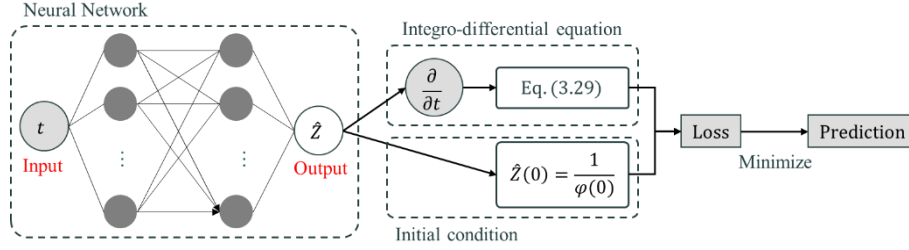


Figure 3.5 PINN for predicting early-age prestress loss

3.3.2 Camber Reconstruction

The second implementation of the proposed PINN focuses on reconstructing the camber of a PSC beam. The camber behavior is governed by the normalized curvature–deflection relationship derived in Subsection 3.2.2, which considers the effects of both the self-weight and the effective prestress in defining the curvature–deflection compatibility. To incorporate this physical relationship, the governing differential equation and boundary conditions are embedded into the loss function of the neural network. Figure 3.6 illustrates the architecture of the proposed PINN for camber reconstruction. The neural network takes the normalized coordinate ξ as input to predict the curvature ϕ_0 and the corresponding deflection w . The predicted curvature is incorporated into the loss function as a physical constraint based on the governing differential equation and its integral terms, while the deflection is simultaneously constrained by the specified boundary conditions. In this process, the effective prestress force predicted by the PINN model in Figure 3.5 is used as a given parameter, enabling a reasonable reconstruction of camber at each time step.

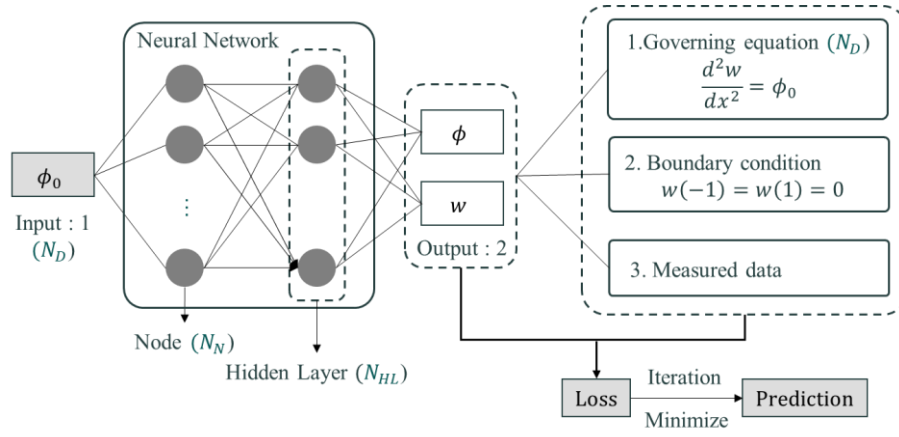


Figure 3.6 PINN for predicting camber of a PSC beam

3.4 Optimal Hyperparameters

Table 3.3 summarizes the representative architecture and components of the PINN. By incorporating prior knowledge directly into the domain of structured artificial neural networks with network structures, layer activation functions, and optimization functions, the PINN can be trained.

Table 3.3 Representative architecture and components of the PINN

Aspect	Category	Components
Network architecture	Network type	FNN, PFNN, ResNet, CNN-PINN
	Layer structure	Node, Hidden Layer
	Optimizer	Adam, L-BFGS
	Activation function	Tanh, ELU, ReLU, Sine
	Training configuration	Learning rate, Batch size, Epochs, Weight coefficients
Prior knowledge	Domain	Initial condition, Boundary condition Governing Equations

The PINN architecture shown in Table 3.3 has three primary hyperparameters: the number of nodes (N_N), the number of hidden layers (N_{HL}), and the number of domains (N_D). Here, N_N and N_{HL} account for the model complexity which enables capturing complicated physical behaviors embedded in the governing equation while N_D

accounts for the amount of prior information provided by the governing equation. Therefore, these three primary hyperparameters determine the model fidelity of the PINN. As these hyperparameters increase, the model fidelity of the PINN improves meaning that the PINN predicts the solution of the governing equation more accurately and consistently. In contrast, the computing cost is likely to decrease as the hyperparameters decreases. In this regard, considering the trade-off between the accuracy error and computing cost, optimal hyperparameter combinations for the PINN should be determined. Figure 3.7 illustrates a schematic of determining optimal hyperparameter combinations considering this trade-off. The accuracy error monotonically decreases with the primary hyperparameters while the computing cost monotonically increases. By introducing a weight factor α that balances the importance of the solution accuracy and the computing cost, optimal hyperparameters can be determined so that the trade-off curve π_T is minimized.

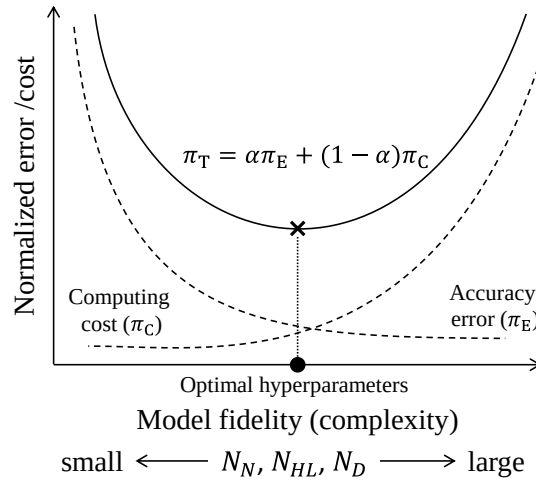


Figure 3.7 Schematic of determining optimal hyperparameters for the PINN using a trade-off curve (π_T) between accuracy error and computing cost where $\pi_T = \alpha\pi_E + (1 - \alpha)\pi_C$ and $0 \leq \alpha \leq 1$

The optimization of N_D is particularly relevant when the PINN is extended to inverse analysis, wherein the model is repeatedly invoked within an iterative inference procedure to estimate unknown structural parameters from field measurements. In this context, the accumulated training cost becomes a critical concern, and a judicious selection of N_D is therefore necessary to balance computational efficiency and predictive accuracy.

3.5 Numerical Verification and Results

3.5.1 Rectangular Cross Section

Figure 3.8 presents a 40m long simply-supported PSC beam with a rectangular cross section (width 0.6m $\times$ height 1.2m). The specification of the PSC beam is provided in Table 3.4. Wet curing is conducted until $t_s=3$ days after which shrinkage begins. The relative humidity of ambient environment $RH=70\%$. The PSC beam is post-tensioned to initial prestressing force such that the initial prestressing ratio $f_{pi}/f_{py}=0.7$. It is assumed that the losses of prestressing force due to friction and anchorage draw-in have been taken-into-account in the initial prestressing ratio. The profile of tendon is straight with a constant eccentricity ratio $e/h=0.3$. The time-dependent properties of concrete are calculated by using CEB-FIP Model Code 1990 [CEB90]. The relaxation of the tendon is calculated by modifying the intrinsic stress relaxation function proposed by Mag62 for low relaxation steel.

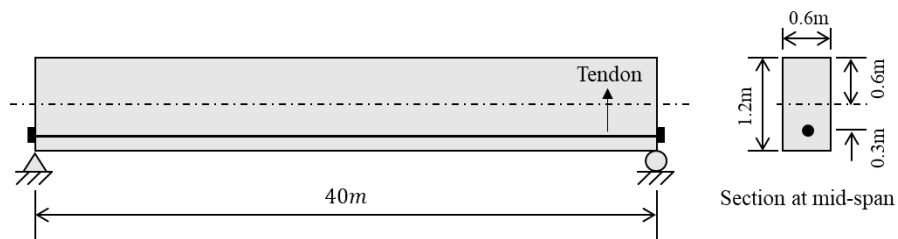


Figure 3.8 40m long simply-supported PSC beam with a rectangular cross section

Table 3.4 Specification of the PSC beam in Figure 3.8

Material properties	Specification
E_{ci}	33,600 MPa
E_p	200 GPa
A_c	0.72 m ²
I_c	0.0864 m ⁴
A_p/A_c	1 %
e/h	0.3
f_{ck}	32 MPa
f_{cu}	36 MPa
f_{py}	1,580 MPa
f_{pi}/f_{py}	0.7
w_c	25 kN/m ³

The forward finite difference method described in Appendices (A6) and (A7) is adopted to obtain reference solutions of Eq. (3.29). The analysis period is from $d_0 = 28$ days on which the initial prestress transfer is carried out to 365 days, i.e. $0 \leq t \leq 337$. Time increment Δt is 0.01685 day in the finite difference equation. The middle of the PSC beam is a section of interest for time-dependent analysis.

Table 3.5 presents hyperparameter combination for varying N_N , N_{HL} and N_D which are three primary hyperparameters described in Section 3.4. N_D is also known as the number of residual points. The residual points are randomly selected at the beginning of training and remain static during the training process. Once the residual points are selected, the deep neural networks are optimized to satisfy the physics imposed by the integro-differential governing equation at the selected residual points. The total of 144 combinations of primary hyperparameters are considered. ‘elu’ activation function is adopted for the activation function and the Adam optimizer followed by

L-BFGS is employed for optimization. Eq. (3.31) is accurately approximated using 40 Gauss quadrature points. All computing simulations were conducted on a workstation equipped with an Intel Core i9-10900X CPU and an NVIDIA Quadro RTX 6000 GPU, running a Windows 10 Pro operating system. The DeepXDE [Lu21a] was adopted to implement the PINNs with hyperparameter combinations for solving Eq. (3.29) in Table 3.5.

Table 3.5 Hyperparameter combinations of the PINN for solving Eq. (3.29)

Hyperparameter	Combinations
N_N	16, 32, 64, 128
N_{NL}	5, 6, 7, 8
N_D	2, 4, 8, 16, 32, 64, 128, 256, 512
Activation function	elu
Optimizer	Adam + L-BFGS
Learning rate	0.001
Maximum iteration	10,000 for Adam optimizer
Number of Gauss quadrature points	40

For each hyperparameter combination outlined in Table 3.5, accuracy errors are calculated through 30 times Monte-Carlo simulations to confirm the consistency of the predicted solutions. Here, the accuracy error is defined as the relative root mean square error between the predicted solution and the reference solution calculated from Eq. (3.28). Figure 3.9 presents a box plot that exhibits the 16 mean values of the 30 accuracy errors, each corresponding to one of the 16 hyperparameter combinations per number of domain. The symbols ' ' and 'o' represent the mean values and the outliers of 16 mean values of the 30 accuracy errors, respectively. The median values in the box plot align closely with the mean values despite some outliers.

Three implications can be drawn from Figure 3.9: Firstly, the most critical hyperparameter is N_D ; the accuracy error decreases rapidly with $N_D < 32$ and then slows down with $N_D \geq 32$. Secondly, when N_D is small, the accuracy error deviates to some extent with varying N_N and N_{HL} . As N_D increases, however, the impact of N_N and N_{HL} on the accuracy error decreases quickly. Thirdly, when N_D is very small, the predicted solution from PINN is likely to poorly fit the reference solution as shown in Figure 3.10. In contrast, when $N_D \geq 32$, the predicted solutions agree with the reference solution very well, as shown in Figure 3.11. The upper and lower bounds depicted in Figures 3.10 and 3.11, which are set at a standard deviation away from the predicted solution, represent the solution precision for 30 times trials during the Monte-Carlo simulation. As observed in Figure 3.10, the solution precision is low when N_D is small, regardless of large N_N and N_{HL} . Conversely, when $N_D \geq 32$, the solution accuracy improves significantly for the identical N_N and N_{HL} as shown in Figure 3.11. The point-wise errors of the well-fitted results from PINN in Figure 3.11 are presented in Figure 3.12.

Similar to Figure 3.9, Figure 3.13 illustrates a box plot that provides the mean values of the 30 computing times, each corresponding to one of the 16 hyperparameter combinations per number of domain. Similar to Figure 3.9, the median values in the box plot are almost identical to the mean values though some outliers are observed. The computing time increases monotonically as N_D increases. Up to $N_D = 64$, the computation time remains constant regardless of N_N and N_{HL} , while for $N_D \geq 128$, the computation time increases with N_N and N_{HL} .

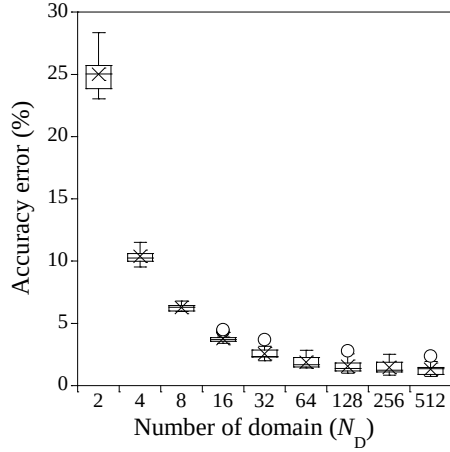


Figure 3.9 Accuracy error of PINN for solving Eq. (3.29)

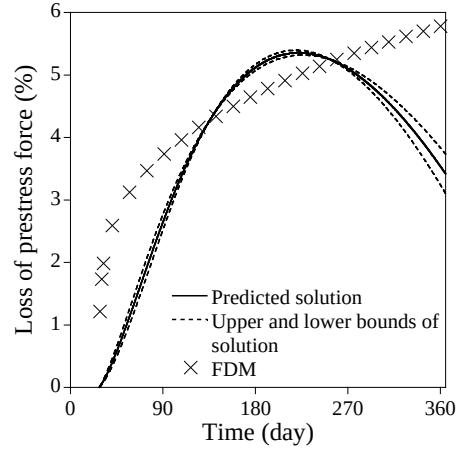


Figure 3.10 Poorly fitted results from PINN for solving Eq. (3.29) [$N_N = 128$, $N_{HL} = 7$, $N_D = 2$]

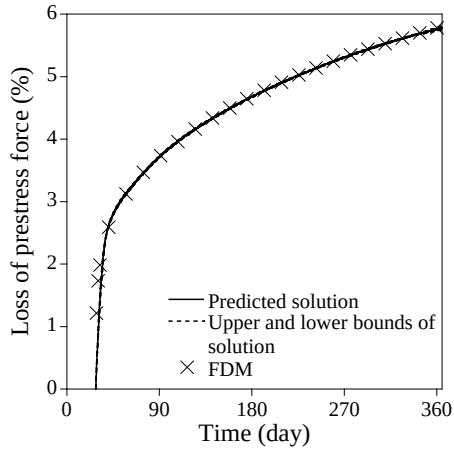


Figure 3.11 Well-fitted results from PINN for solving Eq. (3.29) [$N_N = 128$, $N_{HL} = 7$, $N_D = 32$]

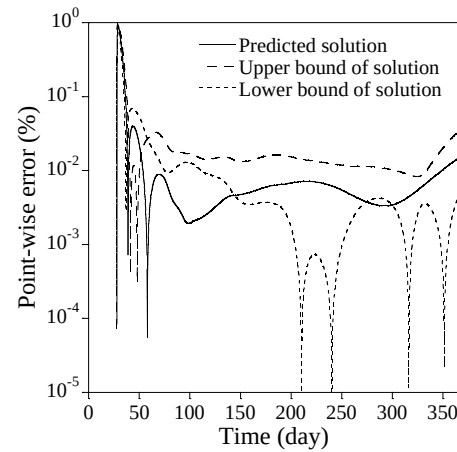


Figure 3.12 Point-wise errors of the well-fitted results from PINN in Figure 3.11

The optimal N_D is determined by using trade-off curve between accuracy error and computing cost in Figure 3.7. The accuracy error π_E and the computing cost π_C in Figure 3.7 are calculated with respect to N_D by using the associated mean values in Figures 3.9 and 3.13, respectively. Note that the mean values in Figures 3.9 and 3.13

are normalized respectively so that the maximum values of π_E and π_C become unit values. Figure 3.14 illustrates trade-off curves to determine an optimal hyperparameter N_D for three different weight factors $\alpha = 0.5, 0.7$, and 0.8 . When $\alpha = 0.5$, the solution accuracy and the computing cost are equally important while the solution accuracy is four times more important than the computing cost when $\alpha = 0.8$. Observing Figure 3.14, the optimal N_D corresponds to 16, 32 and 64 for $\alpha = 0.5, 0.7$, and 0.8 , respectively.

Figure 3.15 presents the predicted solutions from Eq. (3.29) for $N_D = 16, 32$ and 64 , respectively with $N_N = 64$ and $N_{HL} = 8$. Note that each predicted solution is the mean of 30 solutions from Monte-Carlo simulation for each combination of N_D , N_N and N_{HL} . For the comparison purpose, the predicted solution corresponding to $N_D = 512$, $N_N = 128$ and $N_{HL} = 8$ is presented standing for the most accurate solution available from the PINN while the numerical solution from finite difference method is provided as the reference solution. Overall, the three predicted solutions for $N_D = 16, 32$ and 64 agrees very well with that for $N_D = 512$ and the reference solution after 60 days though there exist slight differences from 28 days to 50 days. As N_D increases, these differences vanish as shown at the right bottom in Figure 3.15. The corresponding point-wise errors from 28 days to 50 days are presented in Figure 3.16.

Based on the observation of Figures 3.14 and 3.15, the optimal combinations of the primary hyperparameters in the PINNs are determined as $N_D = 32$, $N_N = 64$ and $N_{HL} = 8$. Note that the number of nodes (N_N) and the number of hidden layers (N_{HL}) affect neither the accuracy error nor the computation time when $N_D = 32$. Therefore,

this hyperparameter combination is adopted in PINNs to solve Eq. (3.30) throughout the numerical verification otherwise mentioned hereafter.

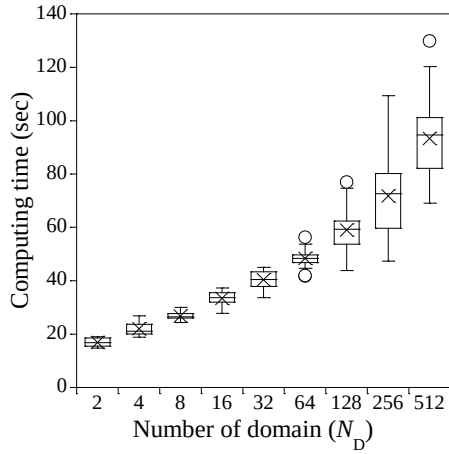


Figure 3.13 Computing time of PINN for solving Eq. (3.29)

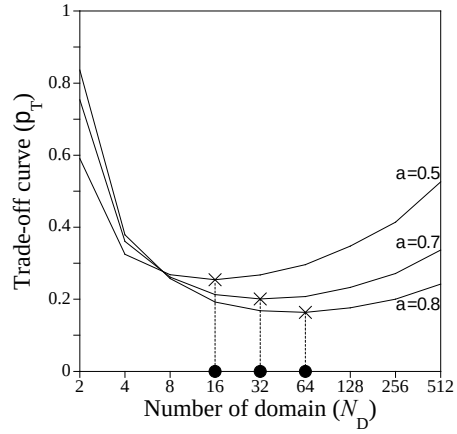


Figure 3.14 Determining optimal N_D through trade-off curves for different α as described in Figure 3.7

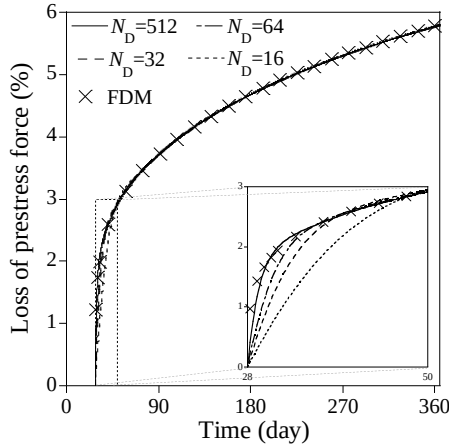


Figure 3.15 The predicted solutions using optimal N_D for different α from Figure 3.14 with $N_N = 64$ and $N_{HL} = 8$

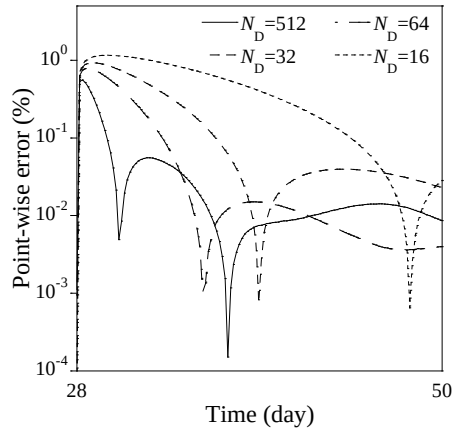


Figure 3.16 Point-wise errors of the predicted solutions from 28 days to 50 days in Figure 3.15

Figure 3.17(a) shows the comparison of the numerical results from the proposed PINN to forward finite difference method regarding the loss of prestress force in case all time-dependent factors of the PSC beam are accounted for. The full coupling among the elastic modulus, creep and shrinkage of concrete and the relaxation of tendon affects loss of prestress force with time. The loss of prestress force surges to 3.7% at 90 days and approaches 5.8% at 365 days. Figure 3.17(b) compares the stresses at the top and the bottom of the beam and at the tendon location. The final fractional magnitudes of the stresses at the top and the bottom of the beam, and at the tendon location become 103%, 83.3% and 69.8%, respectively compared to those on 28 days. The point-wise errors of the numerical results from the PINNs in Figure 3.17 are presented in Figure 3.18. The proposed method yields satisfactory results compared to the forward finite difference method. The absolute difference at 365 days in loss of prestress force is 0.027% while that in all stresses at the top and the bottom of the beam and the tendon location is less than 0.20%.

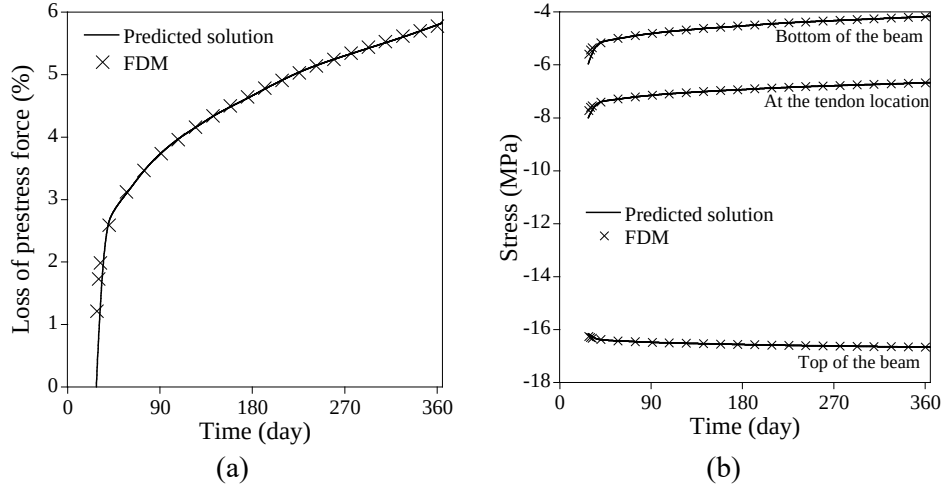


Figure 3.17 Comparison of numerical results from the PINNs ($N_N = 64$, $N_{HL} = 8$, $N_D = 32$) to the forward finite difference method: (a) Loss of prestress force; (b) Stress at the top and the bottom of the beam and at the tendon location

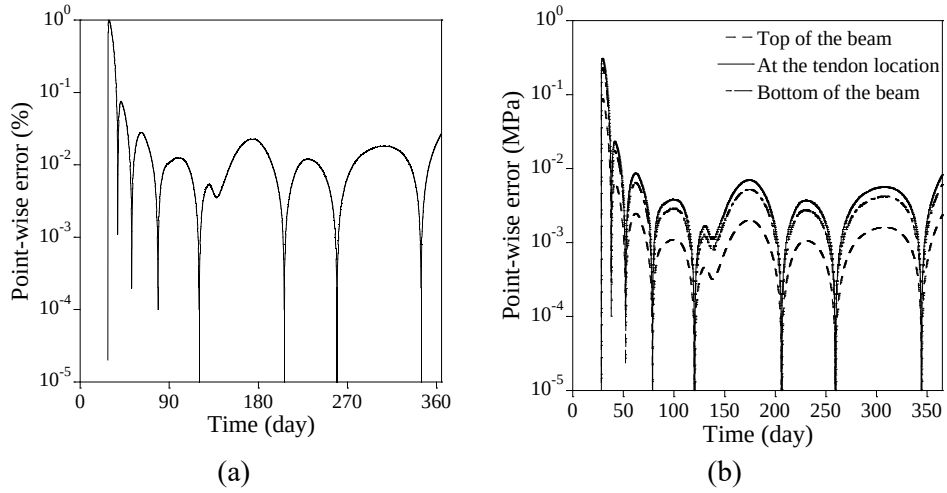


Figure 3.18 Point-wise error of the numerical results from the PINNs in Figure 3.17: (a) Loss of prestress force; (b) Stress at the top and the bottom of the beam and at the tendon location

3.5.2 Conventional PSC I-Beam Section

Figure 3.19 presents a 45m long simply-supported PSC beam with a I-shaped cross section. The specification of the PSC beam is provided in Table 3.6 All design parameters not present in Table 3.4 are identical to those in Subsection 3.5.1 except for the tendon profile, which is parabolic with a maximum eccentricity ratio $e/h = 0.445$ at the middle of the beam.

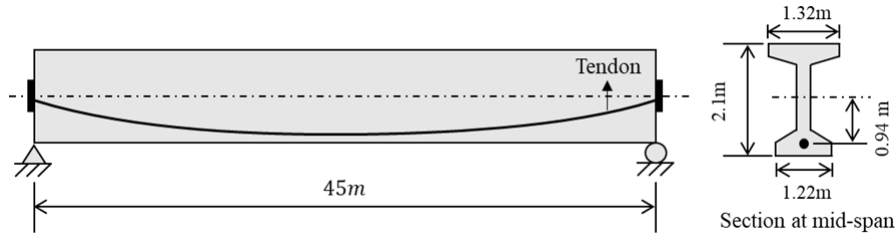


Figure 3.19 45m long simply-supported PSC beam with a I-shaped cross section

Table 3.6 Specification of the PSC I-beam in Figure 3.19

Material properties	Specification
E_{ci}	35,385 MPa
E_p	200 GPa
A_c	1.084 m ²
I_c	0.6454 m ⁴
A_p/A_c	0.972 %
e/h	0.445
f_{ck}	40 MPa
f_{cu}	44 MPa
f_{py}	1,580 MPa
f_{pi}/f_{py}	0.7
w_c	25 kN/m ³

The analysis of time-dependent behaviors of the PSC beam is conducted by using the forward finite difference method in the same way described in Subsection 3.5.1 to obtain the reference solution. The analysis period is from the $d_0 = 7$ days on

which the initial prestress transfer is conducted to 365 days, i.e. $0 \leq t \leq 365$. The time increment Δt is 0.01685 day in the finite difference equation. The middle of the PSC beam is a section of interest for time-dependent analysis. The optimal combination of the primary hyperparameters in the PINNs is identical to that used in Subsection 3.5.1: $N_N=64$, $N_{HL}=8$, $N_D=32$.

Figure 3.20(a) illustrates the comparison between the proposed PINNs and the forward finite difference method for loss of prestress force considering all time-dependent factors of the PSC beam. The complete interaction among the elastic modulus, concrete creep and shrinkage, and tendon relaxation influences the prestress loss over time. The loss of pre-stress force surges to 8.0% at 90 days and reaches 12% at 365 days. Figure 3.20(b) contrasts the stress levels at the beam's top, bottom, and tendon location. The final fractional stress magnitudes at these locations are 119%, 80.6% and 80.0%, respectively compared to those on 28 days. The point-wise errors of the numerical results from the PINNs in Figure 3.20 are presented in Figure 3.21. The proposed method yields satisfactory outcomes when compared to the forward finite difference method. The absolute difference at 365 days in prestress force loss is 0.042% while that in all stresses at the top and the bottom of the beam and the tendon location remains below 0.0075%.

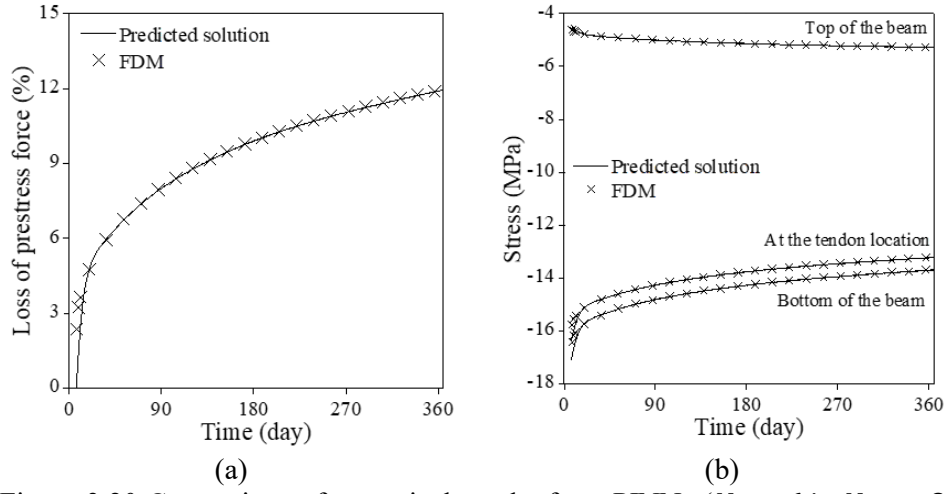


Figure 3.20 Comparison of numerical results from PINNs ($N_N = 64$, $N_{HL} = 8$, $N_D = 32$) to the forward finite difference method: (a) Loss of prestress force; (b) Stress at the top and the bottom of the beam and at the tendon location

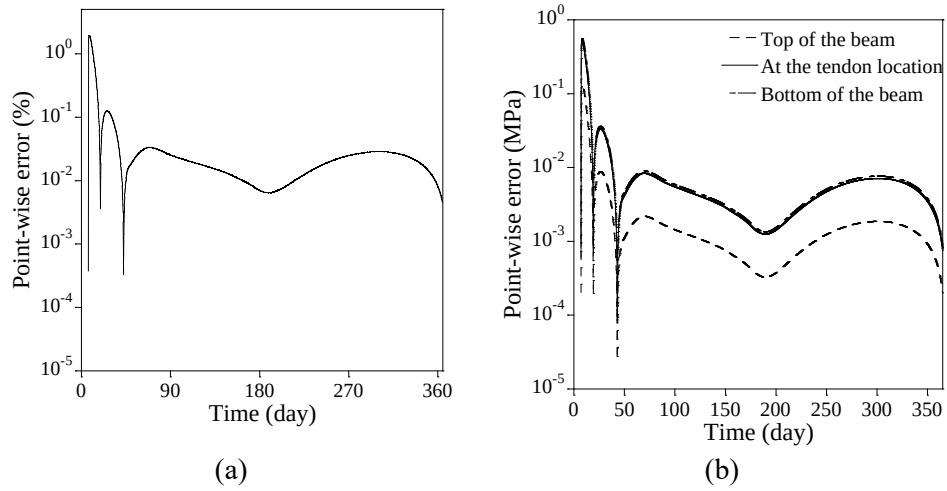


Figure 3.21 Point-wise error of the numerical results from the PINNs in Figure 3.20 (a) Loss of prestress force; (b) Stress at the top and the bottom of the beam and at the tendon location

3.5.3 Camber Reconstruction (TY7)

Figure 3.22 presents an 11.9m long post-tensioned PSC girder with 700mm height TY7 section. The specification of the PSC beam is provided in Table 3.7 But16 reconstructed the camber of a full-scale PSC girder from strains measured after the introduction of prestress. Fiber Bragg Grating (FBG) strain sensors were employed for this purpose. The measurement points were positioned at ten locations with 1 m intervals, as shown in Figure 3.22.

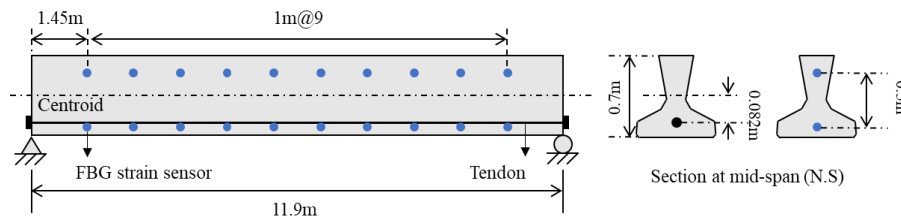


Figure 3.22 11.9m long simply-supported PSC beam with TY7 cross section

Table 3.7 Specification of the TY7 type PSC beam in Figure 3.22

Material properties	Specification
L	11.9m
E	35,300 MPa
I_g	0.01257
q	6.81 KN/m
A_g	272,280 mm ²
A_{strand}	150x16 mm ²

At each point, strains at the top and bottom surfaces of the girder were measured using paired sensors spaced 0.5 m apart vertically. The curvature at each section was calculated from the paired strain data, and the overall camber was reconstructed from

the curvature distribution along the span. The measured cambers were used as reference data for validating the proposed PINN model for camber reconstruction introduced in Subsection 3.3.2. The optimal hyperparameter combination for the PINN was determined using the procedure described in Section 3.4, which balances solution accuracy and computational cost through trade-off analysis. The hyperparameters used for this determination were identical to those listed in Table 3.5 for the prestress-loss prediction model, except that the Gauss quadrature points were not required for camber reconstruction. The optimal set was adopted for the analysis. $N_N=64$, $N_{HL}=8$, $N_D=128$ was adopted for the analysis.

In this study, the camber predicted by the proposed PINN was compared with the result reported by But16. Furthermore, using the released strain measurement data, the PINN model was trained and tested under the three data scenarios defined by Kar21, as illustrated in Figure 1.2: (1) only data, (2) only physics, and (3) physics with data. In the only data scenario, the condition corresponding to conventional data-driven machine learning was simulated by setting $N_D=0$. In this case, camber regression was performed using only the curvature data calculated from the strain measured at the paired sensor points located at the top and bottom of the girder in But16, without incorporating any physical prior knowledge. In the only physics scenario, training was performed using the governing equations without any measurement data. In this case, the process relied entirely on the physical constraints imposed by the equations. In the physics with data scenario, training was performed using both the governing equations and the measured strain data, representing an

ideal PIML framework.

Figure 3.23 presents the curvature and camber reconstruction results under the only data condition ($N_D=0$), where training was performed using only the measured curvature data derived from strain measurements, as no direct camber measurement data are available in this study. Figure 3.23(a) illustrates that the predicted curvature agrees well with the measured values. However, as evident from Figure 3.23(b), the reconstructed camber, obtained by integrating the predicted curvature twice, significantly overestimates the theoretical value, because the measurement noise is progressively amplified through the double integration process and no governing equation constraints are available to regularize this error propagation. The camber values indicated by open circles in Figure 3.23(b) are reconstructed values reported by But16 using their measured strain data, included here for reference only.

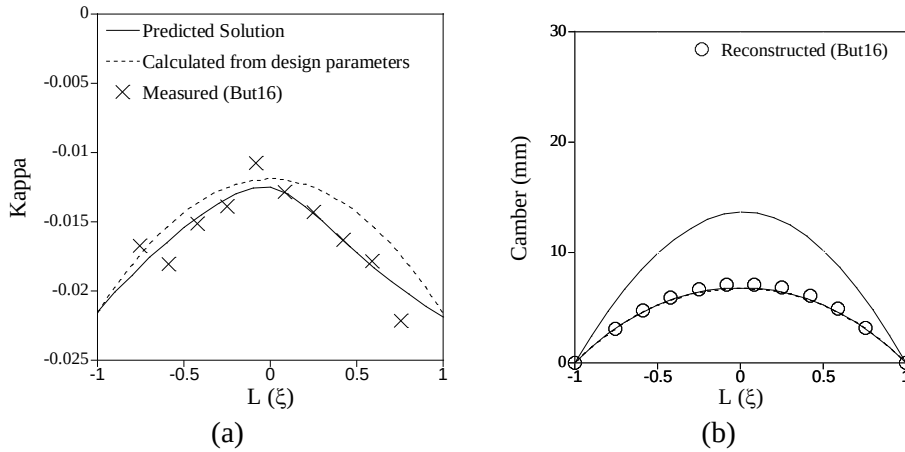


Figure 3.23 Curvature and camber reconstruction results from the PINN model under only data condition. ($N_N=64$, $N_{HL}=8$, $N_D=0$): (a) Curvature; (b) Camber

Figure 3.24 presents the result obtained under the only physics condition, where training was performed using the governing equations derived in Eq. (3.33) without any measurement data. Figure 3.24(a) indicates that the predicted curvature closely follows the analytical solution derived from the design parameters. As presented in Figure 3.24(b), the reconstructed camber also agrees well with the analytical solution. However, since no measurement data are incorporated, the predicted results reflect only the theoretical behavior based on design parameters and show discrepancy with the reconstructed values reported by But16.

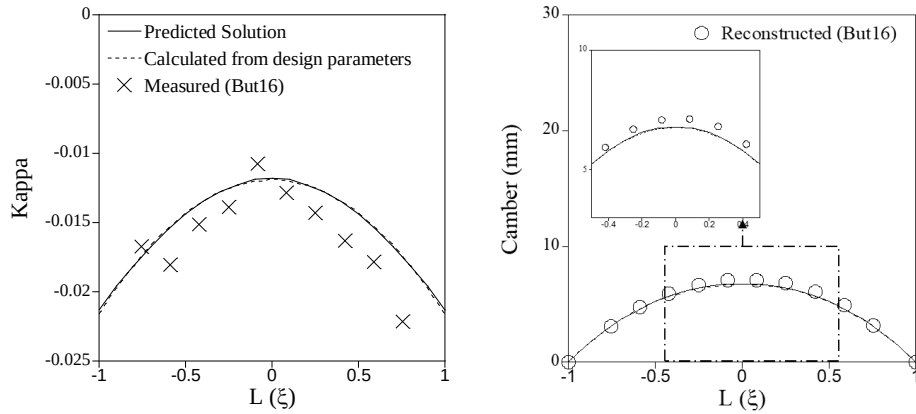


Figure 3.24 Curvature and camber reconstruction results from the PINN model under only physics condition. ($N_N=64$, $N_{HL}=8$, $N_D=128$): (a) Curvature; (b) Camber

Figure 3.25 presents the result obtained under the physics with data condition, where training was performed using both the governing equations and the measured data. Figure 3.25(a) demonstrates that the predicted curvature closely follows the measured values. It can be observed from Figure 3.25(b) that the reconstructed camber agrees well with the reconstructed values reported by But16. Unlike the only data condition, the physical constraints suppress the noise amplification during the

integration process, enabling stable camber reconstruction. Furthermore, unlike the only physics condition, incorporating measured data allows the model to capture the actual structural behavior beyond the theoretical prediction based on design parameters. This confirms that the proposed physics with data approach effectively combines the strengths of both physical knowledge and measurement data, providing reliable camber reconstruction for actual PSC girders.

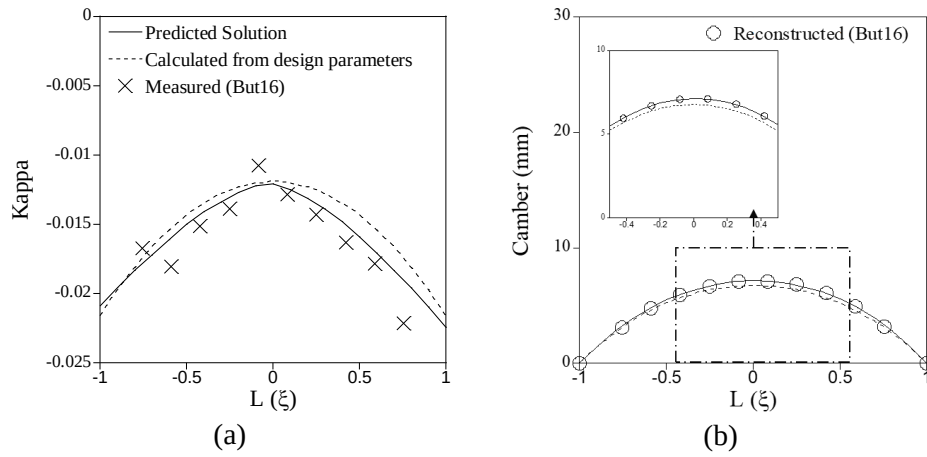


Figure 3.25 Curvature and camber reconstruction results from the PINN model under physics with data condition. ($N_N=64$, $N_{HL}=8$, $N_D=128$): (a) Curvature; (b) Camber

3.6 Chapter Summary

In this chapter, a PINN approach was proposed to predict the early-age time-dependent behaviors of the PSC beam and was also employed to reconstruct its camber. The original contributions of this study are presented below:

- The dimensionless integro-differential governing equation is derived for the PINN. The governing equation accounts for time-dependent coupling among the effective prestress force and several factors including concrete creep and shrinkage, tendon relaxation, and changes in concrete elastic modulus.
- The fidelity of the PINN is determined by three primary hyperparameters: the number of nodes, the number of hidden layers, and the number of domains. Among these primary hyperparameters, the number of domains is the most critical factor for the accuracy and the consistency of predicted solutions from the PINN. Optimal hyperparameter combinations for the PINN was determined by considering the trade-off between solution accuracy and computation time.
- The numerical results from the PINN with an optimal hyperparameter combination yielded satisfactory results, with absolute differences at 365 days in loss of prestress force being 0.027% and in all stresses at the top and bottom of the beam and the tendon location being less than 0.2% for PSC beams with a rectangular and a I-shaped section.

- The proposed PINN approach can effectively predict time-dependent behaviors of PSC beams, offering a promising alternative to conventional numerical methods. The proposed PINN can address the limitations of conventional machine learning by training deep neural networks that satisfy the governing equation associated with the complicated time-dependent behaviors of the PSC beam.
- The proposed PINN framework was also employed for camber reconstruction by embedding the curvature–deflection relationship into the learning process. The model was evaluated under three data scenarios: only data, only prior knowledge, and prior knowledge with data. The results showed that the PINN accurately reproduced the analytical camber under the only prior knowledge scenario, and closely matched the camber reconstructed from measured strain data when both physics and data were incorporated. These findings confirm that the physics-informed approach enables accurate and physically consistent reconstruction of camber profiles, while effectively integrating limited experimental data.

The PINN framework proposed in this chapter requires separate models for predicting time-dependent behavior and camber reconstruction, which may reduce generalization capability and computational efficiency. Therefore, a unified governing equation will be developed to enable the simultaneous prediction of time-dependent behavior and camber, leading to a single PINN model capable of capturing the spatiotemporal behavior of PSC beams. In addition, the present study

employs a basic vanilla PINN architecture, which may limit model performance. Accordingly, advanced PINN architectures, such as adaptive activation functions [Jag20a; Jag 20b; Jag22c; Jag22d], and operator-learning approaches including DeepONet [Kov23; Lu21b], will be incorporated to improve accuracy and convergence. Finally, the proposed framework is intended to be extended to inverse problems, such as estimating time-dependent design parameters associated with concrete creep, shrinkage, and tendon relaxation from limited field measurement data of PSC beams. This extension can serve as a basis for the condition assessment and quality evaluation of PSC girders. In addition, model updating and calibration using actual field measurement data will be investigated to further enhance the practical applicability of the proposed framework to actual PSC structural components

IV. Conclusion

This study presents a unified PIML framework for addressing both classification and regression problems in structural engineering, with a particular focus on data-scarce structural elements. The proposed framework integrates physical knowledge with limited measurement data to enable reliable and physically consistent assessment and prediction of structural behavior. Its effectiveness is demonstrated through two representative applications: the assessment of multiple incipient cracks using broadband modal frequencies, and the prediction of time-dependent behavior and camber reconstruction of PSC girders using a PINN. The results confirm that the proposed PIML framework remains robust and applicable even under practical conditions involving limited and noisy measurement data.

In the classification problem, a PIML-assisted framework was proposed to identify the number of cracks and assess the corresponding locations and depths using NMFS derived from broadband modal frequencies. As demonstrated in Chapter 2, the use of broadband modal frequencies overcomes the fundamental limitation of conventional narrowband-based crack assessment studies, in which multiple cracks cannot be reliably identified in practice, by significantly increasing the amount of available modal information and thereby improving the identifiability of multiple incipient cracks. To address the limitation of insufficient labeled data, a surrogate model based on a closed-form frequency equation derived from the phase-

closure principle was introduced, enabling the efficient generation of large-scale NMFS datasets with minimal computational cost.

The trained PIML model was validated using both finite element-based data and experimentally measured data obtained from surface-bonded piezoelectric wafers, achieving crack-count identification accuracies of 81.1% and 54.1%, respectively. Although the performance decreased under experimental conditions due to the Sim-to-Real gap, the predicted crack-count information effectively constrained the subsequent inverse problem, enabling reliable crack assessment. Based on the identified crack count, crack locations and depths were further assessed using grid optimization, yielding localization accuracies of 94.2% in numerical validation and over 80% in experimental validation. Notably, the assessment performance remained stable even when the crack count was estimated rather than known a priori, demonstrating the robustness of the overall framework against classification errors. Further analysis revealed that most assessment errors were associated with shallow incipient cracks, indicating that the limitation arises from the physical observability of the system rather than from deficiencies in the proposed method.

In the regression problem, as presented in Chapter 3, a PINN-based framework was introduced for predicting the time-dependent behavior of PSC girders and reconstructing the camber at specific time instances. By incorporating the integro-differential-form governing equation for the time-dependent behavior of PSC girders into the neural network architecture, the first model enabled physically consistent prediction of the structural response without relying on conventional numerical

discretization. In addition, a second model based on the curvature–deflection relationship was developed to reconstruct the camber profile at specific time instances. The optimal hyperparameter combination of the PINN was systematically determined by considering the trade-off between prediction accuracy and computational cost, with the number of domains identified as a key factor influencing both accuracy and consistency.

The results demonstrated that the proposed PINN framework produced predictions that closely matched the reference solutions obtained from the finite difference method, with very small discrepancies in prestress loss and stress responses throughout the structure. In addition, the camber profiles at specific time instances were accurately reconstructed. A comparative analysis under different data scenarios showed that purely data-driven models failed to reproduce physically consistent behavior, whereas the integration of governing equations enabled accurate and stable predictions even with limited data.

In conclusion, the proposed PIML-based framework demonstrates that integrating physics-based knowledge with machine learning enables reliable, interpretable, and physically consistent solutions for both structural assessment and prediction. Although the two applications differ fundamentally in problem type, one addressing a classification problem in structural health monitoring and the other a regression problem in structural performance prediction, both share a common underlying challenge: the governing physics are inherently incomplete, and reliable measurement data are severely limited. In both cases, this challenge was overcome

by compensating for data scarcity through the systematic incorporation of physical knowledge, either via physics-informed data augmentation or direct embedding of governing equations. By addressing these two representative problem categories within a unified framework, this study establishes physics-informed learning as a practical and scalable approach for data-scarce civil engineering problems.

Future research should focus on the incorporation of advanced PINN architectures, such as operator learning approaches and adaptive activation functions, to improve model generalization and computational efficiency, thereby enabling the extension of the proposed framework to other structural members and components in civil engineering. In addition, integrating uncertainty quantification and enhancing robustness under varying environmental conditions remain important challenges for real-world applications. Further investigation is also required to address the effects of measurement noise, sensor placement, and scalability to large-scale structures

Appendices

APPENDIX (A1)

Appendix (A1) provides the crack scenarios used to generate the blind-test data for numerical and experimental validations. These scenarios were designed by considering the actual fabrication of cracked beam specimens and the need to use experimentally measured data for crack assessment.

All crack locations were randomly determined in millimeters, taking manufacturing constraints into account, as the fabrication process allowed a positioning accuracy of up to 0.1 mm. For consistency, the generated locations were normalized relative to the total beam length. Crack locations were randomly generated between $0.05L$ and $0.95L$ to exclude regions near the free ends where cracks have minimal influence on the NMFS. For multiple-crack scenarios, a minimum spacing of 5% of the beam length was applied between adjacent cracks to reflect the practical fabrication constraints. Additionally, all cracks located beyond $0.5L$ were mirrored about the beam's centerline in Appendix (A1), considering the geometric symmetry imposed by the free-free boundary condition.

Table A1.1 One-crack scenarios.

<i>Case No.</i>	a_1/L	d_1/h	<i>Case.no</i>	a_1/L	d_1/h
Case 1	0.201	0.15	Case 7	0.355	0.20
Case 2	0.218	0.10	Case 8	0.385	0.15
Case 3	0.271	0.20	Case 9	0.394	0.15
Case 4	0.271	0.15	Case 10	0.397	0.20
Case 5	0.289	0.15	Case 11	0.429	0.20
Case 6	0.293	0.10	Case 12	0.483	0.20

Note: a_m/L and d_m/h are the normalized crack location and depth of the m^{th} crack, relative to the total beam length (L) and height (h), respectively.

Table A1.2. Two-crack scenarios.

<i>Case No.</i>	a_1/L	d_1/h	a_2/L	d_2/h
Case 1	0.223 *0.667	0.20	0.297	0.10
Case 2	0.212	0.15	0.246 *0.754	0.20
Case 3	0.224	0.15	0.442	0.15
Case 4	0.231 *0.769	0.10	0.469	0.20
Case 5	0.204 *0.796	0.20	0.377 *0.623	0.20
Case 6	0.343 *0.657	0.15	0.412	0.20
Case 7	0.327 *0.673	0.15	0.444 *0.556	0.20
Case 8	0.216	0.15	0.256 *0.744	0.10

Note: a_m/L and d_m/h are the normalized crack location and depth of the m^{th} crack, relative to the total beam length (L) and height (h), respectively.

* Actual fabricated crack locations; unmarked values are ground truth used for crack assessment.

Table A1.3. Three-crack scenarios.

<i>Case No.</i>	a_1/L	d_1/h	a_2/L	d_2/h	a_3/L	d_3/h
Case 1	0.208 *0.792	0.20	0.242	0.15	0.425	0.15
Case 2	0.147 *0.853	0.20	0.334	0.10	0.417 *0.583	0.15
Case 3	0.239	0.20	0.370	0.20	0.497	0.20
Case 4	0.102	0.20	0.335	0.10	0.394 *0.606	0.10
Case 5	0.168 *0.832	0.20	0.352 *0.648	0.15	0.480	0.10
Case 6	0.198	0.20	0.380	0.20	0.484 *0.516	0.15
Case 7	0.201	0.10	0.439 *0.561	0.15	0.452	0.15
Case 8	0.274 *0.726	0.15	0.364	0.20	0.495	0.15
Case 9	0.155	0.10	0.272 *0.728	0.10	0.378 *0.622	0.15
Case 10	0.108 *0.892	0.15	0.410 *0.590	0.15	0.487	0.15

Note: a_m/L and d_m/h are the normalized crack location and depth of the m^{th} crack, relative to the total beam length (L) and height (h), respectively.

* Actual fabricated crack locations; unmarked values are ground truth used for crack assessment.

Table A1.4. Four-crack scenarios.

<i>Case No.</i>	a_1/L	d_1/h	a_2/L	d_2/h	a_3/L	d_3/h	a_4/L	d_4/h
Case 1	0.219	0.20	0.308 *0.692	0.10	0.369 *0.631	0.10	0.445	0.15
Case 2	0.221	0.15	0.329 *0.671	0.15	0.393	0.20	0.483	0.20
Case 3	0.255	0.20	0.313	0.15	0.373	0.15	0.454	0.10
Case 4	0.254	0.15	0.326 *0.674	0.10	0.391	0.20	0.480 *0.520	0.20
Case 5	0.211 *0.789	0.15	0.266 *0.734	0.10	0.339 *0.661	0.10	0.451 *0.549	0.15
Case 6	0.230	0.20	0.316	0.20	0.417	0.15	0.494	0.20
Case 7	0.215	0.10	0.295 *0.705	0.10	0.411 *0.589	0.15	0.480	0.15

Note: a_m/L and d_m/h are the normalized crack location and depth of the m^{th} crack, relative to the total beam length (L) and height (h), respectively.

* Actual fabricated crack locations; unmarked values are ground truth used for crack assessment.

APPENDIX (A2)

Appendix (A2) compares the performance of the various classification models tested during the implementation of the PIML model. All models from the MATLAB Classification Learner were trained on the same NMFS training dataset described in Section 2.4.3. The comparison employed five-fold cross-validation for the reliable evaluation of the classification performance. Models that could not be trained with the prepared dataset were excluded from comparison. Among those tested, the cubic SVM demonstrated the highest accuracy, leading to its selection for the PIML-based crack-count identification. The classification performances of all tested models are summarized in Table A2.1

Table A2.1 Comparison of model performance using the MATLAB Classification Learner.

Category	Model Name	Performance (%)
Decision Trees	Fine Tree	72.9
	Medium Tree	66.2
	Coarse Tree	58.7
Discriminant	Linear Discriminant	53.6
Naive Bayes	Kernel Naive Bayes	50.8
SVM	Linear SVM	62.1
	Quadratic SVM	92.1
	Cubic SVM	96.0
	Fine Gaussian SVM	82.9
	Medium Gaussian SVM	89.6
	Coarse Gaussian SVM	68.3
k-Nearest Neighbors (kNN)	Fine kNN	82.6
	Medium kNN	77.6
	Coarse kNN	70.0
	Cosine kNN	71.3
	Cubic kNN	77.3
	Weighted kNN	81.4
Ensemble	Boosted Trees	73.6
	Bagged Trees	83.7
	Subspace Discriminant	53.5
	Subspace kNN	85.3
	RUSBoosted Trees	66.2

APPENDIX (A3)

Appendix (A3) quantitatively evaluates the engineering judgment regarding ICD and ICS introduced in Section 2.4.2 through a sensitivity analysis based on Monte Carlo simulations. The objective is to verify whether the selected ICD and ICS values correspond to practical observability limits within the NMFS feature space used by the trained cubic SVM.

For each candidate combination of ICD and ICS defined on a predefined grid, 30 independent MC realizations were performed. In each realization, a training dataset of 25,000 multi-crack scenarios was randomly generated with crack locations uniformly distributed in $0.01L$ - $0.99L$ while enforcing the minimum separation constraint (ICS) and crack depths uniformly sampled from ICD to $0.2h$. A cubic SVM was then trained for each realization, and the resulting distribution of training accuracies was used to quantify the sensitivity of classification performance to ICD and ICS.

- Sensitivity to ICD (Figure A3.1(a)): The analysis indicates that $ICD=0.1h$ represents a practical threshold. As the crack depth falls below this value, the median accuracy drops sharply, and the variability (box length) significantly increases. This confirms that the NMFS patterns become insufficiently separable for reliable crack-count classification in this regime. In contrast, the compact box distributions observed for $ICD \geq 0.1h$ indicate that classification accuracy remains robust with respect to variations in the secondary parameter, ICS.

- Sensitivity to ICS (Figure A3.1(b)): Unlike ICD, classification accuracy exhibits a saturation behavior with respect to crack separation. The median accuracy is noticeably lower at $ICS = 0$, reflecting the difficulty in distinguishing closely spaced cracks due to wave interference effects. However, the performance improves and stabilizes (plateaus) once the separation reaches $0.05L$. Beyond this point, increasing the separation yields negligible gains in median accuracy. Accordingly, $ICS = 0.05L$ is identified as the minimum separation required to ensure stable NMFS-based discrimination.

Based on the sensitivity analysis presented in Figure A3.1, $ICD = 0.1h$ and $ICS = 0.05L$ are confirmed as practical observability limits for the proposed NMFS-based PIML classifier.

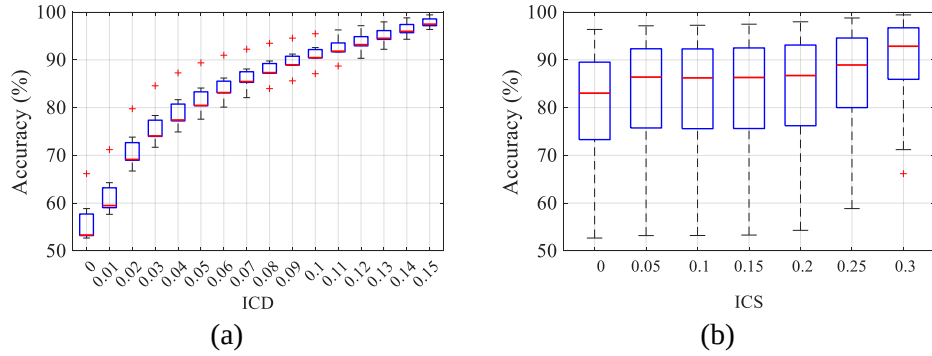


Figure A3.1 Sensitivity analysis of classification accuracy using box plots derived from Monte Carlo simulations. (a) Sensitivity to ICD: Each box represents the distribution of accuracies for a fixed ICD value, aggregated over varying ICS values ($0 < ICS < 0.3L$); (b) Sensitivity to ICS: Each box represents the distribution of accuracies for a fixed ICS value, aggregated over varying ICD values ($0 < ICD < 0.2h$).

APPENDIX (A4)

Appendix (A4) provides the raw data from the crack assessment results obtained from the FE-based blind-test cases. For each crack-count scenario, the results are presented under two conditions: one assuming that the true number of cracks is known a priori and another where it is estimated using the PIML model. The true crack locations, predicted locations, and tolerance ranges for accurate assessment are visually represented. The black vertical lines indicate true crack locations, while shaded areas represent the tolerance range for the correct assessment ($\pm 2.5\%$ from the true location). Symbols ‘○’ and ‘×’ indicate the predicted crack locations when the true number of cracks is known and when it is estimated using the PIML model, respectively. The results demonstrate the performance of the proposed grid optimization method under ideal numerical conditions without measurement uncertainties.

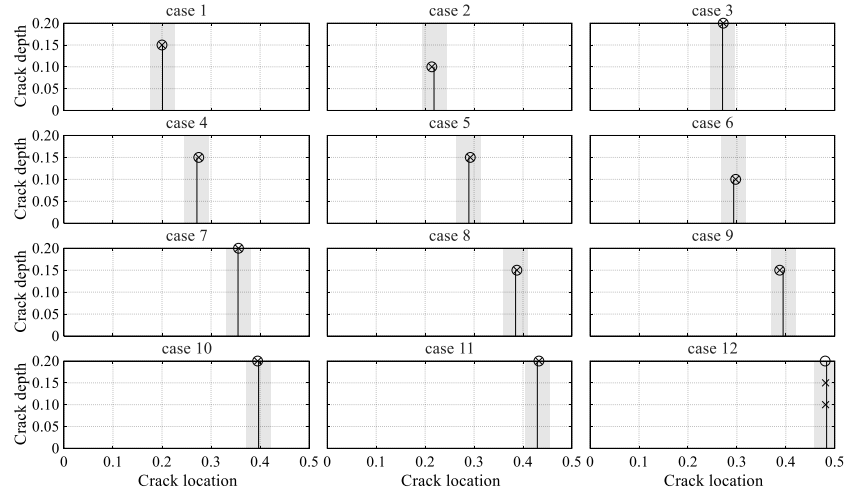


Figure A4.1 Raw data from FE-based blind-test crack assessment for one-crack scenarios (○: predicted crack when the number of cracks is assumed to be known a priori; ×: predicted crack when the number of cracks is identified using the PIML; —: true crack; shaded area: tolerance range for correct assessment)

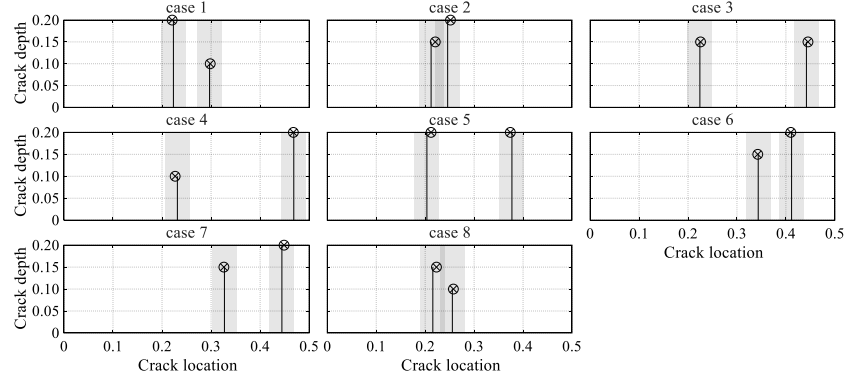


Figure A4.2 Raw data from FE-based blind-test crack assessment for two-crack scenarios (○: predicted crack when the number of cracks is assumed to be known a priori; ×: predicted crack when the number of cracks is identified using the PIML; —: true crack; shaded area: tolerance range for correct assessment)

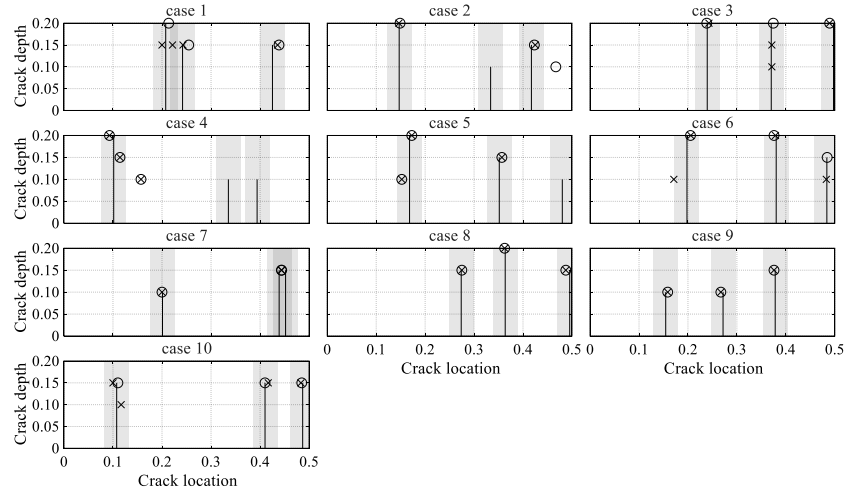


Figure A4.3 Raw data from FE-based blind-test crack assessment for three-crack scenarios (○: predicted crack when the number of cracks is assumed to be known a priori; ×: predicted crack when the number of cracks is identified using the PIML; —: true crack; shaded area: tolerance range for correct assessment)

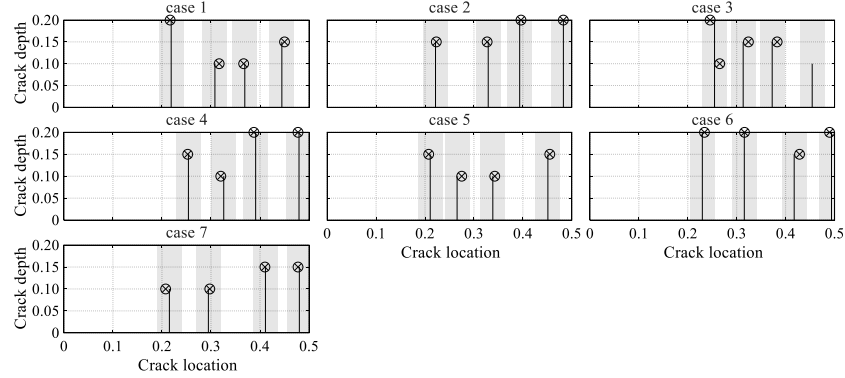


Figure A4.4 Raw data from FE-based blind-test crack assessment for four-crack scenarios (○: predicted crack when the number of cracks is assumed to be known a priori; ×: predicted crack when the number of cracks is identified using the PIML; —: true crack; shaded area: tolerance range for correct assessment)

APPENDIX (A5)

Appendix (A5) presents the raw assessment results from the experimental blind-test cases on the fabricated beam specimens. For each crack-count scenario, the results are presented for conditions when the true number of cracks is assumed to be known a priori and when it is estimated using the PIML model. The true crack locations, predicted locations, and tolerance ranges for a correct assessment are visually depicted. Black vertical lines represent the true crack locations, while shaded areas indicate the tolerance range for correct assessment ($\pm 2.5\%$ from the true location). Symbols ‘o’ and ‘x’ indicate the predicted crack locations when the true number of cracks is known and when it is estimated using the PIML model, respectively. Unlike the FE-based results in Appendix (A4), the experimental results inherently reflect the influence of manufacturing tolerances, measurement noise, and other sources of experimental uncertainty. The results demonstrate the robustness and practical applicability of the proposed grid optimization method under real-world measurement conditions

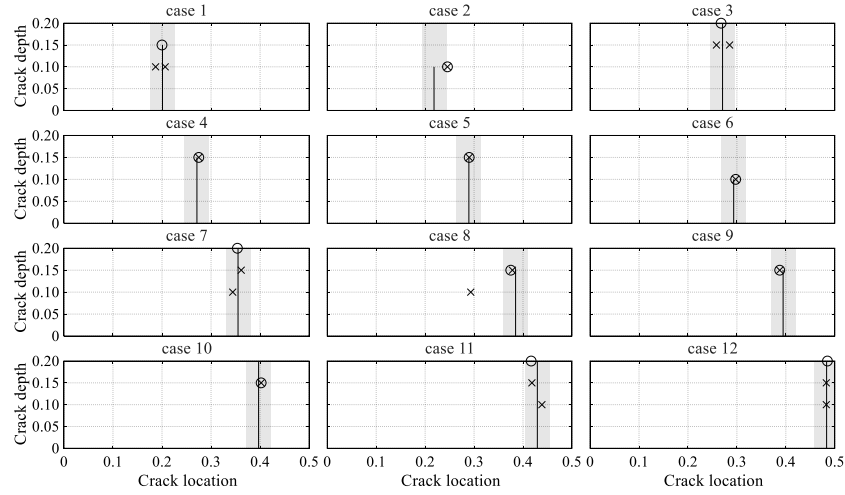


Figure A5.1 Raw data from experimental blind-test crack assessment for one-crack scenarios (○: predicted crack when the number of cracks is assumed to be known a priori; ×: predicted crack when the number of cracks is identified using the PIML; —: true crack; shaded area: tolerance range for correct assessment)

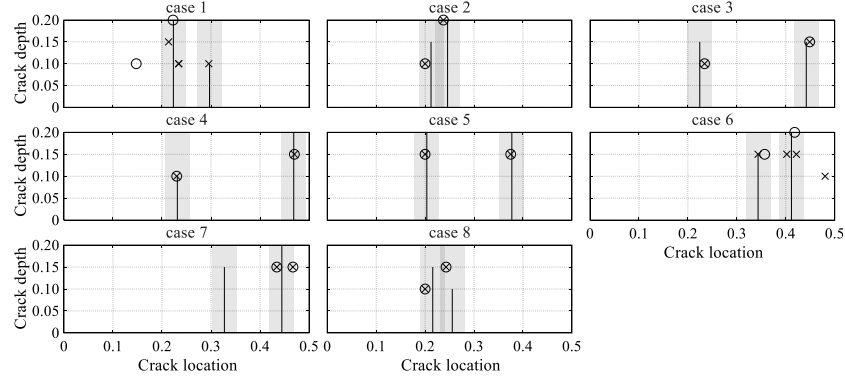


Figure A5.2 Raw data from experimental blind-test crack assessment for two-crack scenarios (○: predicted crack when the number of cracks is assumed to be known a priori; ×: predicted crack when the number of cracks is identified using the PIML; —: true crack; shaded area: tolerance range for correct assessment)

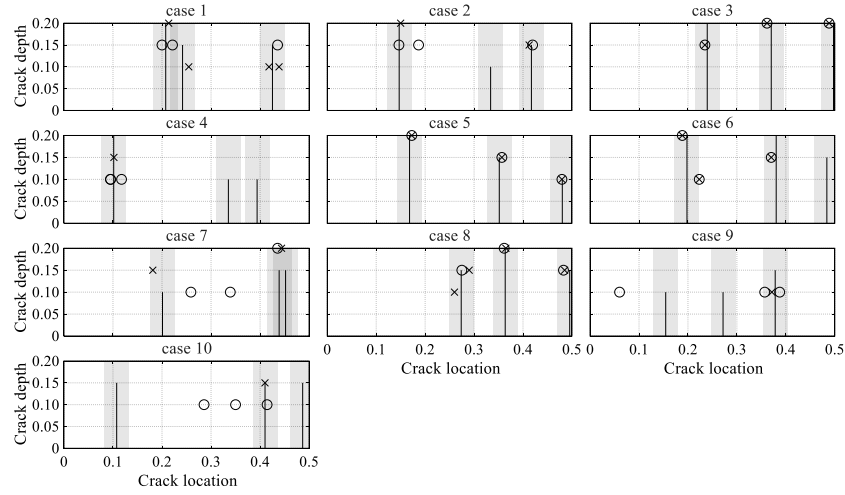


Figure A5.3 Raw data from experimental blind-test crack assessment for three-crack scenarios (○: predicted crack when the number of cracks is assumed to be known a priori; ×: predicted crack when the number of cracks is identified using the PIML; —: true crack; shaded area: tolerance range for correct assessment)

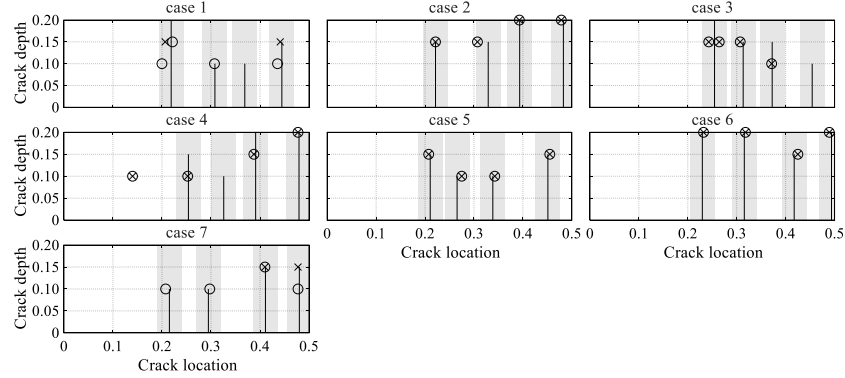


Figure A5.4 Raw data from experimental blind-test crack assessment for four-crack scenarios (○: predicted crack when the number of cracks is assumed to be known a priori; ×: predicted crack when the number of cracks is identified using the PIML; —: true crack shaded area: tolerance range for correct assessment)

APPENDIX (A6)

The numerical solution of Eq. (3.28) for the dimensionless effective prestress force $\hat{P}(t)$ can be calculated by the finite difference method. First, the integral part in Eq. (3.28) is rewritten as in terms of $\frac{\partial \hat{P}(t)}{\partial t}$ as

$$\int_0^t \Gamma \Phi_{\text{cr}}(t, \tau) \frac{\partial \hat{P}(\tau)}{\partial \tau} d\tau = \sum_{i=1}^{n-1} \int_{t_i}^{t_{i+1}} \Gamma \Phi_{\text{cr}}(t, \tau) \frac{\partial \hat{P}(\tau)}{\partial \tau} d\tau \quad (\text{A6.1})$$

where $t_1 = 0$ and $t_n = t$.

Letting constant time increment $\Delta t = t_{i+1} - t_i = \frac{t}{n-1}$ for all i , the finite integrals from t_i to t_{i+1} in Eq. (A6.1) can be approximated through the trapezoidal rule as follows

$$\begin{aligned} \int_{t_i}^{t_{i+1}} \Gamma \Phi_{\text{cr}}(t_n, \tau) \frac{\partial \hat{P}(\tau)}{\partial \tau} d\tau \\ \approx \frac{1}{2} \left\{ \chi_{n,i} \frac{\partial \hat{P}(\tau)}{\partial \tau} \Big|_{\tau=t_i} + \chi_{n,i+1} \frac{\partial \hat{P}(\tau)}{\partial \tau} \Big|_{\tau=t_{i+1}} \right\} \Delta t \end{aligned} \quad (\text{A6.2})$$

where $\chi_{n,i} = \Gamma \Phi_{\text{cr}}(t_n, t_i)$.

The derivative $\frac{\partial \hat{P}(\tau)}{\partial \tau}$ in Eq. (A6.2) can be approximated through forward finite differences as

$$\frac{\partial \hat{P}(\tau)}{\partial \tau} \Big|_{\tau=t_i} \approx \frac{\hat{P}(t_{i+1}) - \hat{P}(t_i)}{\Delta t} \quad (\text{A6.3})$$

Substituting Eq. (A6.3) into Eq. (A6.2) produces

$$\begin{aligned} \int_{t_i}^{t_{i+1}} \Gamma \Phi_{\text{cr}}(t_n, \tau) \frac{\partial \hat{P}(\tau)}{\partial \tau} d\tau \\ \approx \frac{1}{2} [\chi_{n,i+1} \hat{P}(t_{i+2}) + \{\chi_{n,i} - \chi_{n,i+1}\} \hat{P}(t_{i+1}) - \chi_{n,i} \hat{P}(t_i)] \end{aligned} \quad (\text{A6.4})$$

Substituting Eq. (A6.1) incorporated with Eq. (A6.4) into Eq. (3.28) yields the finite difference equation of Eq. (3.28)

$$\begin{aligned} \hat{P}(t_n) \\ = \frac{\varphi(t_n)}{\varphi(t_1)} \\ + \varphi(t_n) [\Phi_{\text{cr}}(t_n, t_1) \{\hat{\sigma}_{\text{pd}} - \Gamma\} - \Phi_{\text{re}}(t_n, t_1) \{1 + \hat{\varepsilon}_{\text{pd}}(t_1)\} + \hat{\varepsilon}_{\text{sh}}(t_n)] \\ - \varphi(t_n) \sum_{i=1}^{n-1} \psi_{n,i} \end{aligned} \quad (\text{A6.5})$$

where $\psi_{n,i} = \frac{1}{2} [\chi_{n,i+1} \hat{P}(t_{i+2}) + \{\chi_{n,i} - \chi_{n,i+1}\} \hat{P}(t_{i+1}) - \chi_{n,i} \hat{P}(t_i)]$.

The numerical procedures to solve Eq. (A6.5) for $\hat{P}(t_n)$ is provided in Appendix (A7)

APPENDIX (A7)

(i) For $t_n = t_1 = 0$, Eq. (A6.5) is expressed for $\hat{P}(t_1)$ as follows

$$\begin{aligned} & \hat{P}(t_1) \\ &= \frac{\varphi(t_1)}{\varphi(t_1)} \\ &+ \varphi(t_1) \left[\underbrace{\Phi_{cr}(t_1, t_1)}_0 \{\hat{\sigma}_{p-d} - \Gamma\} - \underbrace{\Phi_{re}(t_1, t_1)}_0 \{1 + \hat{\varepsilon}_{p-d}(t_1)\} + \underbrace{\hat{\varepsilon}_{sh}(t_1)}_0 \right] \\ &= 1 \end{aligned} \quad (\text{A7.1})$$

Note that $\hat{P}(t_1)$ in Eq. (A7.1) is identical to the initial condition of the dimensionless effective prestress force, i.e. $\hat{P}(0) = 1$.

(ii) For $t_n = t_2 = \Delta t$, Eq. (A6.5) is written for $\hat{P}(t_2)$ as

$$\begin{aligned} & \hat{P}(t_2) \\ &= \frac{\varphi(t_2)}{1 + \frac{1}{2}\varphi(t_2)\chi_{2,1}} \left\{ \frac{1}{\varphi(t_1)} \right. \\ &+ \left[\phi_{2,1}\{\hat{\sigma}_{p-d} - \Gamma\} - \rho_{2,1}\{1 + \hat{\varepsilon}_{p-d}(t_1)\} + \hat{\varepsilon}_{sh}(t_2) \right] + \frac{1}{2}\chi_{2,1}\hat{P}(t_1) \left. \right\} \end{aligned} \quad (\text{A7.2})$$

(iii) $t_n \geq t_3 = 2\Delta t$, Eq. (A6.5) is expressed for $\hat{P}(t_n)$ as follows

$$\begin{aligned} & \hat{P}(t_n) \\ &= \frac{\varphi(t_n)}{1 + \varphi(t_n)\chi_{n,n-1}} \left\{ \frac{1}{\varphi(t_1)} \right. \\ &+ \left[\phi_{n,1}\{\hat{\sigma}_{p-d} - \Gamma\} - \rho_{n,1}\{1 + \hat{\varepsilon}_{p-d}(t_1)\} + \hat{\varepsilon}_{sh}(t_n) \right] \\ &- \frac{1}{2} \left[\{\chi_{n,n-2} - 2\chi_{n,n-1}\}\hat{P}(t_{n-1}) - \chi_{n,n-2}\hat{P}(t_{n-2}) \right] - \sum_{i=1}^{n-3} \psi_{n,i} \left. \right\} \end{aligned} \quad (\text{A7.3})$$

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국문초록

보형 구조물을 위한 물리 정보 기반 기계학습: 금속보의 다중 초기 균열 진단 및 프리스트레스트 콘크리트 보의 시간의존적 거동 예측

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머신러닝(machine learning)은 기존 수치해석 방법을 대체하는 접근법으로서 구조물의 역학적 거동 예측에 활발히 적용되어 왔으나, 그 성능은 대규모의 고품질 데이터 확보 여부에 크게 의존한다. 그러나 측정 방법의 한계와 시험편 수의 제약으로 인해, 실제 구조 부재에서 이러한 양질의 대용량 데이터를 구성하는 것은 불가능에 가깝다. 이러한 데이터 기반(data-driven) 머신러닝의 한계를 극복하기 위해, 물리 지식을 사전 정보로서 학습 과정에 통합하는 물리 지식 기반 머신러닝(Physics-Informed Machine Learning; PIML)이 대안으로 등장하였다. PIML은 데이터가 부족한 조건에서도 물리적 제약 조건을 학습에 추가로 반영함으로써 신뢰할 수 있는 예측을 가능하게 한다. 본 연구는 측정 데이터가 부족하고 이용 가능한 물리 모델이 필연적으로 단순화되어 실제 구조 거동을 완전히 표현할 수 없는 조건에서, 두 가지

기본 문제 유형인 분류(classification)와 회귀(regression) 모두에 적용 가능한 통합 PIML 프레임워크를 제시하고, 각 문제 유형에 대한 구조공학의 대표적인 적용 사례를 통해 그 유효성을 검증하였다.

첫 번째 적용으로, 구조 건전성 모니터링 분야의 대표적인 분류 문제로서 보 구조물의 다중 초기 균열 평가를 위한 PIML 기반 프레임워크를 제안하였다. 위상 폐합 원리에 기초한 닫힌 형태의 주파수 방정식을 대리 모델(surrogate model)로 활용하여 대규모의 광대역 모드 진동수 감소비 학습 데이터셋을 효율적으로 증강하였으며, 증강된 데이터는 레이블 모호성을 제거하는 데이터 정제(data refinement) 과정을 거쳐 학습에 활용되었다. 정제된 데이터를 기반으로 분류 학습기를 통해 최적 알고리즘을 선정하여 균열 개수 식별이 가능한 PIML 모델을 구축하였으며, 식별된 균열 개수를 기반으로 격자 탐색(grid search) 기법을 적용하여 균열 위치 및 깊이를 진단하였다. 제안된 프레임워크는 유한요소 기반 데이터와 부착형 압전소자를 통해 측정된 실험 데이터를 활용한 블라인드 테스트를 통해 검증되었으며, 균열 수에 대한 사전 정보 없이도 신뢰할 수 있는 균열 평가 성능을 입증하였다.

두 번째 적용으로, 구조 성능 예측 분야의 대표적인 회귀 문제로서 PSC 거더의 초기 재령 시간의존적 거동 예측 및 변형 형상 재구성을 위한 물리 지식 기반 인공신경망(Physics-Informed Neural Network; PINN)을 도입하였다. 콘크리트 크리프, 건조 수축, 긴장재 릴렉сей션 및 탄성 계수 변화등 시간의존적 거동을 고려한 지배 방정식을

무차원화하여 신경망 과 결합하여 시간에 따른 긴장력 손실을 예측하기 위한 PINN 모델을 구현하였다. 또한, 곡률-처짐 관계식을 신경망에 통합한 PINN 모델을 별도로 구성하였으며, 앞서 예측된 유효 긴장력을 설계변수로 입력하면 특정 시점에서의 캠버 형상을 재구성할 수 있다. 긴장력 손실 예측은 유한차분법으로부터 도출된 해와 비교하여 검증하였으며, 캠버 형상 재구성은 실제 PSC 거더의 측정변형률로부터 재구성된 캠버 데이터와의 비교를 통해 검증하였다. 제한된 데이터 조건에서도 정확한 긴장력 손실 예측과 물리적으로 일관된 변형 형상 재구성 성능을 입증하였다.

두 가지 적용 사례는 물리 지식 기반 머신러닝이 데이터가 부족한 구조공학 문제에 대한 실용적이고 통합적인 접근법임을 공통적으로 확립하며, 서로 다른 두 가지 문제 유형에 걸쳐 일관된 유효성을 입증하였다.

주요어: 물리지식기반 머신러닝(PIML); 물리 지식기반 인공신경망 (PINN); 분류; 회귀; 광대역 모드 진동수; 다중 초기 균열; 균열 진단; 시간 의존적 거동; 캠버 재구성; 구조 건전성 모니터링

VITA

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